

Vapor–Liquid Critical Properties of Elements and Compounds. 12. Review of Recent Data for Hydrocarbons and Non-hydrocarbons

Douglas Ambrose,[#] Constantine Tsonopoulos,[†] Eugene D. Nikitin,[‡] David W. Morton,[§] and Kenneth N. Marsh^{*,†,¶,||}

[†]18 Dorothy Drive, Morristown, New Jersey 07960, United States,

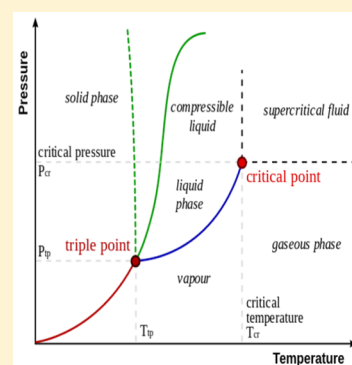
[‡]Institute of Thermal Physics, Ural Branch of the Russian Academy of Sciences, Amundsena Street 106, 620016 Ekaterinburg, Russia

[§]School of Pharmacy and Applied Sciences, La Trobe Institute of Molecular Sciences, La Trobe University, Edwards Road, Bendigo, Victoria 3550, Australia

[‡]Department of Mechanical and Chemical Engineering, University of Western Australia, Crawley, Western Australia 6009, Australia

^{||}Australia and Department of Chemical and Process Engineering, University of Canterbury, Christchurch 8140, New Zealand

ABSTRACT: This review is Part 12 of a series of contributions initially sponsored by the critical properties group of the previous IUPAC Commission I.2 on Thermodynamics, Subcommittee on Thermodynamic Data and then by IUPAC Project #2000-026-1-100, Critical Compilation of Vapor Liquid Critical Properties for the Physical and Biophysical Chemistry Division as a project of the International Association of Chemical Thermodynamics. It presents experimental data for the critical constants of organic compounds that are either new data published since, or data that was not included in, the initial reports. Where necessary revised recommendations are given, together with their uncertainties. Critical temperatures have been converted, where warranted, to the ITS-90 scale.



INTRODUCTION

Part 11, organic compounds containing B + O; halogen + N, + O, + O + S; + Si; N + O; O + S, + Si completed the evaluation of experimental data for the critical properties of organic compounds by examining compounds containing two or more “heteroatoms” (halogens counted as one). Part 12 is the final paper in the series and includes all the published critical property data since the first 11 reviews were published. This review also includes data that was not included in the initial reviews. The presentation and evaluation of the experimental data follow the guidelines of Ambrose et al. in parts 1 and 2 of this series: [95-amb/you] (introductory survey) and [95-amb/tso] (normal alkanes). Succeeding parts have been Part 3 by Tsonopoulos and Ambrose [95-tso/amb] (aromatic hydrocarbons), Part 4 by Gude and Teja [95-gud/tej] (aliphatic alkanols), Part 5 by Daubert [96-dau] (branched alkanes and cycloalkanes), Part 6 by Tsonopoulos and Ambrose [96-tso/amb] (unsaturated aliphatic hydrocarbons), Part 7 by Kudchadker et al. [2001-kud/amb] (oxygen compounds other than alkanols and cycloalkanols), Part 8 by Tsonopoulos and Ambrose [2001-tso/amb] (organic sulfur, silicon, and tin compounds), Part 9 by Marsh et al. [2006-mar/you] (organic nitrogen compounds), Part 10 by Marsh et al. [2007-mar/abr] (organic halogen compounds), and Part 11 by Ambrose et al. [2009-amb/tso] (organic compounds containing B + O; halogen + N, + O, + O + S; + Si; N + O; O + S, + Si).

The recommended values in Table 1 are for all the compounds listed in Table 2. In Table 2 the previous recommendations are given in bold type, and if there are more than one set of previously unrecorded measurements on a compound the recommendations in this review are also given in bold type. All the recent data as well as data not given in the previous reviews are given in Table 2. Within a group, the compounds are listed in the order used by *Chemical Abstracts* (Hill order). Included in Table 2 are, when given, the authors estimate of the uncertainty of the value and the method of measurement. The keys to methods of measurement of the critical properties are given in Table 3. Critical temperatures enclosed in parentheses are not new measurements; they are previous literature values at which investigators determined the critical pressure (with method 5 or 6) and/or the critical density (with method 7 or 9). The references follow the format [year-first three letters of first author/first three letters of second author and, where required, a sequence number]. All four digits are given for the year starting with 2000 and for pre-1900 sources. References in this format will not be linked to the

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Table 1. Recommended Critical Properties of Organic Compounds

	molar mass M g·mol ^{-1a}	T_c K ^b	(±)	p_c MPa	(±)	ρ_c g·cm ⁻³	(±)	V_c cm ³ ·mol ⁻¹	Z_c^c
Alkanes									
methane	16.043	190.564	0.015	4.599	0.003	0.1627	0.0005	98.60	0.286
ethane	30.070	305.32	0.04	4.872	0.01	0.2062	0.002	145.8	0.280
propane	44.097	369.83	0.1	4.248	0.01	0.221	0.003	199.5	0.276
butane	58.124	425.10	0.1	3.792	0.01	0.228	0.003	254.9	0.274
pentane	72.151	469.7	0.2	3.370	0.02	0.232	0.003	311.0	0.268
hexane	86.178	507.6	0.2	3.027	0.02	0.234	0.003	368.3	0.264
heptane	100.205	540.2	0.3	2.74	0.03	0.234	0.003	428.2	0.261
octane	114.232	568.7	0.3	2.49	0.02	0.232	0.003	492.4	0.259
nonane	128.259	594.6	0.6	2.29	0.02	0.231	0.005	555.2	0.257
decane	142.286	617.7	0.5	2.11	0.05	0.228	0.005	624.1	0.256
dodecane	170.340	658	1	1.81	0.1	0.226	0.01	753.7	0.249
nonadecane	268.529	756	8	1.16	0.02				
eicosane	282.556	768	8	1.08	0.02				
heneicosane	296.583	779	10	1.04	0.05				
docasane	310.610	787	8	0.99	0.02				
tricosane	324.637	791	10	0.92	0.05				
tetracosane	338.664	800	8	0.87	0.02				
hexacosane	366.718	819	10	0.81	0.05				
octacosane	394.772	836	10	0.75	0.05				
triacontane	422.826	850	10	0.69	0.05				
hexatriacontane	506.988	888	15	0.58	0.05				
tetracontane	563.096	904	15	0.52	0.05				
tetratetracontane	619.204	919	15	0.48	0.05				
hexacontane	843.636	974	15	0.36	0.05				
Branched Alkanes									
2-methylpropane	58.124	407.81	0.05	3.633	0.005	0.226	0.005	257.2	0.276
2-methylbutane	72.151	460.4	0.5	3.41	0.03	0.236	0.002	305.7	0.272
2,2-dimethylpropane	72.151	433.8	0.1	3.196	0.01	0.235	0.003	307.0	0.272
2,2-dimethylbutane	86.178	489.0	0.3	3.10	0.02	0.241	0.004	357.6	0.273
2-methylheptane	114.232	559.7	0.1	2.50	0.02	0.234	0.002	488.2	0.262
2,2,4-trimethylpentane	114.232	544.0	0.3	2.57	0.02	0.244	0.002	468.2	0.266
2-methyloctane	128.258	582.8	0.1	2.31	0.02				
2,6,10,15,19,23-hexamethyltetracosane	422.826	810	15	0.728	0.03				
Cycloalkanes									
cyclopentane	70.135	511.7	0.2	4.51	0.04	0.270	0.004	259.8	0.275
cyclohexane	84.162	553.6	0.5	4.08	0.05	0.273	0.002	308.3	0.273
cyclooctane	112.216	647.4	0.5	3.56	0.04	0.274	0.005	409.5	0.271
methylcyclohexane	98.189	572.2	0.2	3.48	0.02	0.267	0.002	367.7	0.268
ethylcyclohexane	112.216	606.5	0.5	3.13	0.05	0.259	0.005	433.3	0.269
(Z)-1,2-dimethylcyclohexane	112.216	607.3	0.5	3.35	0.05	0.276	0.005	406.6	0.270
(Z)-1,3-dimethylcyclohexane	112.216	587.0	0.5	3.03	0.05	0.265	0.005	423.5	0.263
(Z)-1,4-dimethylcyclohexane	112.216	603.2	0.2	3.16	0.02	0.258	0.002	434.9	0.274
(E)-1,2-dimethylcyclohexane	112.216	596.3	0.5	3.08	0.05	0.258	0.005	434.9	0.333
(E)-1,4-dimethylcyclohexane	112.216	587	1	3.06	0.1	0.267	0.005	420.3	0.264
propylcyclohexane	126.243	631.0	1	2.86	0.10				
(1-methylethyl)cyclohexane	126.243	632.0	0.5	2.95	0.05	0.260	0.005	485.6	0.273
butylcyclohexane	140.270	650	2	2.53	0.05	0.250	0.01	561.1	0.263
(2-methylpropyl)cyclohexane	140.270	642.0	2						
(1,1-dimethylethyl)cyclohexane	140.270	648	2	2.77	0.05	0.278	0.01	504.6	0.259
1,1'-bicyclopentyl	138.254	690	2	3.3	0.1	0.27	0.02	512.1	0.295
1,3-dimethyltricyclo[3.3.1.1(3,7)]decane	164.292	707	2	3.0	0.2	0.29	0.02	566.5	0.289
1,3,5-trimethyltricyclo[3.3.1.1(3,7)]decane	178.319	702	2						
Unsaturated Aliphatic Hydrocarbons									
ethene	28.054	282.34	0.02	5.041	0.004	0.214	0.002	131.1	0.282
propene	42.081	364.9	0.4	4.60	0.03	0.228	0.005	184.6	0.280
1-butene	56.108	419.4	0.2	4.015	0.02	0.235	0.007	238.8	0.275
1-hexene	84.162	504.0	0.3	3.21	0.03	0.237	0.005	355.1	0.272
1-heptene	98.189	537.3	0.4	2.92	0.04	0.24	0.01	409.1	0.267

Table 1. continued

	molar mass M g·mol ^{-1a}	T_c K ^b	(±)	p_c MPa	(±)	ρ_c g·cm ⁻³	(±)	V_c cm ³ ·mol ⁻¹	Z_c^c
Unsaturated Aliphatic Hydrocarbons									
1-tridecene	182.351	673	7	1.73	0.03				
1-tetradecene	196.378	691	7	1.59	0.03				
1-pentadecene	210.405	705	7	1.54	0.03				
1-hexadecene	224.432	718	7	1.39	0.03				
1-heptadecene	238.459	734	7	1.34	0.03				
1-octadecene	252.486	748	7	1.30	0.03				
1-nonadecene	266.513	755	8	1.19	0.02				
1-eicosene	280.540	772	10	1.14	0.1				
(Z)-2-butene	56.108	435.65	0.10	4.215	0.010	0.239	0.002	234.8	0.273
(E)-2-butene	56.108	428.6	0.10	4.07	0.04	0.236	0.003	237.7	0.271
2-methyl-1-propene	56.108	418.0	0.1	4.005	0.010	0.234	0.001	239.8	0.276
(Z)-2-hexene	84.162	513.4	1						
(Z)-3-hexene	84.162	510.2	1						
(E)-2-hexene	84.162	509.0	1						
(E)-3-hexene	84.162	507.4	1						
2-methyl-1-pentene	84.162	521.0	1						
4-methyl-1-pentene	84.162	494	2	3.29	0.1	0.238	0.01	353.6	0.283
2-methyl-2-pentene	84.162	509.3	1						
(Z)-4-methyl-2-pentene	84.162	496.3	1						
2,3-dimethyl-1-butene	84.162	497.7	1						
3,3-dimethyl-1-butene	84.162	477.4	1						
2,3-dimethyl-2-butene	84.162	521.0	1						
(Z)-2-heptene	98.189	548.5	1						
(E)-2-heptene	98.189	542.8	1						
(E)-3-heptene	98.189	538.6	1						
2-methyl-1-hexene	98.189	541.8	1						
5-methyl-1-hexene	98.189	528.7	1						
2,3-dimethyl-1-pentene	98.189	533.6	1						
4,4-dimethyl-1-pentene	98.189	516	2						
(E)-2-octene	112.216	569.8	1						
(E)-4-octene	112.216	566.3	1						
2-methyl-1-heptene	112.216	567.5	1						
2-methyl-2-heptene	112.216	568.9	1						
2,4,4-trimethyl-1-pentene	112.216	550.7	2.0	2.73	0.06				
cyclopentane	68.119	506.5	0.5	4.80	0.05	0.278	0.010	245.0	0.279
cyclohexene	82.146	560.4	0.1	4.45	0.03	0.28	0.01	293.4	0.280
Aromatic Hydrocarbons									
benzene	78.114	562.05	0.07	4.895	0.006	0.305	0.004	256.1	0.268
methylbenzene	92.141	591.75	0.15	4.108	0.010	0.292	0.004	315.6	0.263
ethylbenzene	106.168	617.15	0.1	3.609	0.010	0.284	0.004	373.8	0.263
butylbenzene	134.222	660.5	0.5	2.89	0.04	0.270	0.010	497.1	0.262
pentylbenzene	148.249	675	7	2.58	0.08				
hexylbenzene	162.276	695	7	2.35	0.07				
heptylbenzene	176.303	708	7	2.14	0.06				
octylbenzene	190.330	725	7	1.98	0.06				
decylbenzene	218.384	752	8	1.72	0.05				
undecylbenzene	232.411	763	8	1.64	0.05				
tridecylbenzene	260.465	790	8	1.54	0.05				
1,2-dimethylbenzene	106.168	630.3	0.5	3.732	0.04	0.287	0.004	369.9	0.263
1,3-dimethylbenzene	106.168	617.0	0.5	3.541	0.04	0.283	0.004	375.2	0.259
1,4-dimethylbenzene	106.168	616.2	0.2	3.511	0.04	0.281	0.004	377.8	0.259
(1-methylethyl)benzene	120.195	631.0	0.5	3.209	0.04	0.282	0.005	426.2	0.261
(1-methylpropyl)benzene	134.222	652.8	0.4	3.027	0.04	0.278	0.005	482.8	0.269
(1,1-dimethylethyl)benzene	134.222	648	1	2.95	0.05	0.280	0.005	479.4	0.262
1,4-bis(1-methylethyl)benzene	162.276	676	2	2.35	0.10	0.263	0.010	617.0	0.258
1,3,5-triethylbenzene	162.276	679	1	2.435	0.10	0.260	0.010	624.1	0.269
1,3-bis(1,1-dimethylethyl)benzene	190.330	687.6	2						
1,4-bis(1,1-dimethylethyl)benzene	190.330	708	2	2.30	0.10	0.260	0.010	732.0	0.286

Table 1. continued

	molar mass M g·mol ^{-1a}	T_c K ^b	(±)	p_c MPa	(±)	ρ_c g·cm ⁻³	(±)	V_c cm ³ ·mol ⁻¹	Z_c^c
Aromatic Hydrocarbons									
1,3-bis(1,1-dimethylethyl)-5-methylbenzene	204.357	695.5	2						
1,3,5-tris(1,1-dimethylethyl)benzene	246.438	700.5	2						
1,2,4,5-tetrakis(1-methylethyl)benzene	246.438	703	2	1.6	0.3	0.27	0.03	912.7	0.250
ethenylbenzene	104.152	637	2	3.95	0.05	0.295	0.010	353.1	0.263
fluorene	166.223	836	8	3.18	0.1				
diphenylmethane	168.239	770	10	2.9	0.3	0.30	0.03	560.8	0.256
1,3-diphenylbenzene	230.310	882	10	2.2	0.3	0.32	0.04	719.7	0.216
naphthalene	128.174	748.4	0.5	4.05	0.05	0.315	0.005		
1-methylnaphthalene	142.201	772	1	3.57	0.10				
2-methylnaphthalene	142.201	761	1	3.30	0.10	0.308	0.002	461.7	0.241
1,3-dimethylnaphthalene	156.228	772	8	2.74	0.08				
1,4-dimethylnaphthalene	156.228	778	8	2.79	0.08				
2-ethylnaphthalene	156.228	763	8	2.79	0.08				
acenaphthene	154.212	820	8	3.70	0.1				
anthracene	178.234	894	9	3.17	0.1				
phenanthrene	178.234	893	9	3.25	0.1				
pyrene	202.256	974	10	3.03	0.09				
1,2,3,4-tetrahydronaphthalene	132.206	720	1	3.65	0.1	0.324	0.010	621.2	0.379
Aliphatic C—H—O compounds									
Aliphatic Alkanols									
methanol	32.042	512.5	0.2	8.084	0.02	0.273	0.002	117.4	0.223
ethanol	46.069	513.9	0.1	6.14	0.02	0.275	0.002	167.5	0.241
1-propanol	60.096	536.8	0.3	5.175	0.007	0.276	0.003	217.7	0.252
1-butanol	74.123	563.1	0.1	4.41	0.01	0.270	0.002	274.5	0.259
1-pentanol	88.150	588.1	0.5	3.897	0.02	0.270	0.003	326.5	0.260
1-hexanol	102.177	610.3	0.5	3.417	0.02	0.264	0.005	387.0	0.261
1-heptanol	116.204	632.6	0.5	3.058	0.02	0.267	0.005	435.2	0.253
1-tridecanol	200.366	732	7	1.79	0.05				
1-tetradecanol	214.393	743	7	1.67	0.05				
1-pentadecanol	228.420	757	8	1.60	0.05				
1-hexadecanol	242.447	770	8	1.46	0.04				
1-heptadecanol	256.474	780	8	1.35	0.04				
1-octadecanol	270.501	790	8	1.28	0.04				
1-eicosanol	298.555	808	8	1.15	0.03				
1-docosanol	326.609	827	8	1.07	0.03				
Other Alkanols									
2-propanol	60.096	508.3	0.3	4.764	0.02	0.271	0.004	221.8	0.250
2-propen-1-ol	58.080	543	5	5.70	0.2				
2-butanol	74.123	536.2	0.3	4.202	0.02	0.276	0.004	268.6	0.253
2-methyl-2-propanol	74.123	506.2	0.3	3.972	0.02	0.270	0.005	274.5	0.259
2-pentanol	88.150	560.3	0.5	3.675	0.02	0.268	0.015	328.9	0.259
3-methyl-1-butanol	88.150	577.2	0.5	3.93	0.02				
3-hexanol	102.177	582.4	0.5	3.33	0.02	0.271	0.005	377.0	0.259
cyclohexanol	100.161	648	1	4.25	0.1	0.30	0.01	333.9	0.263
1-(1,1-dimethylethoxy)-2-propanol	132.203	600.7	2	2.72	0.05				
1,2-propanediol	76.095	676.4	0.6	6.0	0.01	0.31	0.03	245.5	0.262
1,3-propanediol	76.095	720	3	6.4	0.10				
1,3-butanediol	76.095	690	8	5.0	0.3	0.30	0.02	253.7	0.221
1,4-butanediol	76.095	725	2	5.8	0.4				
2-methyl-1,3-propanediol	90.122	708	2	5.4	0.4				
1,5-pentanediol	90.122	731	11	4.00	0.16				
2,2-dimethyl-1,3-propanediol	90.122	687	10	4.20	0.2				
3-oxa-1,5-pentanediol	104.149	687	10	4.20	0.2				
1,6-hexanediol	118.176	740	3	4.0	0.1				
4-oxa-1,7-heptanediol	134.175	705	3	3.4	0.1				
1,8-octanediol	146.230	752	11	2.70	0.14				
1,9-nonanediol	160.257	760	11	2.40	0.10				
1,10-decanediol	174.284	770	12	2.20	0.09				

Table 1. continued

	molar mass M g·mol ^{-1a}	T_c K ^b	(±)	p_c MPa	(±)	ρ_c g·cm ⁻³	(±)	V_c cm ³ ·mol ⁻¹	Z_c^c
Other Alkanols									
1,12-dodecanediol	202.338	798	12	1.93	0.08				
Alkanals									
propanal	58.080	504	2	5.15	0.10	0.285	0.005	203.8	0.250
butanal	72.107	537	3	4.32	0.2	0.280	0.005	257.5	0.249
2-methylbutanal	86.134	532	8	3.71	0.2				
paraldehyde	132.159	534.43	8	4.04	0.2				
Alkanones									
2-propanone	58.080	508.1	0.1	4.70	0.01	0.273	0.005	212.7	0.237
2-butanone	72.107	536.7	0.1	4.207	0.01	0.270	0.025	267.1	0.252
5-hexene-2-one	98.145	593	1	4.6	0.1				
Acids									
ethanoic acid	60.052	590.7	1	5.78	0.02	0.351	0.005	171.1	0.201
propanoic acid	74.079	598.5	1	4.67	0.05	0.318	0.005	233.0	0.219
hexanoic acid	116.160	655	2	3.38	0.1	0.281	0.005	413.4	0.257
octanoic acid	144.214	693	2	2.87	0.1	0.278	0.005	518.8	0.258
undecanoic acid	186.295	728	7	2.13	0.06				
dodecanoic acid	200.322	743	7	1.93	0.06				
tetradecanoic acid	228.376	763	8	1.64	0.05				
pentadecanoic acid	242.403	777	8	1.57	0.05				
hexadecanoic acid	256.430	785	8	1.49	0.05				
heptadecanoic acid	270.457	792	8	1.37	0.05				
octadecanoic acid	284.484	803	8	1.33	0.04				
eicosanoic acid	312.538	820	8	1.20	0.04				
docosanoic acid	340.592	837	8	1.11	0.04				
butanedioic acid	118.088	851	20	6.59	0.03				
pentanedioic acid	132.115	840	13	4.27	0.30				
hexanedioic acid	146.142	841	13	3.85	0.20				
heptanedioic acid	160.169	842	13	3.28	0.15				
octanedioic acid	174.196	843	13	2.97	0.13				
nonanedioic acid	188.223	844	13	2.72	0.12				
decanedioic acid	202.250	845	13	2.50	0.11				
dodecanedioic acid	230.304	859	13	2.15	0.10				
tetradecanedioic acid	258.358	862	13	1.90	0.09				
Esters									
ethyl methanoate	74.079	507.5	1	4.74	0.05	0.323	0.005	229.3	0.258
propyl methanoate	88.106	538.0	1	4.06	0.05	0.309	0.005	285.1	0.259
1-methylethyl methanoate	88.106	520	3	3.9	0.1				
butyl methanoate	102.133	566	2						
2-methylpropyl methanoate	102.133	552	1						
pentyl methanoate	116.160	589	8	3.46	0.05	0.29	0.01	400.6	0.283
hexyl methanoate	130.187	614	8						
heptyl methanoate	144.214	636	8						
octyl methanoate	158.241	654	8						
ethyl ethanoate	88.106	523.3	0.1	3.84	0.01	0.308	0.005	286.1	0.252
ethenyl ethanoate	86.090	519.2	0.1	4.16	0.02				
1,1-dimethylethyl ethanoate	116.160	541.2	2	3.04	0.06				
2-(2-ethoxyethoxy)ethyl ethanoate	176.212	668	5	2.63	0.10				
2-(2-butoxyethoxy)ethyl ethanoate	204.266	688	6	2.35	0.1				
ethyl propanoate	102.133	546.3	0.4	3.39	0.2	0.30	0.01	340.4	0.254
ethyl-2,2-dimethyl propanoate	130.187	566	2	2.95	0.1	0.28	0.01	465.0	0.291
ethyl butanoate	116.160	569.6	0.3	2.90	0.05	0.28	0.02	414.9	0.254
ethyl pentanoate	130.187	594.0	0.1	2.73	0.02				
methyl hexanoate	130.187	612	6	2.88	0.09				
ethyl hexanoate	144.214	615.2	0.1	2.53	0.02				
methyl heptanoate	144.214	628	1	2.53	0.08				
ethyl heptanoate	158.241	635	0.5	2.36	0.02				
methyl octanoate	158.241	646	6	2.34	0.07				
ethyl octanoate	172.268	655	1	2.16	0.02				

Table 1. continued

	molar mass M g·mol ^{-1a}	T_c K ^b	(±)	p_c MPa	(±)	ρ_c g·cm ⁻³	(±)	V_c cm ³ ·mol ⁻¹	Z_c^c
Esters									
methyl nonanoate	172.268	665	7	2.06	0.06				
methyl decanoate	186.295	675	7	1.93	0.06				
methyl undecanoate	200.322	694	7	1.75	0.05				
methyl dodecanoate	214.349	712	2	1.52	0.05				
methyl tetradecanoate	242.403	730	7	1.32	0.04				
methyl hexadecanoate	270.457	760	8	1.17	0.04				
methyl octadecanoate	298.511	785	8	1.08	0.04				
(E)-ethyl 2-butenolate	114.14	593.8	0.5	3.67	0.04				
butyl propenoate	128.171	598	1	2.91	0.04				
methyl 2-methyl-2-propenoate	100.117	540.3	0.5	3.87	0.04				
(Z)-9-methyl 2-octadecenoate	296.495	777	8	1.21	0.04				
dimethyl ethanedioate	118.088	637	6	4.00	0.15				
dimethyl propanedioate	132.115	646	4	3.56	0.15	0.35	0.02	377.5	0.250
dimethyl butanedioate	146.142	662	7	3.06	0.09				
dimethyl pentanedioate	160.169	682	7	2.77	0.08				
diethyl ethanedioate	146.142	646	2	3.1	0.1	0.33	0.02	442.9	0.256
diethyl propanedioate	160.169	652	7	2.67	0.08				
dimethyl hexanedioate	174.196	692	7	2.50	0.08				
diethyl butanedioate	174.196	665	7	2.20	0.04				
dimethyl heptanedioate	188.223	711	7	2.37	0.07				
diethyl pentanedioate	188.223	683	7	2.11	0.04				
dimethyl octanedioate	202.250	723	7	2.30	0.07				
diethyl hexanedioate	202.250	696	7	2.05	0.04				
dimethyl decanedioate	230.304	742	7	2.08	0.06				
1,1-diacetoxyethane	146.142	618.2	2	2.97	0.06				
Ethers									
dimethyl ether	46.069	403.3	0.1	5.34	0.02	0.274	0.004	168.1	0.268
diethyl ether	74.123	466.8	0.1	3.63	0.02	0.261	0.003	284.0	0.266
2-methoxy-2-methylpropane	80.150	497.1	0.2	3.43	0.01				
1,2-dimethoxyethane	90.122	537	3	3.88	0.1	0.292	0.008	308.6	0.268
dipropyl ether	102.177	531	1	3.03	0.05				
2-methoxy-2-methylbutane	102.77	535	1	3.20	0.05	0.27	0.02	380.6	0.274
dihexyl ether	186.339	665	7	1.44	0.04				
dioctyl ether	242.447	723	7	1.19	0.04				
didecyl ether	298.555	768	8	1.03	0.03				
Aromatic C–H–O Compounds									
phenol	94.113	694.2	0.2	5.93	0.05				
1,3-benzenediol	110.112	836	2	6.24	0.12				
phenylmethanol	108.140	714	3	4.3	0.1				
2-phenyl-1-ethanol	122.167	724	2	3.95	0.08				
3-phenyl-1-propanol	136.194	732	7	3.46	0.10				
4-phenyl-1-butanol	150.221	746	7	3.07	0.1				
4-(methylphenyl)methanol	122.167	719	7	3.61	0.1				
phenylmethanoic acid	122.123	755	8	4.53	0.1				
phenylethanoic acid	136.150	766	8	3.95	0.14				
2-methylbenzoic acid	136.150	763	8	3.92	0.12				
3-methylbenzoic acid	136.150	771	8	3.85	0.12				
4-methylbenzoic acid	136.150	775	8	3.80	0.10				
3-phenylpropanoic acid	150.177	776	8	3.46	0.12				
4-phenylbutanoic acid	164.204	783	8	3.21	0.11				
5-phenylpentanoic acid	178.231	790	8	2.95	0.10				
6-phenylhexanoic acid	192.258	794	8	2.60	0.09				
7-phenylheptanoic acid	206.285	798	8	2.47	0.09				
2-carboxybenzaldehyde	150.133	864	9	4.80	0.15				
methyl benzoate	136.150	700	5	3.6	0.2				
ethyl benzoate	150.177	700	7	3.05	0.09				
propyl benzoate	164.204	710	7	2.60	0.08				
butyl benzoate	178.231	725	7	2.40	0.07				

Table 1. continued

	molar mass M g·mol ^{-1a}	T_c K ^b	(±)	p_c MPa	(±)	ρ_c g·cm ⁻³	(±)	V_c cm ³ ·mol ⁻¹	Z_c^c
Aromatic C—H—O Compounds									
pentyl benzoate	192.258	736	7	2.19	0.07				
hexyl benzoate	206.285	748	7	1.98	0.06				
phenyl ethanoate	136.150	686	2	3.59	0.07				
dimethyl-1,2-benzenedicarboxylate	194.186	772	9	2.77	0.1				
diethyl-1,2-benzenedicarboxylate	222.240	776	9	2.26	0.08				
dipropyl-1,2-benzenedicarboxylate	250.294	784	9	1.90	0.07				
dibutyl-1,2-benzenedicarboxylate	278.348	797	9	1.66	0.06				
dipentyl-1,2-benzenedicarboxylate	306.402	811	9	1.43	0.05				
dihexyl-1,2-benzenedicarboxylate	334.456	817	9	1.34	0.05				
diheptyl-1,2-benzenedicarboxylate	362.510	830	9	1.24	0.04				
dioctyl-1,2-benzenedicarboxylate	390.564	840	9	1.08	0.04				
bis(2-ethylhexyl)-1,2-benzenedicarboxylate	390.564	835	9	1.07	0.04				
dinonyl-1,2-benzenedicarboxylate	418.618	858	9	1.02	0.04				
didecyl-1,2-benzenedicarboxylate	446.672	870	10	0.94	0.03				
diundecyl-1,2-benzenedicarboxylate	474.726	886	10	0.89	0.03				
Benzaldehyde	106.124	693	7	4.41	0.1				
4-methylbenzaldehyde	120.151	712	7	3.80	0.1				
Heterocyclic C—H—O Compounds									
1,2-epoxy-2-methylpropane	72.107	500	2	4.35	0.09				
γ -butyrolactone	86.090	731.3	0.04	5.11	0.04				
propylene carbonate	102.089	763	2	4.14	0.08				
Organic Sulfur (C + H + S) Compounds									
dimethyl sulfide	62.130	503	1	5.53	0.10	0.305	0.005	203.7	0.269
dimethyl disulfide	94.190	608	3	5.07	0.10				
1-dodecanethiol	202.400	734	2	1.81	0.04				
methylthiobenzene	124.201	706	2	4.05	0.08				
Organic Ferrum									
bis(η -cyclopentadienyl) iron	186.035	785	8	3.61	0.10				
1-acetylferrocene	228.072	847	13	3.28	0.13				
Butylferrocene	242.143	784	12	2.18	0.09				
benzoylferrocene	290.143	886	13	1.59	0.06				
Organic Compounds Containing Nitrogen									
benzonitrile	103.124	700	15	4.2	0.5				
1-methylpyrrolidine	85.150	548.6	0.5	4.00	0.02	0.282	0.005	302.0	0.265
piperazine	86.138	657	3	5.5	0.2	0.32	0.03	269.2	0.271
2, 6-dimethylquinoline	157.216	786	2	3.48	0.05	0.308	0.003	510.4	0.272
Halogenated Alkanes									
chlorotrifluoromethane	104.456	301.95	0.05	3.88	0.01	0.580	0.002	180.1	0.278
chlorodifluoromethane	86.465	369.25	0.15	4.99	0.01	0.523	0.005	165.3	0.269
difluoromethane	52.023	351.25	0.03	5.783	0.005	0.425	0.004	122.4	0.242
pentafluoroethane	120.020	339.30	0.05	3.625	0.006	0.570	0.004	210.6	0.271
1,1,1,2-tetrafluoroethane	102.030	374.14	0.05	4.050	0.006	0.510	0.004	200.1	0.260
1,1,1-trifluoroethane	84.040	345.86	0.02	3.768	0.006	0.434	0.003	193.6	0.254
1,1-difluoroethane	66.050	386.43	0.04	4.52	0.01	0.368	0.002	179.5	0.252
hexafluoroethane	138.012	293.02	0.02	3.04	0.02	0.619	0.01	223.0	0.278
1,1,1,3,3-pentafluoropropane	134.047	427.10	0.10	3.655	0.004	0.520	0.004	257.8	0.265
octafluoropropane	188.017	345.0	0.2	2.66	0.02	0.627	0.005	299.9	0.278
1,1,1,2,3,3,3-heptafluoropropane	170.027	374.96	0.05	2.930	0.005	0.595	0.005	285.8	0.269
1,1,1,2,3,3-hexafluoropropane	152.037	412.44	0.02	3.42	0.04	0.565	0.003	269.1	0.268
1,1,1,3,3-pentafluoropropane	134.047	427.09	0.08	3.648	0.005	0.517	0.006	259.3	0.266
decafluorobutane	238.024	386.33	0.05	2.302	0.006	0.63	0.01	377.8	0.271
1,1,1,3,3-pentafluorobutane	148.074	459.9	0.2	3.20	0.01				
octadecafluorooctane	438.052	497.0	0.2	1.50	0.05	0.600	0.005	730.1	0.265
Other Halogenated Compounds									
hexafluoropropene	150.021	359	0.08	3.134	0.005	0.579	0.002	259.1	0.272
3,3,3-trifluoro-1-propene	96.051	377.8	0.8	3.65	0.08	0.46	0.01	208.8	0.243
(E)-1-chloro-3,3,3-trifluoropropene	130.493	438.75	0.1	3.77	0.03				
(E)-1,3,3,3-tetrafluoropropene	114.041	382.4	0.02	3.640	0.020	0.486	0.003	234.7	0.269

Table 1. continued

	molar mass M g·mol ^{-1a}	T_c K ^b	(±)	p_c MPa	(±)	ρ_c g·cm ⁻³	(±)	V_c cm ³ ·mol ⁻¹	Z_c^c
Other Halogenated Compounds									
(Z)-1,3,3,3-tetrafluoropropene	114.041	423.3	0.03	3.533	0.006	0.470	0.008	242.6	0.244
2,3,3,3-tetrafluoro-1-propene	114.041	367.85	0.06	3.381	0.004	0.476	0.002	239.6	0.265
hexafluorobenzene	186.054	516.7	0.1	3.28	0.01	0.552	0.003	337.1	0.257
pentafluorobenzene	168.064	531.0	0.1	3.53	0.03	0.518	0.003	324.4	0.259
1,4-dichlorobenzene	184.865	669	15						
trifluoro(trifluoromethyl)oxirane	166.020	359.4	0.1	2.930	0.007	0.585	0.008	283.8	0.278
Organic Compounds Containing Si + O									
tetraisopropoxysilane	264.437	587	6	1.31	0.04				
Organic Compounds Containing Ti + O									
tetraethoxytitanium	228.111	647	6	1.40	0.04				
tetrapropoxytitanium	284.219	685	7	1.15	0.04				
tetraisopropoxytitanium	284.219	641	6	1.15	0.04				
tetrabutoxytitanium	340.327	719	7	1.02	0.03				
tetra-tert-butoxytitanium	340.327	657	7	1.07	0.03				
tetrapentaoxytitanium	396.435	751	8	0.91	0.03				
tetrahexaoxytitanium	452.543	780	8	0.78	0.02				
tetraheptaoxytitanium	508.651	808	8	0.73	0.02				
tetraoctoxytitanium	564.759	831	8	0.66	0.02				
tetranonaoxytitanium	620.867	855	9	0.63	0.02				
tetradecoxytitanium	676.975	882	9	0.61	0.02				

^aMolar masses are based on the following relative atomic masses: carbon, 12.010 7; hydrogen, 1.007 94; boron, 10.811; chlorine, 35.453; fluorine, 18.998 403 2; nitrogen, 14.006 7; oxygen, 15.999 4; sulfur, 32.065; silicon, 28.085 5; titanium, 204.383 3 from *Pure Appl. Chem.* **2006**, 78, 2051–2066. ^bTemperatures are expressed in ITS-90. ^c $Z_c = p_c V_c / RT_c$, where $R = 8.314\,472\text{ J}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$.

references in the reference section and hence will not connect to the article.

Uncertainties in the experimental values given in Table 2 are those given by authors or have been inferred from related work. As noted in earlier parts, such uncertainties are qualitative and frequently are low estimates. The uncertainties given in the recommended values in Table 2 (and in Table 1) are generally greater than those appearing with each experimental value in Table 2. When only a single investigation has been made and no uncertainties reported, the recommended value and estimated uncertainties are given only in Table 1. Generally, the values have been adjusted upward from the uncertainties given by the authors for single investigations to bring them in line with the uncertainties for compounds investigated by several authors using a variety of techniques—including compounds examined in earlier parts of this series. In addition, we have tried to take into account information on the purity of the compound (before and after the measurement), relative merits of the technique used, and, in a few cases, the reputation of the laboratory and its investigators.

The criteria used to determine which values should be included based on correlation, extrapolations, etc. have been discussed in detail in the earlier parts, especially in Part 1. Table 3 lists the only methods used to determine the critical properties. All values are experimentally supported, preferably those directly measured. In some cases values were obtained indirectly, for example from equation of state measurements or evaluations in the critical region.

The values included in Table 2 are primarily based on experimental measurements made on pure samples. Until the 1960s it was not possible to quantify purity and investigators could only offer qualitative indications. Today, when methods are available for determining purity or, more correctly, the amount of impurity in a sample, a minimum mole fraction

purity of 99% is desirable, but a slightly lower purity may be acceptable if the impurities do not react with the major component and the boiling range of the sample is at most about 0.2 K. However, these criteria can be deceiving as some material sold as pure materials can consist of several major close boiling components.

Only critical temperatures determined according to the methods 1, 2, 3, or 4 of Table 3 are included in Table 2, although in some cases it was not possible to confirm the method used. No estimated values of T_c (or p_c) are included even if such estimates agree with experimental results. The critical pressures in Table 2 may have been determined with methods 1, 2, 3, or 4, but method 5 provides a confirmation of the value by combining it with vapor pressure measurements up to T_c —perhaps within at most 5 K of T_c —coupled with a fit of the data to a reliable vapor pressure equation; the best possible value for p_c is then the value given by the equation at T_c . Some values were determined by the indirect method 6 as discussed in part 8 [2001-tso/amb].

Because the fluid is infinitely compressible at the critical point, direct determination of the critical density is difficult, and only a few such determinations have been made with any claim of accuracy. Critical density is normally obtained indirectly by study of the vapor–liquid coexistence curve (method 7); the orthobaric densities are obtained and then the critical density is that given by the rectilinear diameter at the critical temperature. In the absence of other calculation methods specifically for ρ_c in Table 3, method 7 may be used even when the orthobaric density measurements do not come close to T_c . Other methods for deriving the critical density have been discussed extensively in the earlier parts.

Table 2. Critical Properties from the Literature*

ref	values reported in nonstandard units	T_{90}/K	p/MPa	$\rho/\text{g}\cdot\text{cm}^{-3}$	method	authors
NORMAL ALKANES						
METHANE: Molar Mass, 16.043 g; CH_4 ; CASRN 74-82-8						
95-amb/tso		190.564 ± 0.015	4.599 ± 0.003	0.1627 ± 0.0005	10	Ambrose and Tsonopoulos
96-wag/der		190.564 ± 0.010	4.5992 ± 0.002	0.16266 ± 0.0002	8	Wagner and de Reuck
2003-abd/ste		$(190.410)^a$	4.610 ± 0.003		6	Abdulagatov et al.
recommended values		190.564 ± 0.015	4.599 ± 0.003	0.1627 ± 0.0005		
ETHANE: Molar Mass, 30.070 g; C_2H_6 ; CASRN 74-84-0						
$T_{90} - T_{68} = -0.008$ at 305 K						
72-efr/shv	32.20 °C	305.34			1	Efremova and Shvarts
73-ber/git		305.3553 ± 0.0005			4a	Berestov et al.
81-mor	(32.229 ± 0.001) °C	305.371			1	Morrison
89-jan/luk	32.18 °C	305.32	4.881		1	Jangkamolkulchai and Luks
95-amb/tso		305.32 ± 0.04	4.872 ± 0.01	0.2066 ± 0.003	10	Ambrose and Tsonopoulos
2000-hor/fis		305.41 ± 0.1	4.881 ± 0.01		2	Horstmann et al.
2000-sin/luc		305.6 ± 0.2	4.9 ± 0.05		2a	Singh et al.
2001-hor/fis		305.37 ± 0.1	4.878 ± 0.01		2	Horstmann et al.
2002-fun/kle		305.322 ± 0.015	4.8722 ± 0.0016	0.20618 ± 0.00015	3,5,7	Funke et al.
2008-gil/oti		305.57 ± 0.08	4.882 ± 0.007		2	Gil et al.
recommended values		305.32 ± 0.04	4.872 ± 0.01	0.2062 ± 0.002		
PROPANE: Molar Mass, 44.097 g; C_3H_8 ; CASRN 74-98-6						
$T_{90} - T_{68} = -0.025$ at 370 K						
72-glo	96.60 °C, 42.96 atm, $0.199 \text{ L}\cdot\text{mol}^{-1}$	369.73	4.353	0.222	1,7	Glowka
95-amb/tso		369.83 ± 0.1	4.248 ± 0.01	0.220 ± 0.003	10	Ambrose and Tsonopoulos
95-jou/car	96.6 °C	369.75	4.270		2	Jou et al.
2001-hor/fis		369.70 ± 0.1	4.250 ± 0.01		2	Horstmann et al.
2001-yos		369.811	4.248	0.222	1,5	Yoshii
2003-abd/ste		$(369.700)^a$	4.230 ± 0.004		6	Abdulagatov et al.
2004-hig		369.818 ± 0.010	4.2465 ± 0.0029	0.227 ± 0.002	2	Higashi
2005-yas/uch		369.84 ± 0.01	4.247 ± 0.0005	0.216 ± 0.001	2	Yasumoto et al.
2006-ji/qin		370.1 ± 0.3	4.24 ± 0.03	0.217 ± 0.003	2	Ji et al.
2008-gil/oti		369.84 ± 0.08	4.253 ± 0.007		2	Gil et al.
2008-hon/sat		370.01 ± 0.01	4.260 ± 0.001	0.2215 ± 0.001	2,5	Honda et al.
2009-lem/mcl		369.89 ± 0.03	4.2512 ± 0.005	0.2205 ± 0.002	8	Lemmon et al.
2010-soo/thé		369.85 ± 0.01	4.243 ± 0.001		2	Soo et al.
2011-soo		369.85 ± 0.03	4.243 ± 0.001		2	Soo
2013-jun/bel		369.84 ± 0.03	4.2426 ± 0.0027		2	Juntarachat et al.
recommended values		369.83 ± 0.1	4.248 ± 0.01	0.221 ± 0.003		
BUTANE: Molar Mass, 58.124 g; C_4H_{10} ; CASRN 106-97-8						
95-amb/tso		425.12 ± 0.1	3.796 ± 0.01	0.228 ± 0.003	10	Ambrose and Tsonopoulos
95-deá/vic		425.03 ± 0.05	3.782 ± 0.010		1,5	Deák et al.
95-hol/mag		$(425.12)^b$		0.2292	7	Holcomb et al.
96-war		425.08	3.794	0.2275	1a	Warowny
2001-hor/fis		424.92 ± 0.1	3.790 ± 0.01		2	Horstmann et al.
2005-yas/uch		425.16 ± 0.01	3.787 ± 0.0005	0.225 ± 0.001	2	Yasumoto et al.
2007-miy/uem		$(425.125)^c$	3.785		6	Miyamoto and Uematsu
2010-soo/thé		425.22 ± 0.02	3.787 ± 0.001		2	Soo et al.
2011-soo		425.22 ± 0.03	3.787 ± 0.001		2	Soo
2013-jun/bel		424.82 ± 0.03	3.7983 ± 0.0023		2	Juntarachat et al.
recommended values		425.10 ± 0.1	3.792 ± 0.01	0.228 ± 0.003		
PENTANE: Molar Mass, 72.151 g; C_5H_{12} ; CASRN 109-66-0						
92-ma/fan		469.99	3.368	0.2318	1,7	Ma et al.
92-zha/zha		470.30 ± 0.11	3.383 ± 0.020	0.2301 ± 0.0017	1,7	Zhang et al.
95-amb/tso		469.7 ± 0.2	3.370 ± 0.02	0.232 ± 0.003	10	Ambrose and Tsonopoulos
95-ma/wan	(196.76 ± 0.12) °C	469.91 ± 0.12	3.381 ± 0.02	0.2317 ± 0.0019	1,5,7	Ma et al.
2000-lia/ma		469.90	3.380		1	Liang et al.
2003-abd/ste		$(469.951)^a$	3.400 ± 0.003		6	Abdulagatov et al.
2003-spa/wag		469.70	3.370	0.23200	8	Span and Wagner
2006-ewi/och-1		$(469.7)^b$	3.3674		6	Ewing and Ochoa
2010-ras/abd		469.90 ± 0.2	3.36 ± 0.01		3	Rasulov and Abdulagatov
2010-soo/thé		470.10 ± 0.02	3.379		2	Soo et al.
2011-soo		470.85 ± 0.12	3.379 ± 0.001		2	Soo
2012-jun/mor		469.70 ± 0.09	3.3696 ± 0.005		2a	Juntarachat et al.
recommended values		469.7 ± 0.2	3.370 ± 0.02	0.232 ± 0.003		

Table 2. continued

ref	values reported in nonstandard units	T_{90}/K	p/MPa	$\rho/\text{g}\cdot\text{cm}^{-3}$	method	authors
HEXANE: Molar Mass, 86.178 g; C_6H_{14} ; CASRN 110-54-3						
$T_{90} - T_{68} = -0.040 \text{ K}$; $T_{90} - T_{48} = 0.016 \text{ K}$ at 508 K						
13-sch	234.6 °C, 30.15 atm	507.8	3.055		1	Scheffer
88-del/poo		507.34 $\pm$ 0.20	3.025 $\pm$ 0.020		1	de Loos et al.
93-gud/tej		507.45 $\pm$ 0.2	3.038 $\pm$ 0.015		1c	Gude and Teja
95-amb/tso		507.6 $\pm$ 0.2	3.025 $\pm$ 0.02	0.234 $\pm$ 0.003	10	Ambrose and Tsonopoulos
2000-lia/ma		507.35	3.041		1	Liang et al.
2001-gao/hou		507.2 $\pm$ 0.3	3.003 $\pm$ 0.03	0.231 $\pm$ 0.005	1	Gao et al.
2002-ma/gao		507.35 $\pm$ 0.1	3.041 $\pm$ 0.1		1	Ma et al.
2003-abd/ste		(507.670) ^a	3.050 $\pm$ 0.005		6	Abdulagatov et al.
2003-liu/qin		507.4 $\pm$ 0.3	3.02 $\pm$ 0.03	0.231	2a	Liu et al.
2003-mor/lui		507.7 $\pm$ 0.3			1	Morton et al.
2003-spa/wag		507.82	3.034	0.23318	8	Span and Wagner
2006-ewi/och		(507.49) ^d	3.027		5	Ewing and Ochoa
2008-gil/oti		507.78 $\pm$ 0.08	3.046 $\pm$ 0.007		2	Gil et al.
2010-soo/thé		507.6 $\pm$ 0.04	3.031 $\pm$ 0.001		2	Soo et al.
2011-soo		507.61 $\pm$ 0.12	3.031 $\pm$ 0.001		2	Soo
2012-gil/bla		507.64 $\pm$ 0.14	3.044 $\pm$ 0.011		2	Gil et al.
2013-han/xia		507.9 $\pm$ 1.2	3.030 $\pm$ 0.015		2	Han et al.
2014-liu/zha		507.79 $\pm$ 0.4	3.033 $\pm$ 0.01		2	Liu et al.
recommended values		507.6 $\pm$ 0.2	3.027 $\pm$ 0.02	0.234 $\pm$ 0.003		
HEPTANE: Molar Mass, 100.205 g; C_7H_{16} ; CASRN 142-82-5						
$T_{90} - T_{68} = -0.039$ at 540 K						
88-del/poo		539.9	2.75		1	de Loos et al.
95-amb/tso		540.2 $\pm$ 0.3	2.74 $\pm$ 0.03	0.234 $\pm$ 0.003	10	Ambrose and Tsonopoulos
2000-web		(540.13) ^d	2.7397 $\pm$ 0.0025		6	Weber
2003-abd/ste		(540.100) ^a	2.768 $\pm$ 0.004		6	Abdulagatov et al.
2003-mor/lui		540.3 $\pm$ 0.3			1	Morton et al.
2003-spa/wag		540.13	2.736	0.23200	8	Span and Wagner
2005-ewi/och		(540.13) ^d	2.734		5	Ewing and Ochoa
2006-naz/sar-1		540.8 $\pm$ 0.3			1	Nazmutdinov et al.
2012-jun/mor		540.28 $\pm$ 0.09	2.7457 $\pm$ 0.005		2a	Juntarachat et al.
2013-han/xia		540.0 $\pm$ 1.2	2.761 $\pm$ 0.015		2	Han et al.
2014-liu/zha		540.14 $\pm$ 0.4	2.736 $\pm$ 0.01		2	Liu et al.
recommended values		540.2 $\pm$ 0.3	2.74 $\pm$ 0.03	0.234 $\pm$ 0.003		
OCTANE: Molar Mass, 114.232 g; C_8H_{18} ; CASRN 111-65-9						
90-smi/neg		569.2 $\pm$ 0.5			1	Smith et al.
95-amb/tso		568.7 $\pm$ 0.3	2.49 $\pm$ 0.03	0.232 $\pm$ 0.003	10	Ambrose and Tsonopoulos
2003-abd/ste		(568.936) ^a	2.497 $\pm$ 0.003		6	Abdulagatov et al.
2003-ewi/och		(568.7) ^b	2.482		5	Ewing and Ochoa
2003-mor/lui		568.9 $\pm$ 0.3			1	Morton et al.
2003-spa/wag		569.32	2.497	0.23400	8	Span and Wagner
2008-naz/ale		570.5 $\pm$ 0.5			1	Nazmutdinov et al.
2014-liu/zha		568.92 $\pm$ 0.4	2.496 $\pm$ 0.01		2	Liu et al.
recommended values		568.7 $\pm$ 0.3	2.49 $\pm$ 0.02	0.232 $\pm$ 0.003		
NONANE: Molar Mass, 128.259 g; C_9H_{20} ; CASRN 111-84-2						
95-amb/tso		594.6 $\pm$ 0.6	2.29 $\pm$ 0.05	0.231 $\pm$ 0.005	10	Ambrose and Tsonopoulos
2003-abd/ste		(594.480) ^a	2.270 $\pm$ 0.003		6	Abdulagatov et al.
2003-mor/lui		594.6 $\pm$ 0.5			1	Morton et al.
2014-liu/zha		594.52 $\pm$ 0.4	2.278 $\pm$ 0.01		2	Liu et al.
recommended values		594.6 $\pm$ 0.6	2.29 $\pm$ 0.02	0.231 $\pm$ 0.005		
DECANE: Molar Mass, 142.286 g; $\text{C}_{10}\text{H}_{22}$; CASRN 124-18-5						
95-amb/tso		617.7 $\pm$ 0.6	2.11 $\pm$ 0.05	0.228 $\pm$ 0.005	10	Ambrose and Tsonopoulos
2003-abd/ste		(617.470) ^a	2.140 $\pm$ 0.003		6	Abdulagatov et al.
2003-mor/lui		617.7 $\pm$ 0.5			1	Morton et al.
2008-naz/ale		617.70 $\pm$ 0.5			1	Nazmutdinov et al.
2012-jun/mor		617.92 $\pm$ 0.09	2.1105 $\pm$ 0.005		2a	Juntarachat et al.
2014-liu/zha		617.73 $\pm$ 0.4	2.106 $\pm$ 0.01		2	Liu et al.
recommended values		617.7 $\pm$ 0.5	2.11 $\pm$ 0.05	0.228 $\pm$ 0.005		
DODECANE: Molar Mass, 170.340 g; $\text{C}_{12}\text{H}_{26}$; CASRN 112-40-3						
95-amb/tso		658 $\pm$ 1	1.82 $\pm$ 0.1	0.226 $\pm$ 0.01	10	Ambrose and Tsonopoulos
2002-ste		658 $\pm$ 1	1.8 $\pm$ 0.2	0.225 $\pm$ 0.010	4,6,7	Steele
2003-mor/lui		658.2 $\pm$ 0.5			1	Morton et al.

Table 2. continued

ref	values reported in nonstandard units	T_{90}/K	p/MPa	$\rho/\text{g}\cdot\text{cm}^{-3}$	method	authors
DODECANE: Molar Mass, 170.340 g; $\text{C}_{12}\text{H}_{26}$; CASRN 112-40-3						
recommended values		658 ± 1	1.81 ± 0.1	0.226 ± 0.01		
NONADECANE: Molar Mass, 268.529 g; $\text{C}_{19}\text{H}_{40}$; CASRN 629-92-5						
94-nik/pav		755.3 ± 8	1.16 ± 0.02		4d	Nikitin et al.
97-nik/pav		755 ± 8	1.16 ± 0.02		4d	Nikitin et al.
2014-nik/pop-2		757 ± 10	1.17 ± 0.05		4d	Nikitin and Popov
recommended values		756 ± 8	1.16 ± 0.02			
EICOSANE: Molar Mass, 282.556 g; $\text{C}_{20}\text{H}_{42}$; CASRN 112-95-8						
94-nik/pav		767.5 ± 8	1.07 ± 0.02		4d	Nikitin et al.
97-nik/pav		769 ± 8	1.08 ± 0.02		4d	Nikitin et al.
2014-nik/pop-2		769 ± 10	1.08 ± 0.05		4d	Nikitin and Popov
recommended values		768 ± 8	1.08 ± 0.02			
HENEICOSANE: Molar Mass, 296.583 g; $\text{C}_{21}\text{H}_{44}$; CASRN 629-94-7						
94-nik/pav		777.6 ± 8	1.03 ± 0.02		4d	Nikitin et al.
2014-nik/pop-2		779 ± 10	1.04 ± 0.05		4d	Nikitin and Popov
recommended values		779 ± 10	1.04 ± 0.05			
DOCOSANE: Molar Mass, 310.610 g; $\text{C}_{22}\text{H}_{46}$; CASRN 629-97-0						
94-nik/pav		785.6 ± 8	0.982 ± 0.02		4d	Nikitin et al.
97-nik/pav		787 ± 8	1.00 ± 0.02		4d	Nikitin et al.
2014-nik/pop-2		787 ± 10	0.986 ± 0.05		4d	Nikitin and Popov
recommended values		787 ± 8	0.99 ± 0.02			
TRICOSANE: Molar Mass, 324.637 g; $\text{C}_{23}\text{H}_{48}$; CASRN 638-67-5						
94-nik/pav		789.7 ± 8	0.915 ± 0.02		4d	Nikitin et al.
2014-nik/pop-2		791 ± 10	0.921 ± 0.05		4d	Nikitin and Popov
recommended values		791 ± 10	0.92 ± 0.05			
TETRACOSANE: Molar Mass, 338.664 g; $\text{C}_{24}\text{H}_{50}$; CASRN 646-31-1						
94-nik/pav		799.8 ± 8	0.866 ± 0.02		4d	Nikitin et al.
97-nik/pav		800 ± 8	0.875 ± 0.02		4d	Nikitin et al.
2014-nik/pop-2		801 ± 10	0.871 ± 0.05		4d	Nikitin and Popov
recommended values		800 ± 8	0.87 ± 0.02			
HEXACOSANE: Molar Mass, 366.718 g; $\text{C}_{26}\text{H}_{54}$; CASRN 630-01-3						
97-nik/pav		816 ± 12	0.795 ± 0.03		4d	Nikitin et al.
2014-nik/pop-2		819 ± 10	0.814 ± 0.05		4d	Nikitin and Popov
recommended values		819 ± 10	0.81 ± 0.05			
OCTACOSANE: Molar Mass, 394.772 g; $\text{C}_{28}\text{H}_{58}$; CASRN 630-02-4						
97-nik/pav		824 ± 12	0.744 ± 0.03		4d	Nikitin et al.
2014-nik/pop-2		836 ± 10	0.750 ± 0.05		4d	Nikitin and Popov
recommended values		836 ± 10	0.75 ± 0.05			
TRIACONTANE: Molar Mass, 422.826 g; $\text{C}_{30}\text{H}_{62}$; CASRN 638-68-6						
97-nik/pav		843 ± 12	0.636 ± 0.03		4d	Nikitin et al.
2014-nik/pop-2		850 ± 10	0.690 ± 0.05		4d	Nikitin and Popov
recommended values		850 ± 10	0.69 ± 0.05			
HEXATRIACONTANE: Molar Mass, 506.988 g; $\text{C}_{36}\text{H}_{74}$; CASRN 630-06-8						
97-nik/pav		872 ± 13	0.472 ± 0.02		4d	Nikitin et al.
2014-nik/pop-2		888 ± 15	0.580 ± 0.05		4d	Nikitin and Popov
recommended values		888 ± 15	0.58 ± 0.05			
TETRACONTANE: Molar Mass, 563.096 g; $\text{C}_{40}\text{H}_{82}$; CASRN 4181-95-7						
2014-nik/pop-2		904 ± 15	0.523 ± 0.05		4d	Nikitin and Popov
TETRATETRACONTANE: Molar Mass, 619.204 g; $\text{C}_{44}\text{H}_{90}$; CASRN 7098-22-8						
2014-nik/pop-2		919 ± 15	0.482 ± 0.05		4d	Nikitin and Popov
HEXACONTANE: Molar Mass, 843.636 g; $\text{C}_{60}\text{H}_{122}$; CASRN 7667-80-3						
2014-nik/pop-2		974 ± 15	0.360 ± 0.05		4d	Nikitin and Popov
BRANCHED ALKANES						
2-METHYLPROPANE (Isobutane): Molar Mass, 58.124 g; C_4H_{10} ; CASRN 75-28-5						
96-dau		407.8 ± 0.5	3.640 ± 0.05	0.224 ± 0.003	10	Daubert
2001-yos		407.761	3.631	0.233	1,5	Yoshii
2003-spa/wag		407.817	3.640	0.22436	8	Span and Wagner
2005-yas/uch		407.81 ± 0.01	3.633 ± 0.0005	0.225 ± 0.001	2	Yasumoto et al.
2006-bue/wag-1		407.81	3.629	0.2225	8	Bücker and Wagner
2006-hig		407.795 ± 0.010	3.628 ± 0.0014	0.229 ± 0.002	2,6	Higashi
2006-mas/hon		407.885 ± 0.012	3.6390 ± 0.0016	0.2275 ± 0.0013	2,5	Masui et al.
2007-li/qin		409.9 ± 0.3	3.62 ± 0.03	0.220 ± 0.003	2	Li et al.
2014-liu/fen		(407.81) ^{cc}	3.6318		3	Liu et al.

Table 2. continued

ref	values reported in nonstandard units	T_{90}/K	p/MPa	$\rho/\text{g}\cdot\text{cm}^{-3}$	method	authors
2-METHYLPROPANE (Isobutane): Molar Mass, 58.124 g; C_4H_{10} ; CASRN 75-28-5						
recommended values		407.81 ± 0.05	3.633 ± 0.005	0.226 ± 0.005		
2-METHYLBUTANE: Molar mass, 72.151 g; C_5H_{12} ; CASRN 78-78-4						
91-ewi/goo	(460.43) ^e		3.4322 ± 0.0058		6	Ewing and Goodwin
96-dau	460.4 ± 0.5		3.38 ± 0.05	0.236 ± 0.002	10	Daubert
recommended values	460.4 ± 0.5		3.41 ± 0.03	0.236 ± 0.002		
2,2-DIMETHYLPROPANE: Molar Mass, 72.151 g; C_5H_{12} ; CASRN 463-82-1						
93-mac/bar	(433.71)			0.2318 ± 0.0001	7	McLure and Barbarin-Castillo
96-dau	433.8 ± 0.1		3.196 ± 0.01	0.235 ± 0.003	10	Daubert
2006-lem/spa	433.74		3.196	0.236	8	Lemmon and Span
recommended values	433.8 ± 0.1		3.196 ± 0.01	0.235 ± 0.003		
2,2-DIMETHYLBUTANE: Molar mass, 86.178 g; C_6H_{14} ; CASRN 75-83-2						
96-dau	489.0 ± 0.3		3.10 ± 0.02	0.241 ± 0.004	10	Daubert
2003-mor/lui	488.8 ± 0.3				1	Morton et al.
recommended values	489.0 ± 0.3		3.10 ± 0.02	0.241 ± 0.004		
2-METHYLHEPTANE: Molar Mass, 114.232 g; C_8H_{18} ; CASRN 592-27-8						
96-dau	559.7 ± 0.1		2.50 ± 0.02	0.234 ± 0.002	10	Daubert
2014-nes/vos	559.7 ± 0.1				1	Nesterova et al.
recommended values	559.7 ± 0.1		2.50 ± 0.02	0.234 ± 0.002		
2,2,4-TRIMETHYLPENTANE: Molar Mass, 114.232 g; C_8H_{18} ; CASRN 540-84-1						
96-dau	543.8 ± 0.3		2.57 ± 0.02	0.244 ± 0.002	10	Daubert
2003-mor/lui	544.0 ± 0.3				1	Morton et al.
2007-li/qin	544.1 ± 0.3		2.57 ± 0.03	0.251 ± 0.003	2	Li et al.
recommended values	544.0 ± 0.3		2.57 ± 0.02	0.244 ± 0.002		
2-METHYLOCTANE: Molar Mass, 128.258 g; C_9H_{20} ; CASRN 3221-61-2						
96-dau	582.8 ± 0.1		2.31 ± 0.02		10	Daubert
2014-nes/vos	582.87				1	Nesterova et al.
recommended values	582.8 ± 0.1		2.31 ± 0.02			
2,6,10,15,19,23-HEXAMETHYLTETRACOSANE (Squalane): Molar Mass, 422.826 g; $\text{C}_{30}\text{H}_{62}$; CASRN 111-01-3						
2000-von/wil-1	795.9 ± 1		0.590 ± 0.012		3d	VonNiederhausern et al.
2005-nik/pop	822 ± 12		0.70 ± 0.03		4d	Nikitin and Popov
recommended values	810 ± 15		0.728 ± 0.03			
CYCLOALKANES						
CYCLOPENTANE: Molar Mass, 70.135 g; C_5H_{10} ; CASRN 287-92-3						
$T_{90} - T_{68} = -0.040$ at 512 K						
70-ale/kru	238.65 °C	511.76			2	Alekhnin et al.
96-dau		511.7 ± 0.2	4.51 ± 0.04	0.270 ± 0.004	10	Daubert
recommended values		511.7 ± 0.2	4.51 ± 0.04	0.270 ± 0.004		
CYCLOHEXANE: Molar Mass, 84.162 g; C_6H_{12} ; CASRN 110-82-7						
96-dau		553.8 ± 0.2	4.08 ± 0.03	0.273 ± 0.002	10	Daubert
2000-ewi/och		(553.60) ^f	4.0803		5	Ewing and Ochoa
2003-nik/pop		551 ± 6	4.06 ± 0.12		4	Nikitin et al.
2003-spa/wag		553.60	4.078	0.27302	8	Span and Wagner
2004-mor/lui		553.5 ± 0.5			1	Morton et al.
2005-zha/qin	283.2 °C	556.4 ± 0.3	4.20 ± 0.03	0.280	2a	Zhang et al.
2006-naz/sar		553.5 ± 0.5			1	Nazmutdinov et al.
2008-gil/oti		553.52 ± 0.02	4.099 ± 0.006		2	Gil et al.
2013-han/xia		554.1 ± 1.2	4.110 ± 0.015		2	Han et al.
recommended values		553.6 ± 0.5	4.08 ± 0.05	0.273 ± 0.002		
CYCLOOCTANE: Molar Mass, 112.216 g; C_8H_{16} ; CASRN 292-64-8						
96-dau		647.2 ± 0.5	3.56 ± 0.04	0.274 ± 0.005	10	Daubert
2004-mor/lui		647.5 ± 0.5			1	Morton et al.
recommended values		647.4 ± 0.5	3.56 ± 0.04	0.274 ± 0.005		
METHYLCYCLOHEXANE: Molar Mass, 98.189 g; C_7H_{14} ; CASRN 108-87-2						
96-dau		572.1 ± 0.2	3.48 ± 0.02	0.266 ± 0.002	10	Daubert
2003-nik/pop		569 ± 6		3.47 ± 0.1	4	Nikitin et al.
recommended values		572.2 ± 0.2	3.48 ± 0.02	0.267 ± 0.002		
ETHYLCYCLOHEXANE: Molar Mass, 112.216 g; C_8H_{16} ; CASRN 1678-91-7						
2003-nik/pop		604 ± 6	3.25 ± 0.1		4	Nikitin et al.
2004-mor/lui		606.9 ± 0.2			1	Morton et al.
2008-ihm/wol		605.9 ± 0.1	3.108 ± 0.010	0.2590 ± 0.005	2	Ihmels et al.
recommended values		606.5 ± 0.5	3.13 ± 0.05	0.259 ± 0.005		

Table 2. continued

ref	values reported in nonstandard units	T_{90}/K	p/MPa	$\rho/\text{g}\cdot\text{cm}^{-3}$	method	authors
(Z)-1,2-DIMETHYLCYCLOHEXANE (<i>cis</i> -1,2-Dimethylcyclohexane): Molar Mass, 112.216 g; C_8H_{16} ; CASRN 2207-01-4						
2011-ihm/ste		607.29 ± 0.1	3.354 ± 0.010	0.2762 ± 0.005	2	Ihmels et al.
(Z)-1,3-DIMETHYLCYCLOHEXANE (<i>cis</i> -1,3-Dimethylcyclohexane): Molar Mass, 112.216 g; C_8H_{16} ; CASRN 638-04-0						
92-zha/zha		587.67 ± 0.11	3.007 ± 0.020	0.2696 ± 0.0020	1,7	Zhang et al.
2008-ihm/wol		586.1 ± 0.1	3.042 ± 0.010	0.2588 ± 0.005	2	Ihmels et al.
recommended values		587.0 ± 0.5	3.03 ± 0.05	0.265 ± 0.005		
(Z)-1,4-DIMETHYLCYCLOHEXANE (<i>cis</i> -1,4-Dimethylcyclohexane): Molar Mass, 112.216 g; C_8H_{16} ; CASRN 624-29-3						
95-ma/wan	$(330.04 \pm 0.12)^\circ\text{C}$	603.19	3.157 ± 0.02	0.2584 ± 0.0019	1,5,7	Ma et al.
(E)-1,2-DIMETHYLCYCLOHEXANE (<i>trans</i> -1,2-Dimethylcyclohexane): Molar Mass, 112.216 g; C_8H_{16} ; CASRN 6876-23-9						
2008-ihm/wol		596.3 ± 0.1	3.080 ± 0.010	0.2575 ± 0.005	2	Ihmels et al.
(E)-1,4-DIMETHYLCYCLOHEXANE (<i>trans</i> -1,4-Dimethylcyclohexane): Molar Mass, 112.216 g; C_8H_{16} ; CASRN 2207-04-7						
$T_{90} - T_{68} = -0.039$ at 590 K						
70-pow/swi		587.7			1	Powell et al.
72-you		590.0	2.938		1	Young
96-dau		587.7 ± 0.5			10	Daubert
2011-ihm/ste		585.84 ± 0.1	3.058 ± 0.010	0.2669 ± 0.005	2	Ihmels et al.
recommended values		587 ± 1	3.06 ± 0.1	0.267 ± 0.005		
PROPYLCYCLOHEXANE: Molar Mass, 126.243 g; C_9H_{18} ; CASRN 1678-92-8						
2003-nik/pop		624 ± 6	2.86 ± 0.09		4	Nikitin et al.
2004-mor/lui		630.8 ± 0.4			1	Morton et al.
recommended values		631.0 ± 1	2.86 ± 0.10			
(1-METHYLETHYL)CYCLOHEXANE (Isopropylcyclohexane): Molar Mass, 126.243 g; C_9H_{18} ; CASRN 696-29-7						
2004-mor/lui		632.2 ± 0.2			1	Morton et al.
2008-ihm/wol		635.3 ± 0.1	2.942 ± 0.010	0.2604 ± 0.005	2	Ihmels et al.
recommended values		632.0 ± 0.5	2.95 ± 0.05	0.260 ± 0.005		
BUTYLCYCLOHEXANE: Molar Mass, 140.270 g; $\text{C}_{10}\text{H}_{20}$; CASRN 1678-93-9						
2003-nik/pop		650 ± 6	2.56 ± 0.08		4	Nikitin et al.
2004-mor/lui		653.1 ± 0.4			1	Morton et al.
2009-ihm/wol		649.3 ± 0.1	2.510 ± 0.010	0.2504 ± 0.005	2	Ihmels et al.
recommended values		650 ± 2	2.53 ± 0.05	0.250 ± 0.010		
(2-METHYLPROPYL)CYCLOHEXANE (Isobutylcyclohexane): Molar Mass, 140.270 g; $\text{C}_{10}\text{H}_{20}$; CASRN 1678-98-4						
2004-mor/lui		642.1 ± 0.2			1	Morton et al.
(1,1-DIMETHYLETHYL)CYCLOHEXANE (<i>tert</i> -Butylcyclohexane): Molar Mass, 140.270 g; $\text{C}_{10}\text{H}_{20}$; CASRN 3178-22-1						
2004-mor/lui		652.0 ± 0.2			1	Morton et al.
2009-ihm/wol		646.9 ± 0.1	2.765 ± 0.010	0.2780 ± 0.005	2	Ihmels et al.
recommended values		648 ± 2	2.77 ± 0.05	0.278 ± 0.01		
1,1'-BICYCLOPENTYL (Cyclopentylcyclopentane): Molar Mass, 138.254 g; $\text{C}_{10}\text{H}_{18}$; CASRN 1636-39-1						
2004-chi/ste		690 ± 2	3.30 ± 0.07	0.2736 ± 0.02	4,6,7	Chirico and Steele
1,3-DIMETHYLTRICYCLO[3.3.1.1(3,7)]DECANE (1,3-Dimethyladamantane): Molar Mass, 164.292 g; $\text{C}_{12}\text{H}_{20}$; CASRN 702-79-4						
96-ste/chi		708 ± 1	3.000 ± 0.200	0.288 ± 0.009	4,6,7	Steele et al.
2006-naz/sar		706.7 ± 0.4			1	Nazmutdinov et al.
recommended values		707 ± 2	3.0 ± 0.2	0.29 ± 0.02		
1,3,5-TRIMETHYLTRICYCLO[3.3.1.1(3,7)]DECANE (1,3,5-Trimethyladamantane): Molar Mass, 178.319 g; $\text{C}_{13}\text{H}_{22}$; CASRN 707-35-7						
2006-naz/sar		701.9 ± 0.5			1	Nazmutdinov et al.
UNSATURATED ALIPHATIC HYDROCARBONS						
ETHENE (Ethylene): Molar Mass, 28.054 g; C_2H_4 ; CASRN 74-85-1						
$T_{90} - T_{68} = -0.002$ K; $T_{90} - T_{48} = -0.006$ at 282 K						
48-die/sch		282.64	5.061		1	Diepen and Scheffer
96-tso/amb		282.34 ± 0.02	5.041 ± 0.004	0.214 ± 0.002	10	Tsonopoulos and Ambrose
2001-liu/fu		282.75	5.06	0.210	2a	Liu et al.
2003-spa/wag		282.35	5.042	0.21424	8	Span and Wagner
2010-che/yao		282.3	5.02		2a	Chen et al.
recommended values		282.34 ± 0.02	5.041 ± 0.004	0.214 ± 0.002		
PROPENE (Propylene): Molar Mass, 42.081 g; C_3H_6 ; CASRN 115-07-1						
95-zha/ma		(365.57) ^g	4.57	0.230	6,7	Zhao and Ma
96-tso/amb		364.9 ± 0.4	4.60 ± 0.03	0.228 ± 0.005	10	Tsonopoulos and Ambrose
2006-kur		364.209 ± 0.003	4.555 ± 0.00683	0.23008 ± 0.0001	?	Kurzeja
2010-che/yao		365.1	4.63		2a	Chen et al.
recommended values		364.9 ± 0.4	4.60 ± 0.03	0.228 ± 0.005		
1-BUTENE: Molar Mass, 56.108 g; C_4H_8 ; CASRN 106-98-9						
96-tso/amb		419.5 ± 0.5	4.02 ± 0.05	0.233 ± 0.012	10	Tsonopoulos and Ambrose
2005-ihm/fis		419.29 ± 0.1	4.0051 ± 0.005	0.23788 ± 0.00238	2a	Ihmels et al.
2007-li/qin		421.2 ± 0.3	4.00 ± 0.03	0.233 ± 0.003	2	Li et al.

Table 2. continued

ref	values reported in nonstandard units	T_{90}/K	p/MPa	$\rho/\text{g}\cdot\text{cm}^{-3}$	method	authors
1-BUTENE: Molar Mass, 56.108 g; C_4H_8 ; CASRN 106-98-9						
recommended values		419.4 ± 0.2	4.015 ± 0.02	0.235 ± 0.007		
1-HEXENE: Molar Mass, 84.162 g; C_6H_{12} ; CASRN 592-41-6						
92-ma/fan		504.05 ± 0.12	3.186 ± 0.010	0.2290 ± 0.0009	1,7	Ma et al.
93-li/ma		502.25	3.339	0.229	1	Li et al.
96-tso/amb		504.0 ± 0.3	3.21 ± 0.03	0.237 ± 0.005	10	Tsonopoulos and Ambrose
recommended values		504.0 ± 0.3	3.21 ± 0.03	0.237 ± 0.005		
1-HEPTENE: Molar Mass, 98.189 g; C_7H_{14} ; CASRN 592-76-7						
96-tso/amb		537.3 ± 0.4	2.92 ± 0.04	0.24 ± 0.01	10	Tsonopoulos and Ambrose
2000-lia/ma		537.53 ± 0.1	2.921 ± 0.1		1	Liang et al.
2002-ma/gao		537.53 ± 0.1	2.921 ± 0.1		1	Ma et al.
2005-wan/jia		537.53 ± 0.28	2.921 ± 0.02		1	Wan et al.
recommended values		537.3 ± 0.4	2.92 ± 0.04	0.24 ± 0.01		
1-TRIDECENE: Molar Mass, 182.351 g; $\text{C}_{13}\text{H}_{26}$; CASRN 2437-56-1						
99-nik/pop		673 ± 7	1.73 ± 0.03		4d	Nikitin and Popov
1-TETRADECENE: Molar Mass, 196.378 g; $\text{C}_{14}\text{H}_{28}$; CASRN 26952-13-6						
99-nik/pop		691 ± 7	1.59 ± 0.03		4d	Nikitin and Popov
1-PENTADECENE: Molar Mass, 210.405 g; $\text{C}_{15}\text{H}_{30}$; CASRN 13360-61-7						
99-nik/pop		705 ± 7	1.54 ± 0.03		4d	Nikitin and Popov
1-HEXADECENE: Molar Mass, 224.432 g; $\text{C}_{16}\text{H}_{32}$; CASRN 26952-14-7						
99-nik/pop		718 ± 7	1.39 ± 0.03		4d	Nikitin and Popov
1-HEPTADECENE: Molar Mass, 238.459 g; $\text{C}_{17}\text{H}_{34}$; CASRN 6765-39-5						
99-nik/pop		734 ± 7	1.34 ± 0.03		4d	Nikitin and Popov
1-OCTADECENE: Molar Mass, 252.486 g; $\text{C}_{18}\text{H}_{36}$; CASRN 27070-58-2						
99-nik/pop		748 ± 7	1.30 ± 0.03		4d	Nikitin and Popov
1-NONADECENE: Molar Mass, 266.513 g; $\text{C}_{19}\text{H}_{38}$; CASRN 18435-45-5						
99-nik/pop		755 ± 8	1.19 ± 0.02		4d	Nikitin and Popov
1-EICOSENE: Molar Mass, 280.540 g; $\text{C}_{20}\text{H}_{40}$; CASRN 3452-07-1						
99-nik/pop		772 ± 10	1.14 ± 0.1		4d	Nikitin and Popov
(Z)-2-BUTENE (<i>cis</i> -2-Butene): Molar Mass, 56.108 g; C_4H_8 ; CASRN 590-18-1						
96-tso/amb		435.5 ± 0.1	4.21 ± 0.05	0.240 ± 0.010	10	Tsonopoulos and Ambrose
2005-ihm/fis		435.75 ± 0.1	4.2255 ± 0.005	0.23812 ± 0.00238	2a	Ihmels et al.
recommended values		435.65 ± 0.10	4.215 ± 0.010	0.239 ± 0.002		
(E)-2-BUTENE (<i>trans</i> -2-Butene): Molar Mass, 56.108 g; C_4H_8 ; CASRN 624-64-6						
96-tso/amb		428.6 ± 0.1	4.10 ± 0.02	0.236 ± 0.005	10	Tsonopoulos and Ambrose
2005-ihm/fis		428.61 ± 0.1	4.0273 ± 0.005	0.23640 ± 0.00236	2a	Ihmels et al.
recommended values		428.6 ± 0.1	4.07 ± 0.04	0.236 ± 0.003		
2-METHYL-1-PROPENE (Isobutene): Molar Mass, 56.108 g; C_4H_8 ; CASRN 115-11-7						
96-tso/amb		417.9 ± 0.1	4.000 ± 0.010	0.235 ± 0.004	10	Tsonopoulos and Ambrose
2005-ihm/fis		418.09 ± 0.1	4.0098 ± 0.005	0.23397 ± 0.00234	2a	Ihmels et al.
recommended values		418.0 ± 0.1	4.005 ± 0.010	0.234 ± 0.001		
(Z)-2-HEXENE (<i>cis</i> -2-Hexene): Molar Mass, 84.1626 g; C_6H_{12} ; CASRN 7688-21-3						
2004-mor/lui		513.4 ± 0.4			1	Morton et al.
(Z)-3-HEXENE (<i>cis</i> -3-Hexene): Molar Mass, 84.162 g; C_6H_{12} ; CASRN 7642-09-3						
2004-mor/lui		510.2 ± 0.6			1	Morton et al.
(E)-2-HEXENE (<i>trans</i> -2-Hexene): Molar Mass, 84.162 g; C_6H_{12} ; CASRN 4050-45-7						
2004-mor/lui		509.0 ± 0.4			1	Morton et al.
(E)-3-HEXENE (<i>trans</i> -3-Hexene): Molar Mass, 84.162 g; C_6H_{12} ; CASRN 13269-52-8						
2004-mor/lui		507.4 ± 0.3			1	Morton et al.
2-METHYL-1-PENTENE: Molar Mass, 84.162 g; C_6H_{12} ; CASRN 763-29-1						
2004-mor/lui		521.0 ± 0.4			1	Morton et al.
4-METHYL-1-PENTENE: Molar Mass, 84.162 g; C_6H_{12} ; CASRN 691-37-2						
97-ste/chi-2		495 ± 1	3.290 ± 0.1	0.238 ± 0.01	4,6,7	Steele et al.
2004-mor/lui		493.1 ± 0.2			1	Morton et al.
recommended values		494 ± 2	3.29 ± 0.1	0.238 ± 0.01		
2-METHYL-2-PENTENE: Molar Mass, 84.162 g; C_6H_{12} ; CASRN 625-27-4						
2004-mor/lui		509.3 ± 0.2			1	Morton et al.
(Z)-4-METHYL-2-PENTENE (<i>cis</i> -4-Methyl-2-pentene): Molar Mass, 84.162 g; C_6H_{12} ; CASRN 691-38-3						
2004-mor/lui		496.3 ± 0.2			1	Morton et al.
2,3-DIMETHYL-1-BUTENE: Molar Mass, 84.162 g; C_6H_{12} ; CASRN 563-78-0						
2004-mor/lui		497.7 ± 0.2			1	Morton et al.
3,3-DIMETHYL-1-BUTENE: Molar Mass, 84.162 g; C_6H_{12} ; CASRN 558-37-2						
2004-mor/lui		477.4 ± 0.4			1	Morton et al.

Table 2. continued

ref	values reported in nonstandard units	T_{90}/K	p/MPa	$\rho/\text{g}\cdot\text{cm}^{-3}$	method	authors
2,3-DIMETHYL-2-BUTENE: Molar Mass, 84.162 g; C_6H_{12} ; CASRN 563-79-1						
2004-mor/lui		521.0 ± 0.4			1	Morton et al.
(Z)-2-HEPTENE (<i>cis</i> -2-Heptene): Molar Mass, 98.189 g; C_7H_{14} ; CASRN 6443-92-1						
2004-mor/lui		548.5 ± 0.2			1	Morton et al.
(E)-2-HEPTENE (<i>trans</i> -2-Heptene): Molar Mass, 98.189 g; C_7H_{14} ; CASRN 14686-13-6						
2004-mor/lui		542.8 ± 0.2			1	Morton et al.
(E)-3-HEPTENE (<i>trans</i> -3-Heptene): Molar Mass, 98.189 g; C_7H_{14} ; CASRN 14696-14-7						
2004-mor/lui		538.6 ± 0.2			1	Morton et al.
2-METHYL-1-HEXENE: Molar Mass, 98.189 g; C_7H_{14} ; CASRN 6094-02-6						
2004-mor/lui		541.8 ± 0.9			1	Morton et al.
5-METHYL-1-HEXENE: Molar Mass, 98.189 g; C_7H_{14} ; CASRN 3524-73-0						
2004-mor/lui		528.7 ± 0.5			1	Morton et al.
2,3-DIMETHYL-1-PENTENE: Molar Mass, 98.189 g; C_7H_{14} ; CASRN 3404-72-6						
2004-mor/lui		533.6 ± 0.3			1	Morton et al.
4,4-DIMETHYL-1-PENTENE: Molar Mass, 98.189 g; C_7H_{14} ; CASRN 762-62-9						
2004-mor/lui		516 ± 2			1	Morton et al.
(E)-2-OCTENE (<i>trans</i> -2-Octene): Molar Mass, 112.216 g; C_8H_{16} ; CASRN 13389-42-9						
2004-mor/lui		569.8 ± 0.5			1	Morton et al.
(E)-4-OCTENE (<i>trans</i> -4-Octene): Molar Mass, 112.216 g; C_8H_{16} ; CASRN 14850-23-8						
2004-mor/lui		566.3 ± 0.2			1	Morton et al.
2-METHYL-1-HEPTENE: Molar Mass, 112.216 g; C_8H_{16} ; CASRN 15870-10-7						
2004-mor/lui		567.5 ± 0.3			1	Morton et al.
2-METHYL-2-HEPTENE: Molar Mass, 112.216 g; C_8H_{16} ; CASRN 627-97-4						
2004-mor/lui		568.9 ± 0.4			1	Morton et al.
2,4,4-TRIMETHYL-1-PENTENE: Molar Mass, 112.216 g; C_8H_{16} ; CASRN 107-39-1						
2002-von/gil		550.7 ± 2.0	2.73 ± 0.06		3	VonNiederhausern and Giles
CYCLOPENTENE: Molar Mass, 68.119 g; C_5H_8 ; CASRN 142-29-0						
92-ma/fan		506.24 ± 0.12	4.804 ± 0.024	0.2742 ± 0.0009	1,7	Ma et al.
96-tso/amb		506.5 ± 0.5	4.80 ± 0.05	0.278 ± 0.010	10	Tsonopoulos and Ambrose
recommended values		506.5 ± 0.5	4.80 ± 0.05	0.278 ± 0.010		
CYCLOHEXENE: Molar Mass, 82.146 g; C_6H_{10} ; CASRN 110-83-8						
92-ma/fan		560.46 ± 0.12	4.453 ± 0.024	0.2868 ± 0.0011	1,7	Ma et al.
96-ste/chi-1		$(560.4)^h$	4.905	0.2767	6,7	Steele et al.
96-tso/amb		560.4 ± 0.1			10	Tsonopoulos and Ambrose
recommended values		560.4 ± 0.1	4.45 ± 0.03	0.28 ± 0.01		
AROMATIC HYDROCARBONS						
BENZENE: Molar Mass, 78.114 g; C_6H_6 ; CASRN 71-43-2						
72-hal/tow				0.3037 ± 0.0004	7	Hales and Townsend
92-zha/zha		562.12 ± 0.11	4.893 ± 0.050	0.3052 ± 0.0023	1,7	Zhang et al.
95-tso/amb		562.05 ± 0.07	4.895 ± 0.006	0.305 ± 0.004	10	Tsonopoulos and Ambrose
2001-liu/fu		563.65	5.01	0.297	2a	Liu et al.
2006-naz/sar-1		562.4 ± 0.2			1	Nazmutdinov et al.
2012-tho/lem		$(562.02)^i$	4.894 ± 0.015	0.3048 ± 0.004	8	Thol et al.
recommended values		562.05 ± 0.07	4.895 ± 0.006	0.305 ± 0.004		
METHYLBENZENE (Toluene): Molar Mass, 92.141 g; C_7H_8 ; CASRN 108-88-3						
72-hal/tow				0.2914 ± 0.0012	7	Hales and Townsend
95-tso/amb		591.75 ± 0.15	4.108 ± 0.010	0.292 ± 0.004	10	Tsonopoulos and Ambrose
95-wil/wil	$(0.318 \pm 0.001) \text{ L}\cdot\text{mol}^{-1}$	591.90 ± 0.05	4.111 ± 0.005	0.290 ± 0.001	2a	Wilson et al.
2000-von/wil-1		591.8 ± 1	4.11 ± 0.08		3	VonNiederhausern et al.
2002-nik/pop		588 ± 6	4.09 ± 0.12		4	Nikitin et al.
2003-abd/ste		$(593.871)^a$	2.220 ± 0.004		6	Abdulagatov et al.
2005-zha/qin-1		592.2	4.11		2	Zhang et al.
2008-abd/pol		591.89 ± 0.02		0.2916 ± 0.002	4	Abdulagatov et al.
2008-naz/ale		592.0 ± 0.5			1	Nazmutdinov et al.
recommended values		591.75 ± 0.15	4.108 ± 0.010	0.292 ± 0.004		
ETHYLBENZENE: Molar Mass, 106.168 g; C_8H_{10} ; CASRN 100-41-4						
72-hal/tow				0.2843 ± 0.0008	7	Hales and Townsend
95-wil/wil	$(0.372 \pm 0.001) \text{ L}\cdot\text{mol}^{-1}$	617.26 ± 0.05	3.616 ± 0.005	0.285 ± 0.001	2a	Wilson et al.
95-tso/amb		617.15 ± 0.1	3.609 ± 0.010	0.284 ± 0.004	10	Tsonopoulos and Ambrose
97-chi/kni-3		617.3 ± 1.0			4	Chirico et al.
97-chi/kni-3		$(617.15)^j$	3.630 ± 0.040		6	Chirico et al.
2000-von/wil-1		618.0 ± 1	3.62 ± 0.07		3	VonNiederhausern et al.
2002-nik/pop		614 ± 6	3.60 ± 0.11		4	Nikitin et al.

Table 2. continued

ref	values reported in nonstandard units	T_{90}/K	p/MPa	$\rho/\text{g}\cdot\text{cm}^{-3}$	method	authors
ETHYLBENZENE: Molar Mass, 106.168 g; C_8H_{10} ; CASRN 100-41-4						
recommended values		617.15 ± 0.1	3.609 ± 0.010	0.284 ± 0.004		
BUTYLBENZENE: Molar Mass, 134.222 g; $\text{C}_{10}\text{H}_{14}$; CASRN 104-51-8						
95-tso/amb		660.5 ± 0.5	2.89 ± 0.04	0.270 ± 0.010	10	Tsonopoulos and Ambrose
2002-nik/pop		660 ± 7	2.90 ± 0.09		4	Nikitin et al.
2002-ste/chi		660.5 ± 1	2.990 ± 0.2	0.269 ± 0.009	4,6,7	Steele et al.
2009-ihm/wol-1		660.3 ± 0.1	2.890 ± 0.010	0.2688 ± 0.005	2	Ihmels et al.
recommended values		660.5 ± 0.5	2.89 ± 0.04	0.270 ± 0.010		
PENTYLBENZENE: Molar Mass, 148.249 g; $\text{C}_{11}\text{H}_{16}$; CASRN 538-68-1						
2002-nik/pop		675 ± 7	2.58 ± 0.08		4d	Nikitin et al.
HEXYLBENZENE: Molar Mass, 162.276 g; $\text{C}_{12}\text{H}_{18}$; CASRN 1077-16-3						
2002-nik/pop		695 ± 7	2.35 ± 0.07		4d	Nikitin et al.
HEPTYLBENZENE: Molar Mass, 176.303 g; $\text{C}_{13}\text{H}_{20}$; CASRN 1078-71-3						
2002-nik/pop		708 ± 7	2.14 ± 0.06		4d	Nikitin et al.
OCTYLBENZENE: Molar Mass, 190.330 g; $\text{C}_{14}\text{H}_{22}$; CASRN 2189-60-8						
2002-nik/pop		725 ± 7	1.98 ± 0.06		4d	Nikitin et al.
DECYLBENZENE: Molar Mass, 218.384 g; $\text{C}_{16}\text{H}_{26}$; CASRN 104-72-3						
2002-nik/pop		752 ± 8	1.72 ± 0.05		4d	Nikitin et al.
UNDECYLBENZENE: Molar Mass, 232.411 g; $\text{C}_{17}\text{H}_{28}$; CASRN 6742-54-7						
2002-nik/pop		763 ± 8	1.64 ± 0.05		4d	Nikitin et al.
TRIDECYLBENZENE: Molar Mass, 260.465 g; $\text{C}_{19}\text{H}_{32}$; CASRN 123-02-4						
2002-nik/pop		790 ± 8	1.54 ± 0.05		4d	Nikitin et al.
1,2-DIMETHYLBENZENE (<i>o</i> -Xylene): Molar Mass, 106.168; C_8H_{10} ; CASRN 95-47-6						
72-hal/tow				0.2870 ± 0.0014	7	Hales and Townsend
95-tso/amb		630.3 ± 0.5	3.732 ± 0.04	0.287 ± 0.004	10	Tsonopoulos and Ambrose
97-chi/kni-2		630.5 ± 1.0			4	Chirico et al.
97-chi/kni-2		(630.3) ^f	3.730 ± 0.040		6	Chirico et al.
recommended values		630.3 ± 0.5	3.732 ± 0.04	0.287 ± 0.004		
1,3-DIMETHYLBENZENE (<i>m</i> -Xylene): Molar Mass, 106.168 g; C_8H_{10} ; CASRN 108-38-3						
72-hal/tow				0.2825 ± 0.0013	7	Hales and Townsend
95-tso/amb		617.0 ± 0.5	3.541 ± 0.04	0.283 ± 0.004	10	Tsonopoulos and Ambrose
97-chi/kni-1		617.6 ± 1.0			4	Chirico et al.
97-chi/kni-1		(617.0) ^f	3.540 ± 0.040		6	Chirico et al.
recommended values		617.0 ± 0.5	3.541 ± 0.04	0.283 ± 0.004		
1,4-DIMETHYLBENZENE (<i>p</i> -Xylene): Molar Mass, 106.168 g; C_8H_{10} ; CASRN 106-42-3						
72-hal/tow				0.2815 ± 0.0011	7	Hales and Townsend
95-tso/amb		616.2 ± 0.2	3.511 ± 0.04	0.281 ± 0.004	10	Tsonopoulos and Ambrose
97-chi/kni		616.4 ± 1.0			4	Chirico et al.
97-chi/kni		(616.2) ^f	3.510 ± 0.040		6	Chirico et al.
recommended values		616.2 ± 0.2	3.511 ± 0.04	0.281 ± 0.004		
(1-METHYLETHYL)BENZENE (Isopropylbenzene, Cumene): Molar Mass, 120.195; C_9H_{12} ; CASRN 98-82-8						
72-hal/tow				0.2790 ± 0.0007	7	Hales and Townsend
95-tso/amb		631.0 ± 0.5	3.209 ± 0.04		10	Tsonopoulos and Ambrose
95-wil/wil	(0.422 ± 0.001) L·mol ⁻¹	631.23 ± 0.05	3.217 ± 0.005	0.2850 ± 0.001	2a	Wilson et al.
recommended values		631.0 ± 0.5	3.209 ± 0.04	0.282 ± 0.005		
(1-METHYLPROPYL)BENZENE (<i>sec</i> -Butylbenzene): Molar Mass, 134.222; $\text{C}_{10}\text{H}_{14}$; CASRN 135-98-8						
2002-ste/chi		652.5 ± 1	3.025 ± 0.2	0.274 ± 0.009	4,6,7	Steele et al.
2009-ihm/wol-1		653.0 ± 0.1	3.027 ± 0.010	0.2800 ± 0.005	2	Ihmels et al.
recommended values		652.8 ± 0.4	3.027 ± 0.04	0.278 ± 0.005		
(1,1-DIMETHYLETHYL)BENZENE (<i>tert</i> -Butylbenzene): Molar Mass, 134.222; $\text{C}_{10}\text{H}_{14}$; CASRN 98-06-06						
92-zha/zha		648.31 ± 0.51	2.979 ± 0.020	0.2846 ± 0.0021	1,7	Zhang et al.
2002-ste/chi		647.5 ± 1	2.900 ± 0.200	0.277 ± 0.009	4,6,7	Steele et al.
2009-ihm/wol-1		646.9 ± 0.1	2.903 ± 0.010	0.2769 ± 0.005	2	Ihmels et al.
recommended values		648 ± 1	2.95 ± 0.05	0.280 ± 0.005		
1,4-Bis(1-METHYLETHYL)BENZENE: Molar Mass, 162.276 g; $\text{C}_{12}\text{H}_{18}$; CASRN 100-18-5						
2002-ste/chi-4		675 ± 1	2.33 ± 0.10	0.263 ± 0.010	4,6,7	Steele et al.
2006-von/wil-2		676.2 ± 2	2.36 ± 0.05		3d	VonNiederhausern et al.
recommended values		676 ± 2	2.35 ± 0.10	0.263 ± 0.010		
1,3,5-TRIETHYLBENZENE: Molar Mass, 162.276 g; $\text{C}_{12}\text{H}_{18}$; CASRN 102-25-0						
97-ste/chi		679 ± 1	2.435 ± 0.10	0.260 ± 0.010	4,6,7	Steele et al.
1,3-Bis(1,1-DIMETHYLETHYL)BENZENE: Molar Mass, 190.330 g; $\text{C}_{14}\text{H}_{22}$; CASRN 1014-60-4						
2006-nes/nes		687.6			1	Nesterov et al.

Table 2. continued

ref	values reported in nonstandard units	T_{90}/K	p/MPa	$\rho/\text{g}\cdot\text{cm}^{-3}$	method	authors
1,4-Bis(1,1-DIMETHYLETHYL)BENZENE: Molar Mass, 190.330 g; $\text{C}_{14}\text{H}_{22}$; CASRN 1012-72-2						
97-ste/chi-1		708 ± 1	2.300 ± 0.100	0.260 ± 0.010	4,6,7	Steele et al.
2006-nes/nes		702.9			1	Nesterov et al.
recommended values		708 ± 2	2.300 ± 0.100	0.260 ± 0.010		
1,3-Bis(1,1-DIMETHYLETHYL)-5-METHYLBENZENE: Molar Mass, 204.357 g; $\text{C}_{15}\text{H}_{24}$; CASRN 15181-11-0						
2006-nes/nes		695.5			1	Nesterov et al.
1,3,5-Tris(1,1-DIMETHYLETHYL)BENZENE: Molar Mass, 246.438 g; $\text{C}_{18}\text{H}_{30}$; CASRN 1460-02-2						
2006-nes/nes		700.5			1	Nesterov et al.
1,2,4,5-Tetrakis(1-METHYLETHYL)BENZENE: Molar Mass, 246.438 g; $\text{C}_{18}\text{H}_{30}$; CASRN 635-11-0						
2002-ste/chi-4		703 ± 2	1.600 ± 0.300	0.270 ± 0.030	4,6,7	Steele et al.
ETHENYLBENZENE(Styrene): Molar Mass, 104.152 g; C_8H_8 ; CASRN 100-42-5						
2000-von/wil-1		635.2 ± 1	3.87 ± 0.08		3d	VonNiederhausern et al.
2003-ste		638	4.000	0.295	4,6,7	Steele
recommended values		637 ± 2	3.95 ± 0.05	0.295 ± 0.010		
FLUORENE: Molar Mass, 166.223 g; $\text{C}_{13}\text{H}_{10}$; CASRN 86-73-7						
2015-nik/pop		836 ± 8	3.18 ± 0.1		4d	Nikitin and Popov
DIPHENYLMETHANE: Molar Mass, 168.239 g; $\text{C}_{13}\text{H}_{12}$; CASRN 101-81-5						
95-tso/amb		760 ± 8	2.71 ± 0.3	0.299 ± 0.030	10	Tsonopoulos and Ambrose
2002-von/gil		770.0 ± 1.0	2.92 ± 0.03		3	VonNiederhausern and Giles
2005-chi/ste		778 ± 2	3.280	0.2996	4d,6,7	Chirico and Steele.
recommended values		770 ± 10	2.9 ± 0.3	0.30 ± 0.03		
1,3-DIPHENYLBENZENE (<i>m</i> -Terphenyl): Molar Mass, 230.310 g; $\text{C}_{18}\text{H}_{14}$; CASRN 92-06-8						
95-tso/amb		883 ± 10	2.48 ± 0.5	0.318 ± 0.040	10	Tsonopoulos and Ambrose
97-nik/pav		882 ± 9	2.16 ± 0.04		4d	Nikitin et al.
recommended values		882 ± 10	2.2 ± 0.3	0.32 ± 0.04		
NAPHTHALENE: Molar Mass, 128.174 g; C_{10}H_8 ; CASRN 91-20-3						
$T_{90} - T_{68} = 0.006$ at 749 K						
49-gol/rav	468 °C	741			1	Golik and Ravikovich
72-hal/tow				0.3201 ± 0.0008	7	Hales and Townsend
95-tso/amb		748.4 ± 0.5	4.05 ± 0.05	0.315 ± 0.005	10	Tsonopoulos and Ambrose
97-nik/pav		747 ± 7	3.96 ± 0.08		4d	Nikitin et al.
2006-von/wil		750.0 ± 2	4.13 ± 0.08		3d	VonNiederhausern et al.
recommended values		748.4 ± 0.5	4.05 ± 0.05	0.315 ± 0.005		
1-METHYLNAPHTHALENE: Molar Mass, 142.201 g; $\text{C}_{11}\text{H}_{10}$; CASRN 90-12-0						
95-tso/amb		772 ± 1	3.60 ± 0.1		10	Tsonopoulos and Ambrose
2013-nik/pop-1		773 ± 8	3.54 ± 0.11		4d	Nikitin et al.
recommended values		772 ± 1	3.57 ± 0.10			
2-METHYLNAPHTHALENE: Molar Mass, 142.201 g; $\text{C}_{11}\text{H}_{10}$; CASRN 91-57-6						
72-hal/tow				0.3083 ± 0.0005	7	Hales and Townsend
95-tso/amb		761 ± 1			10	Tsonopoulos and Ambrose
2013-nik/pop-1		755 ± 8	3.30 ± 0.10		4d	Nikitin et al.
recommended values		761 ± 1	3.30 ± 0.10	0.308 ± 0.002		
1,3-DIMETHYLNAPHTHALENE: Molar Mass, 156.228 g; $\text{C}_{12}\text{H}_{12}$; CASRN 575-41-7						
2013-nik/pop-1		772 ± 8	2.74 ± 0.08		4d	Nikitin et al.
1,4-DIMETHYLNAPHTHALENE: Molar Mass, 156.228 g; $\text{C}_{12}\text{H}_{12}$; CASRN 571-58-4						
2013-nik/pop-1		778 ± 8	2.79 ± 0.08		4d	Nikitin et al.
2-ETHYLNAPHTHALENE: Molar Mass, 156.228 g; $\text{C}_{12}\text{H}_{12}$; CASRN 939-27-5						
2013-nik/pop-1		763 ± 8	2.79 ± 0.08		4d	Nikitin et al.
ACENAPHTHENE: Molar Mass, 154.212 g; $\text{C}_{12}\text{H}_{10}$; CASRN 83-32-9						
2015-nik/pop		820 ± 8	3.70 ± 0.1		4d	Nikitin and Popov
ANTHRACENE: Molar Mass, 178.234 g; $\text{C}_{14}\text{H}_{10}$; CASRN 120-12-7						
2015-nik/pop		894 ± 9	3.17 ± 0.1		4d	Nikitin and Popov
PHENANTHRENE: Molar Mass, 178.234 g; $\text{C}_{14}\text{H}_{10}$; CASRN 85-01-8						
63-che	596.1 °C	869.2			1	Cheng
2015-nik/pop		893 ± 9	3.25 ± 0.1		4d	Nikitin and Popov
recommended values		893 ± 9	3.25 ± 0.1			
PYRENE: Molar Mass, 202.256 g; $\text{C}_{16}\text{H}_{10}$; CASRN 129-00-0						
2015-nik/pop		974 ± 10	3.03 ± 0.09		4d	Nikitin and Popov
1,2,3,4-TETRAHYDRONAPHTHALENE (Tetralin): Molar Mass, 132.206 g; $\text{C}_{10}\text{H}_{12}$; CASRN 119-64-2						
94-gud/tej		719.5 ± 0.4	3.63 ± 0.02		2c	Gude and Teja
95-tso/amb		720 ± 1	3.65 ± 0.1	0.324 ± 0.010	10	Tsonopoulos and Ambrose
recommended values		720 ± 1	3.65 ± 0.1	0.324 ± 0.010		

Table 2. continued

ref	values reported in nonstandard units	T_{90}/K	p/MPa	$\rho/\text{g}\cdot\text{cm}^{-3}$	method	authors
LINEAR 1-ALKANOLS						
METHANOL: Molar Mass, 32.042 g; CH_4O ; CASRN 67-56-1						
85-zaw		512.54	8.096		1,7	Zawisza
95-gud/tej		512.5 $\pm$ 0.2	8.084 $\pm$ 0.02	0.273 $\pm$ 0.002	10	Gude and Teja
2003-liu/qin		512.6 $\pm$ 0.3	8.12 $\pm$ 0.03	0.276	2a	Liu et al.
2006-pol/rad		512.78		0.277	4	Polikhronidi et al.
2007-baz/abd		512.75 $\pm$ 0.1	8.120 $\pm$ 0.02	0.2716 $\pm$ 0.003	3	Bazaev et al.
2008-gil/oti		512.90 $\pm$ 0.08	8.094 $\pm$ 0.007		2	Gil et al.
2008-naz/ale		513.7 $\pm$ 0.5			1	Nazmutdinov et al.
2010-ale/naz		513.2			1	Alekina et al.
2012-gil/bla		512.90 $\pm$ 0.14	8.094 $\pm$ 0.011		2	Gil et al.
2013-han/xia		512.3 $\pm$ 1.2	8.128 $\pm$ 0.015		2	Han et al.
recommended values		512.5 $\pm$ 0.2	8.084 $\pm$ 0.02	0.273 $\pm$ 0.002		
ETHANOL: Molar Mass, 46.069 g; $\text{C}_2\text{H}_6\text{O}$; CASRN 64-17-5						
95-deá/vic		513.88 $\pm$ 0.05	6.136 $\pm$ 0.010		1,5	Deák et al.
95-gud/tej		514.0 $\pm$ 0.2	6.137 $\pm$ 0.02	0.275 $\pm$ 0.002	10	Gude and Teja
2003-mor/lui		513.9 $\pm$ 0.3			1	Morton et al.
2004-hu/qin		516.7 $\pm$ 0.3	6.27 $\pm$ 0.03	0.280	2a	Hu et al.
2007-baz/abd-1		514.71 $\pm$ 0.2	6.268 $\pm$ 0.008	0.27321 $\pm$ 0.002	3	Bazaev et al.
2007-pol/abd		514.44 $\pm$ 0.2		0.2837 $\pm$ 0.002	4,7	Polikhronidi et al.
2008-naz/ale		514.2 $\pm$ 0.5			1	Nazmutdinov et al.
2008-gil/oti		514.25 $\pm$ 0.08	6.168 $\pm$ 0.007		2	Gil et al.
2010-ale/naz		514.2			1	Alekina et al.
2010-soo/thé		513.90 $\pm$ 0.01	6.128		2	Soo et al.
2011-soo		513.90 $\pm$ 0.12	6.128 $\pm$ 0.001		2	Soo
2012-gil/bla		514.25 $\pm$ 0.14	6.168 $\pm$ 0.011		2	Gil et al.
2013-han/xia		514.2 $\pm$ 1.2	6.135 $\pm$ 0.015		2	Han et al.
recommended values		513.9 $\pm$ 0.1	6.14 $\pm$ 0.02	0.275 $\pm$ 0.002		
1-PROPANOL: Molar Mass, 60.096 g; $\text{C}_3\text{H}_8\text{O}$; CASRN 71-23-8						
95-gud/tej		536.8 $\pm$ 0.3	5.169 $\pm$ 0.02	0.276 $\pm$ 0.003	10	Gude and Teja
2003-abd/ste		(536.780) ^a	5.160 $\pm$ 0.005		6	Abdulagatov et al.
2003-mor/lui		536.8 $\pm$ 0.3			1	Morton et al.
2008-gil/oti		537.06 $\pm$ 0.07	5.179 $\pm$ 0.005		2	Gil et al.
2010-ale/naz		536.9			1	Alekina et al.
2010-soo/thé		536.44 $\pm$ 0.03	5.178 $\pm$ 0.001		2	Soo et al.
2011-soo		536.44 $\pm$ 0.12	5.178 $\pm$ 0.001		2	Soo
2012-gil/bla		537.06 $\pm$ 0.14	5.179 $\pm$ 0.011		2	Gil et al.
2013-han/xia		537.0 $\pm$ 1.2	5.172 $\pm$ 0.015		2	Han et al.
recommended values		536.8 $\pm$ 0.3	5.175 $\pm$ 0.007	0.276 $\pm$ 0.003		
1-BUTANOL: Molar Mass, 74.123 g; $\text{C}_4\text{H}_{10}\text{O}$; CASRN 71-36-3						
$T_{90} - T_{68} = -0.039$ at 563 K						
80-nau/tyv	289.9 °C, 275 $\text{cm}^3\cdot\text{mol}^{-1}$	563.0		0.270	1	Naumova et al.
82-vas/tyv	290 °C	563			1	Vasyl'eva and Tyvina
94-sha/naz		562.17	4.3914	0.2702034	3	Shakhverdiev et al.
95-gud/tej		563.0 $\pm$ 0.3	4.414 $\pm$ 0.02	0.271 $\pm$ 0.003	10	Gude and Teja
2003-abd/ste		(563.020) ^a	4.420 $\pm$ 0.003		6	Abdulagatov et al.
2008-gil/oti		563.42 $\pm$ 0.03	4.425 $\pm$ 0.003		2	Gil et al.
2008-naz/ale		563.0 $\pm$ 0.5			1	Nazmutdinov et al.
2012-gil/bla		563.42 $\pm$ 0.14	4.425 $\pm$ 0.011		2	Gil et al.
2013-ras/rad		562.95 $\pm$ 0.3	4.405 $\pm$ 0.01	0.270 $\pm$ 0.003	3	Rasulov et al.
2013-ras/rad		563.05 $\pm$ 0.02		0.270 $\pm$ 0.001	4	Rasulov et al.
recommended values		563.1 $\pm$ 0.1	4.41 $\pm$ 0.01	0.270 $\pm$ 0.002		
1-PENTANOL: Molar Mass, 88.150 g; $\text{C}_5\text{H}_{12}\text{O}$; CASRN 71-41-0						
95-gud/tej		588.1 $\pm$ 0.5	3.897 $\pm$ 0.02	0.270 $\pm$ 0.003	10	Gude and Teja
2003-mor/lui		587.7 $\pm$ 0.5			1	Morton et al.
recommended values		588.1 $\pm$ 0.5	3.897 $\pm$ 0.02	0.270 $\pm$ 0.003		
1-HEXANOL: Molar Mass, 102.177 g; $\text{C}_6\text{H}_{14}\text{O}$; CASRN 111-27-3						
$T_{90} - T_{68} = -0.040$ at 610 K						
80-nau/tyv	337.6 °C, 33.0 atm, 380 $\text{cm}^3\cdot\text{mol}^{-1}$	610.7	3.34	0.269	1	Naumova et al.
95-gud/tej		610.3 $\pm$ 0.5	3.417 $\pm$ 0.02	0.264 $\pm$ 0.005	10	Gude and Teja
2010-ale/naz		610.8			1d	Alekina et al.
recommended values		610.3 $\pm$ 0.5	3.417 $\pm$ 0.02	0.264 $\pm$ 0.005		

Table 2. continued

ref	values reported in nonstandard units	T_{90}/K	p/MPa	$\rho/\text{g}\cdot\text{cm}^{-3}$	method	authors
1-HEPTANOL: Molar Mass, 116.204 g; $\text{C}_7\text{H}_{16}\text{O}$; CASRN 111-70-6						
95-gud/tej		632.6 ± 0.5	3.058 ± 0.02	0.267 ± 0.005	10	Gude and Teja
2008-gil/oti		632.67 ± 0.08	3.060 ± 0.007		2	Gil et al.
recommended values		632.6 ± 0.6	3.058 ± 0.02	0.267 ± 0.005		
1-TRIDECANOL: Molar Mass, 200.366 g; $\text{C}_{13}\text{H}_{28}\text{O}$; CASRN 112-70-9						
98-nik/pav		732 ± 7	1.79 ± 0.05		4d	Nikitin et al.
1-TETRADECANOL: Molar Mass, 214.393 g; $\text{C}_{14}\text{H}_{30}\text{O}$; CASRN 112-72-1						
98-nik/pav		743 ± 7	1.67 ± 0.05		4d	Nikitin et al.
1-PENTADECANOL: Molar Mass, 228.420 g; $\text{C}_{15}\text{H}_{32}\text{O}$; CASRN 629-76-5						
98-nik/pav		757 ± 8	1.60 ± 0.05		4d	Nikitin et al.
1-HEXADECANOL: Molar Mass, 242.447 g; $\text{C}_{16}\text{H}_{34}\text{O}$; CASRN 36653-82-4						
98-nik/pav		770 ± 8	1.46 ± 0.04		4d	Nikitin et al.
1-HEPTADECANOL: Molar Mass, 256.474 g; $\text{C}_{17}\text{H}_{36}\text{O}$; CASRN 1454-85-9						
98-nik/pav		780 ± 8	1.35 ± 0.04		4d	Nikitin et al.
1-OCTADECANOL: Molar Mass, 270.501 g; $\text{C}_{18}\text{H}_{38}\text{O}$; CASRN 112-92-5						
98-nik/pav		790 ± 8	1.28 ± 0.04		4d	Nikitin et al.
1-EICOSANOL: Molar Mass, 298.555 g; $\text{C}_{20}\text{H}_{42}\text{O}$; CASRN 629-96-9						
98-nik/pav		808 ± 8	1.15 ± 0.03		4d	Nikitin et al.
1-DOCOSANOL: Molar Mass, 326.609 g; $\text{C}_{22}\text{H}_{46}\text{O}$; CASRN 661-19-8						
98-nik/pav		827 ± 8	1.07 ± 0.03		4d	Nikitin et al.
OTHER ALKANOLS						
2-PROPANOL: Molar Mass, 60.096 g; $\text{C}_3\text{H}_8\text{O}$; CASRN 67-63-0						
$T_{90} - T_{68} = -0.040$ at 508 K						
85-lyo		508.8	4.807		1a	Lyons
87-dau/jal ^k	$(234.21 \pm 0.1) ^\circ\text{C}, \pm 0.5$ psi	507.3 ± 0.1	4.716 ± 0.003		1a	Daubert et al.
95-gud/tej		508.3 ± 0.3	4.764 ± 0.02	0.271 ± 0.004	10	Gude and Teja
2003-abd/ste		$(508.510)^a$	4.800 ± 0.003		6	Abdulagatov et al.
2003-mor/lui		508.3 ± 0.3			1	Morton et al.
2004-oh/lee		508.15	4.755		2	Oh et al.
2008-naz/ale		508.5			1	Nazmutdinov et al.
2010-ale/naz		508.2			1	Alekina et al.
recommended values		508.3 ± 0.3	4.764 ± 0.2	0.271 ± 0.004		
2-PROPEN-1-OL: Molar Mass, 58.080 g; $\text{C}_3\text{H}_6\text{O}$; CASRN 107-18-6						
$T_{90} - T_{48} = 0.028$ at 545.1 K						
1882-nad	271.9 $^\circ\text{C}$	545.1			1	Nadezhdin
2005-wan/jia		539.85 ± 0.28	5.76 ± 0.02		1d	Wang et al.
2006-von/wil		547.1 ± 2	5.64 ± 0.11		3d	VonNiederhausern et al.
recommended values		543 ± 5	5.70 ± 0.2			
2-BUTANOL: Molar Mass, 74.123 g; $\text{C}_4\text{H}_{10}\text{O}$; CASRN 78-92-2						
95-gud/tej		536.2 ± 0.3	4.202 ± 0.02	0.276 ± 0.004	10	Gude and Teja
97-ste/roe		532.03 ± 0.05	4.226 ± 0.01		1	Stevens et al.
2003-abd/ste		$(536.001)^a$	4.280 ± 0.002		6	Abdulagatov et al.
2003-mor/lui		536.1 ± 0.3			1	Morton et al.
recommended values		536.2 ± 0.3	4.202 ± 0.02	0.276 ± 0.004		
2-METHYL-2-PROPANOL: Molar Mass, 74.123 g; $\text{C}_4\text{H}_{10}\text{O}$; CASRN 75-65-0						
83-hal/gun		$(506.21)^l$		0.270	7	Hales et al.
95-gud/tej		506.2 ± 0.3	3.972 ± 0.02	0.270 ± 0.005	10	Gude and Teja
2003-mor/lui		506.5 ± 0.3			1	Morton et al.
recommended values		506.2 ± 0.3	3.972 ± 0.02	0.270 ± 0.005		
2-PENTANOL: Molar Mass, 88.150 g; $\text{C}_5\text{H}_{12}\text{O}$; CASRN 6032-29-7						
95-gud/tej		560.3 ± 0.5	3.675 ± 0.02	0.268 ± 0.015	10	Gude and Teja
2003-mor/lui		560.4 ± 0.3			1	Morton et al.
recommended values		560.3 ± 0.5	3.675 ± 0.02	0.268 ± 0.015		
3-METHYL-1-BUTANOL: Molar Mass, 88.150 g; $\text{C}_5\text{H}_{12}\text{O}$; CASRN 123-51-3						
95-gud/tej		577.2 ± 0.5	3.93 ± 0.02		10	Gude and Teja
2003-mor/lui		579.4 ± 0.3			1	Morton et al.
recommended values		577.2 ± 0.5	3.93 ± 0.02			
3-HEXANOL: Molar Mass, 102.177 g; $\text{C}_6\text{H}_{14}\text{O}$; CASRN 623-37-0						
92-zha/zha		582.59 ± 0.11	3.288 ± 0.020	0.2755 ± 0.0021	1,7	Zhang et al.
95-gud/tej		582.4 ± 0.5	3.36 ± 0.02	0.267 ± 0.005	10	Gude and Teja
recommended values		582.4 ± 0.5	3.33 ± 0.02	0.271 ± 0.005		

Table 2. continued

ref	values reported in nonstandard units	T_{90}/K	p/MPa	$\rho/g\cdot cm^{-3}$	method	authors
CYCLOHEXANOL: Molar Mass, 100.161 g; $C_6H_{12}O$; CASRN 108-93-0						
95-gud/tej		650.1 ± 2.0	4.26 ± 0.05		10	Gude and Teja
96-wil/wil		647.1 ± 0.1	4.401 ± 0.021		2a,c	Wilson et al.
97-ste/chi-1		648 ± 1	4.075 ± 0.1	0.300	4,6,7	Steele et al.
recommended values		648 ± 1	4.25 ± 0.1	0.30 ± 0.01		
ALKOXYALKANOLS						
1-(1,1-DIMETHYLETHOXY)-2-PROPANOL: Molar Mass, 132.203 g; $C_7H_{16}O_2$; CASRN 57018-52-7						
2006-von/wil		600.7 ± 2	2.72 ± 0.05		3d	VonNiederhausern et al.
POLYOLS						
1,2-PROPANEDIOL (Propylene Glycol): Molar Mass, 76.095 g; $C_3H_8O_2$; CASRN 57-55-6						
2000-von/wil		676.4 ± 0.3	5.941 ± 0.007		2a,c	VonNiederhausern et al.
2002-ste/chi-2	(676) ^m		6.750 ± 0.5	0.308 ± 0.020	4,6,7	Steele et al.
recommended values		676.4 ± 0.6	6.00 ± 0.01	0.31 ± 0.03		
1,3-PROPANEDIOL (Trimethylene Glycol): Molar Mass, 76.095 g; $C_3H_8O_2$; CASRN 504-63-2						
2000-von/wil		722 ± 4	6.3 ± 0.6		2a,c	VonNiederhausern et al.
2002-wil/von		718.2 ± 2	6.55 ± 0.13		3d	Wilson et al.
2013-nik/pop		721 ± 11	6.38 ± 0.26		4d	Nikitin et al.
recommended values		720 ± 3	6.4 ± 0.1			
1,3-BUTANEDIOL (1,3-Butylene Glycol): Molar Mass, 90.122 g; $C_4H_{10}O_2$; CASRN 107-88-0						
96-ste/chi-1		676 ± 1	4.02 ± 0.20	0.295 ± 0.009	4,6,7	Steele et al.
2006-von/wil-2		692.4 ± 2	5.18 ± 0.10		3d	VonNiederhausern et al.
recommended values		690 ± 8	5.0 ± 0.3	0.30 ± 0.02		
1,4-BUTANEDIOL (Tetramethylene Glycol): Molar Mass, 90.122 g; $C_4H_{10}O_2$; CASRN 110-63-4						
96-wil/wil		727 ± 2	6.22 ± 0.14		2a,c	Wilson et al.
2002-wil/von		723.8 ± 2	5.52 ± 0.11		3d	Wilson et al.
2013-nik/pop		728 ± 11	5.40 ± 0.22		4d	Nikitin et al.
recommended values		725 ± 2	5.8 ± 0.4			
2-METHYL-1,3-PROPANEDIOL: Molar Mass, 90.122 g; $C_4H_{10}O_2$; CASRN 2163-42-0						
2002-wil/von		708.0 ± 2	5.35 ± 0.11		3d	Wilson et al.
1,5-PENTANEDIOL (Pentamethylene Glycol): Molar Mass, 104.149 g; $C_5H_{12}O_2$; CASRN 111-29-5						
2013-nik/pop		731 ± 11	4.00 ± 0.16		4d	Nikitin et al.
2,2-DIMETHYL-1,3-PROPANEDIOL: Molar Mass, 104.149 g; $C_5H_{12}O_2$; CASRN 126-30-7						
2014-nik/pop-3		687 ± 10	4.20 ± 0.2		4d	Nikitin and Popov
3-OXA-1,5-PENTANEDIOL (Diethylene Glycol): Molar Mass, 106.121 g; $C_4H_{10}O_3$; CASRN 111-46-6						
2001-kud/amb		750 ± 15	4.7 ± 1		10	Kudchadker et al.
2002-ste/chi-1	(753) ⁿ		4.77 ± 0.2		6	Steele et al.
recommended values		750 ± 15	4.7 ± 1			
1,6-HEXANEDIOL (Hexamethylene Glycol): Molar Mass, 118.176 g; $C_6H_{14}O_2$; CASRN 629-11-8						
2006-von/wil-1		740.8 ± 2	4.08 ± 0.08		3d	VonNiederhausern et al.
2013-nik/pop		734 ± 11	3.40 ± 0.16		4d	Nikitin et al.
recommended values		740 ± 3	4.0 ± 0.1			
4-OXA-1,7-HEPTANEDIOL (Dipropylene Glycol) ^o : Molar Mass, 134.175 g; $C_6H_{14}O_3$; CASRN 25265-71-8						
2006-von/wil-2		705.2 ± 2	3.38 ± 0.07		3d	VonNiederhausern et al.
1,8-OCTANEDIOL (Octamethylene Glycol): Molar Mass, 146.230 g; $C_8H_{18}O_2$; CASRN 629-41-4						
2013-nik/pop		752 ± 11	2.70 ± 0.14		4d	Nikitin et al.
1,9-NONANEDIOL (Nonamethylene Glycol): Molar Mass, 160.257 g; $C_9H_{20}O_2$; CASRN 3937-56-2						
2013-nik/pop		760 ± 11	2.40 ± 0.10		4d	Nikitin et al.
1,10-DECANEDIOL (Decamethylene Glycol): Molar Mass, 174.284 g; $C_{10}H_{22}O_2$; CASRN 112-47-0						
2013-nik/pop		770 ± 12	2.20 ± 0.09		4d	Nikitin et al.
1,12-DODECANEDIOL (Dodecamethylene Glycol): Molar Mass, 202.338 g; $C_{12}H_{26}O_2$; CASRN 5675-51-4						
2013-nik/pop		798 ± 12	1.93 ± 0.08		4d	Nikitin et al.
ALKANALS						
PROPANAL (Propionaldehyde): Molar Mass, 58.080 g; C_3H_6O ; CASRN 123-38-6						
2001-kud/amb		505 ± 2	5.26 ± 0.2	0.285 ± 0.005	10	Kudchadker et al.
2002-ma/gao		503.60 ± 0.1	5.038 ± 0.1		1	Ma et al.
recommended values		504 ± 2	5.15 ± 0.10	0.285 ± 0.005		
BUTANAL (Butyraldehyde): Molar Mass, 72.107 g; C_4H_8O ; CASRN 123-72-8						
2001-kud/amb		537 ± 3	4.32 ± 0.2	0.280 ± 0.005	10	Kudchadker et al.
2002-ma/gao		522.28 ± 0.1	4.410 ± 0.1		1	Ma et al.
recommended values		537 ± 3	4.32 ± 0.2	0.280 ± 0.005		
2-METHYLBUTANAL: Molar Mass, 86.134 g; $C_5H_{10}O$; CASRN 96-17-3						
2002-ma/gao		531.56 ± 0.1	3.712 ± 0.1		1	Ma et al.

Table 2. continued

ref	values reported in nonstandard units	T_{90}/K	p/MPa	$\rho/\text{g}\cdot\text{cm}^{-3}$	method	authors
PARALDEHYDE(ethanal trimer): Molar Mass, 132.159 g; $\text{C}_6\text{H}_{12}\text{O}_3$; CASRN 123-63-7						
2002-ma/gao		534.43 ± 0.1	4.037 ± 0.1		1	Ma et al.
ALKANONES						
2-PROPANONE (acetone): Molar Mass, 58.080 g; CASRN 67-64-1						
2001-kud/amb		508.1 ± 0.1	4.70 ± 0.01	0.273 ± 0.005	10	Kudchadker et al.
2006-lem/spa		508.1	4.70	0.273	8	Lemmon and Span
recommended values		508.1 ± 0.1	4.70 ± 0.01	0.273 ± 0.005		
2-BUTANONE (Methyl Ethyl Ketone): Molar Mass, 72.107 g; CASRN 78-93-3						
2001-kud/amb		536.7 ± 0.1	4.207 ± 0.01	0.270 ± 0.025	10	Kudchadker et al.
2008-naz/ale		536.8 ± 0.5			1	Nazmutdinov et al.
recommended values		536.7 ± 0.1	4.207 ± 0.01	0.270 ± 0.025		
5-HEXENE-2-ONE (Allyl Acetone): Molar Mass, 98.145 g; $\text{C}_6\text{H}_{10}\text{O}$; CASRN 109-49-9						
2005-wan/jia		593.47 ± 0.28	4.59 ± 0.02		1d	Wang et al.
ACIDS						
ETHANOIC ACID (Acetic Acid): Molar Mass, 60.052 g; $\text{C}_2\text{H}_4\text{O}_2$; CASRN 64-19-7						
83-hal/gun		$(592.71)^f$		0.351	7	Hales et al.
2001-kud/amb		590.7 ± 1	5.78 ± 0.02	0.351 ± 0.005	10	Kudchadker et al.
2001-nik/pav		590 ± 6	5.80 ± 0.17		4	Nikitin et al.
recommended values		590.7 ± 1	5.78 ± 0.02	0.351 ± 0.005		
PROPANOIC ACID (Propionic Acid): Molar Mass, 74.079 g; $\text{C}_3\text{H}_6\text{O}_2$; CASRN 79-09-4						
83-hal/gun		$(602.2)^f$		0.334	7	Hales et al.
2001-kud/amb		598.5 ± 1	4.67 ± 0.05	0.318 ± 0.005	10	Kudchadker et al.
recommended values		598.5 ± 1	4.67 ± 0.05	0.318 ± 0.005		
HEXANOIC ACID (Caproic Acid): Molar Mass, 116.160 g; $\text{C}_6\text{H}_{12}\text{O}_2$; CASRN 142-62-1						
2001-kud/amb		655 ± 2	3.38 ± 0.1	0.281 ± 0.005	10	Kudchadker et al.
2001-nik/pav		652 ± 7	3.36 ± 0.10		4d	Nikitin et al.
recommended values		655 ± 2	3.38 ± 0.1	0.281 ± 0.005		
OCTANOIC ACID (Caprylic Acid): Molar Mass, 144.214 g; $\text{C}_8\text{H}_{16}\text{O}_2$; CASRN 124-07-2						
2001-kud/amb		693 ± 2	2.87 ± 0.1	0.278 ± 0.005	10	Kudchadker et al.
2001-nik/pav		690 ± 7	2.85 ± 0.09		4d	Nikitin et al.
recommended values		693 ± 2	2.87 ± 0.1	0.278 ± 0.005		
UNDECANOIC ACID: Molar Mass, 186.295 g; $\text{C}_{11}\text{H}_{22}\text{O}_2$; CASRN 112-37-8						
2001-nik/pav		728 ± 7	2.13 ± 0.06		4d	Nikitin et al.
DODECANOIC ACID (Lauric Acid): Molar Mass, 200.322 g; $\text{C}_{12}\text{H}_{24}\text{O}_2$; CASRN 143-07-7						
2001-nik/pav		743 ± 7	1.93 ± 0.06		4d	Nikitin et al.
TETRADECANOIC ACID (Myristic Acid): Molar Mass, 228.376 g; $\text{C}_{14}\text{H}_{28}\text{O}_2$; CASRN 544-63-8						
2001-nik/pav		763 ± 8	1.64 ± 0.05		4d	Nikitin et al.
PENTADECANOIC ACID: Molar Mass, 242.403 g; $\text{C}_{15}\text{H}_{30}\text{O}_2$; CASRN 1002-84-2						
2001-nik/pav		777 ± 8	1.57 ± 0.05		4d	Nikitin et al.
HEXADECANOIC ACID (Palmitic Acid): Molar Mass, 256.430 g; $\text{C}_{16}\text{H}_{32}\text{O}_2$; CASRN 57-10-3						
2001-nik/pav		785 ± 8	1.49 ± 0.05		4d	Nikitin et al.
HEPTADECANOIC ACID (Margaric Acid): Molar Mass, 270.457 g; $\text{C}_{17}\text{H}_{34}\text{O}_2$; CASRN 506-12-7						
2001-nik/pav		792 ± 8	1.37 ± 0.04		4d	Nikitin et al.
OCTADECANOIC ACID (Stearic Acid): Molar Mass, 284.484 g; $\text{C}_{18}\text{H}_{36}\text{O}_2$; CASRN 57-11-4						
2001-nik/pav		803 ± 8	1.33 ± 0.04		4d	Nikitin et al.
EICOSANOIC ACID (Arachic Acid): Molar Mass, 312.538 g; $\text{C}_{20}\text{H}_{40}\text{O}_2$; CASRN 506-30-9						
2001-nik/pav		820 ± 8	1.20 ± 0.04		4d	Nikitin et al.
DOCOSANOIC ACID (Behenic Acid): Molar Mass, 340.592 g; $\text{C}_{22}\text{H}_{44}\text{O}_2$; CASRN 112-85-6						
2001-nik/pav		837 ± 8	1.11 ± 0.03		4d	Nikitin et al.
BUTANEDIOIC ACID (Succinic Acid): Molar Mass, 118.088 g; $\text{C}_4\text{H}_6\text{O}_4$; CASRN 110-15-6						
2004-nik/pop		851 ± 20	6.59 ± 0.30		4d	Nikitin et al.
PENTANEDIOIC ACID (Glutaric Acid): Molar Mass, 132.115 g; $\text{C}_5\text{H}_8\text{O}_4$; CASRN 110-94-1						
2004-nik/pop		840 ± 13	4.27 ± 0.20		4d	Nikitin et al.
HEXANEDIOIC ACID (Adipic Acid): Molar Mass, 146.142 g; $\text{C}_6\text{H}_{10}\text{O}_4$; CASRN 124-04-9						
2004-nik/pop		841 ± 13	3.85 ± 0.15		4d	Nikitin et al.
HEPTANEDIOIC ACID (Pimelic Acid): Molar Mass, 160.169 g; $\text{C}_7\text{H}_{12}\text{O}_4$; CASRN 111-16-0						
2004-nik/pop		842 ± 13	3.28 ± 0.13		4d	Nikitin et al.
OCTANEDIOIC ACID (Suberic Acid): Molar Mass, 174.196 g; $\text{C}_8\text{H}_{14}\text{O}_4$; CASRN 505-48-6						
2004-nik/pop		843 ± 13	2.97 ± 0.12		4d	Nikitin et al.
NONANEDIOIC ACID (Azelaic Acid): Molar Mass, 188.223 g; $\text{C}_9\text{H}_{16}\text{O}_4$; CASRN 123-99-9						
2004-nik/pop		844 ± 13	2.72 ± 0.11		4d	Nikitin et al.
DECANEDIOIC ACID (Sebacic Acid): Molar Mass, 202.250 g; $\text{C}_{10}\text{H}_{18}\text{O}_4$; CASRN 111-20-6						
2004-nik/pop		845 ± 13	2.50 ± 0.10		4d	Nikitin et al.

Table 2. continued

ref	values reported in nonstandard units	T_{90}/K	p/MPa	$\rho/\text{g}\cdot\text{cm}^{-3}$	method	authors
DODECANEDIOIC ACID: Molar Mass, 230.304 g; $\text{C}_{12}\text{H}_{22}\text{O}_4$; CASRN 693-23-2						
2004-nik/pop		859 ± 13	2.15 ± 0.09		4d	Nikitin et al.
TETRADECANEDIOIC ACID: Molar Mass, 258.358 g; $\text{C}_{14}\text{H}_{26}\text{O}_4$; CASRN 821-38-5						
2004-nik/pop		862 ± 13	1.90 ± 0.08		4d	Nikitin et al.
ESTERS						
ETHYL METHANOATE (Ethyl Formate): Molar Mass, 74.079 g; $\text{C}_3\text{H}_6\text{O}_2$; CASRN 109-94-4						
2001-kud/amb		508.4 ± 1	4.74 ± 0.05	0.323 ± 0.005	10	Kudchadker et al.
2011-naz/sam		506.6 ± 0.3			1	Nazmutdinov et al.
2011-naz/sam		508.5 ± 0.7			1 ^P	Nazmutdinov et al.
recommended values		507.5 ± 1	4.74 ± 0.05	0.323 ± 0.005		
PROPYL METHANOATE (Propyl Formate): Molar Mass, 88.106 g; $\text{C}_4\text{H}_8\text{O}_2$; CASRN 110-74-7						
2001-kud/amb		538.0 ± 1	4.06 ± 0.05	0.309 ± 0.005	10	Kudchadker et al.
2011-naz/sam		536.4 ± 0.5			1	Nazmutdinov et al.
2011-naz/sam		539.0 ± 0.7			1 ^P	Nazmutdinov et al.
recommended values		538.0 ± 1	4.06 ± 0.05	0.309 ± 0.005		
1-METHYLETHYL METHANOATE (Isopropyl Formate): Molar Mass, 88.106 g; $\text{C}_4\text{H}_8\text{O}_2$; CASRN 625-55-8						
95-ma/wan	261.45 °C	534.60	3.946		1	Ma et al.
2011-sam/naz		518.2 ± 0.6			1d	Samarov et al.
2011-sam/naz		520.5 ± 0.7			1d ^P	Samarov et al.
recommended values		520 ± 3	3.9 ± 0.1			
BUTYL METHANOATE (Butyl Formate): Molar Mass, 102.133 g; $\text{C}_5\text{H}_{10}\text{O}_2$; CASRN 592-84-7						
2011-naz/sam		566.0 ± 0.7			1d ^P	Nazmutdinov et al.
2-METHYLPROPYL METHANOATE (Isobutyl Formate): Molar Mass, 102.133 g; $\text{C}_5\text{H}_{10}\text{O}_2$; CASRN 542-55-2						
$T_{90} - T_{48} = 0.030$ at 551.4 K						
1887-nad	278.2 °C, 38.29 atm	551.4	3.880	0.2879	1	Nadezhdin
2011-sam/naz		552.5 ± 0.7			1d ^P	Samarov et al.
recommended values		552 ± 1	3.88 ± 0.05	0.29 ± 0.01		
PENTYL METHANOATE (Pentyl Formate): Molar Mass, 116.160 g; $\text{C}_6\text{H}_{12}\text{O}_2$; CASRN 638-49-3						
$T_{90} - T_{48} = 0.034$ at 551.4 K						
1887-nad	302.6 °C, 34.12 atm	575.8	3.457	0.282	1	Nadezhdin
2011-naz/sam		591.0 ± 0.7			1d ^P	Nazmutdinov et al.
recommended values		587 ± 8	3.46 ± 0.05	0.28 ± 0.01		
HEXYL METHANOATE (Hexyl Formate): Molar Mass, 130.187 g; $\text{C}_7\text{H}_{14}\text{O}_2$; CASRN 629-33-4						
2011-naz/sam		614.0 ± 8			1d ^P	Nazmutdinov et al.
HEPTYL METHANOATE (Heptyl Formate): Molar Mass, 144.214 g; $\text{C}_8\text{H}_{16}\text{O}_2$; CASRN 112-23-2						
2011-naz/sam		635.6 ± 8			1d ^P	Nazmutdinov et al.
OCTYL METHANOATE (Octyl Formate): Molar Mass, 158.241 g; $\text{C}_9\text{H}_{18}\text{O}_2$; CASRN 112-32-3						
2011-naz/sam		653.8 ± 8			1d ^P	Nazmutdinov et al.
ETHYL ETHANOATE: Molar Mass, 88.106 g; $\text{C}_4\text{H}_8\text{O}_2$; CASRN 141-78-6						
2001-kud/amb		523.3 ± 0.1	3.87 ± 0.05	0.308 ± 0.005	10	Kudchadker et al.
2004-hu/qin		523.8 ± 0.3	3.90 ± 0.03	0.308	2a	Hu et al.
2014-jun/pri		523.29 ± 0.08	3.8289 ± 0.0017		2a	Juntarachat et al.
recommended values		523.3 ± 0.1	3.84 ± 0.01	0.308 ± 0.005		
ETHENYL ETHANOATE (Vinyl Acetate): Molar Mass, 86.090 g; $\text{C}_4\text{H}_6\text{O}_2$; CASRN 108-05-4						
97-ste/roe		519.20 ± 0.05	4.140 ± 0.01		1	Stevens et al.
recommended values		519.2 ± 0.1	4.16 ± 0.02			
1,1-DIMETHYLETHYL ETHANOATE (<i>tert</i> -Butyl Acetate): Molar Mass, 116.160 g; $\text{C}_6\text{H}_{12}\text{O}_2$; CASRN 540-88-5						
2006-von/wil-2		541.2 ± 2	3.04 ± 0.06		3d	VonNiederhausern et al.
2-(2-ETHOXYETHOXY)ETHYL ETHANOATE (Diethylene Glycol Monoethyl Ether Acetate): Molar Mass, 176.212 g; $\text{C}_8\text{H}_{16}\text{O}_4$; CASRN 112-15-2						
96-wil/wil		663 ± 2	2.73 ± 0.14		2a,c	Wilson et al.
2002-wil/von		673.5 ± 2	2.59 ± 0.05		3d	Wilson et al.
recommended values		668 ± 5	2.63 ± 0.10			
2-(2-BUTOXYETHOXY)ETHYL ETHANOATE (Diethylene Glycol Monobutyl Ether Acetate): Molar Mass, 204.266 g; $\text{C}_{10}\text{H}_{20}\text{O}_4$; CASRN 124-17-4						
96-wil/wil		681 ± 2	3.15 ± 0.14		2a,c	Wilson et al.
2002-wil/von		693.9 ± 2	2.15 ± 0.04		3d	Wilson et al.
recommended values		692 ± 6	2.35 ± 0.1			
ETHYL PROPANOATE (Ethyl Propionate): Molar Mass, 102.133 g; CASRN 105-37-3						
2001-kud/amb		546.7 ± 1	3.45 ± 0.1	0.299 ± 0.01	10	Kudchadker et al.
2014-jun/pri		546.28 ± 0.02	3.3856 ± 0.0022		2a	Juntarachat et al.
recommended values		546.3 ± 0.4	3.39 ± 0.02	0.30 ± 0.01		
ETHYL-2,2-DIMETHYL PROPANOATE: Molar Mass, 130.187 g; $\text{C}_7\text{H}_{14}\text{O}_2$; CASRN 3938-95-2						
2002-ste/chi-3		566 ± 1	2.95 ± 0.1	0.280 ± 0.01	4,6,7	Steele et al.

Table 2. continued

ref	values reported in nonstandard units	T_{90}/K	p/MPa	$\rho/\text{g}\cdot\text{cm}^{-3}$	method	authors
ETHYL BUTANOATE (Ethyl Butyrate): Molar Mass, 116.160 g; CASRN 105-54-4						
2001-kud/amb		568.8 ± 1	3.1 ± 0.3	0.280 ± 0.02	10	Kudchadker et al.
2014-jun/pri		569.98 ± 0.09	2.8915 ± 0.0037		2a	Juntarachat et al.
recommended values		569.6 ± 0.3	2.90 ± 0.005	0.280 ± 0.02		
ETHYL PENTANOATE (Ethyl Valerate): Molar Mass, 130.187 g; CASRN 539-82-2						
2001-kud/amb		593.3 ± 1			10	Kudchadker et al.
2014-jun/pri		594.02 ± 0.02	2.7304 ± 0.0011		2a	Juntarachat et al.
recommended values		594.0 ± 0.1	2.73 ± 0.002			
METHYL HEXANOATE (Methyl Caproate): Molar Mass, 130.187 g; $\text{C}_7\text{H}_{14}\text{O}_2$; CASRN 106-70-7						
2015-nik/pop-1		612 ± 6	2.88 ± 0.09		4	Nikitin and Popov
ETHYL HEXANOATE (Ethyl Caproate): Molar Mass, 144.214 g; $\text{C}_8\text{H}_{16}\text{O}_2$; CASRN 123-66-0						
2001-kud/amb		615.2 ± 0.3	.			
2014-jun/pri		615.22 ± 0.02	2.5313 ± 0.0011		2a	Juntarachat et al.
recommended values		615.2 ± 0.1	2.53 ± 0.002			
METHYL HEPTANOATE (Methyl Enanthate): Molar Mass, 144.214 g; $\text{C}_8\text{H}_{16}\text{O}_2$; CASRN 106-73-0						
99-mor/lui		628 ± 1			1	Morton et al.
2015-nik/pop-1		626 ± 6	2.53 ± 0.08		4	Nikitin and Popov
recommended values		628 ± 1	2.53 ± 0.08			
ETHYL HEPTANOATE (Ethyl Enanthate): Molar Mass, 158.241 g; CASRN 106-30-9						
2001-kud/amb		634.3 ± 0.3	.			
2014-jun/pri		635.12 ± 0.03	2.3569 ± 0.0036		2a	Juntarachat et al.
recommended values		635.0 ± 0.5	2.36 ± 0.002			
METHYL OCTANOATE (Methyl Caprylate): Molar Mass, 158.241 g; $\text{C}_9\text{H}_{18}\text{O}_2$; CASRN 111-11-5						
2015-nik/pop-1		646 ± 6	2.34 ± 0.07		4	Nikitin and Popov
ETHYL OCTANOATE (Ethyl Caprylate): Molar Mass, 172.268 g; CASRN 106-32-1						
2001-kud/amb		649 ± 3			10	Kudchadker et al.
2014-jun/pri		655.74 ± 0.03	2.1592 ± 0.0032		2a	Juntarachat et al.
recommended values		655 ± 1	2.16 ± 0.02			
METHYL NONANOATE (Methyl Pelargonate): Molar Mass, 172.268 g; $\text{C}_{10}\text{H}_{20}\text{O}_2$; CASRN 1731-84-6						
2015-nik/pop-1		665 ± 7	2.06 ± 0.06		4	Nikitin and Popov
METHYL DECANOATE (Methyl Caprate): Molar Mass, 186.295 g; $\text{C}_{11}\text{H}_{22}\text{O}_2$; CASRN 110-42-9						
2015-nik/pop-1		675 ± 7	1.93 ± 0.06		4	Nikitin and Popov
METHYL UNDECANOATE: Molar Mass, 200.322 g; $\text{C}_{12}\text{H}_{24}\text{O}_2$; CASRN 1731-86-8						
2015-nik/pop-1		694 ± 7	1.75 ± 0.05		4	Nikitin and Popov
METHYL DODECANOATE (Methyl Laurate): Molar Mass, 214.349 g; $\text{C}_{13}\text{H}_{26}\text{O}_2$; CASRN 111-82-0						
63-amb		712 ± 2			1	Ambrose
2015-nik/pop-1		709 ± 7	1.52 ± 0.05		4d	Nikitin and Popov
recommended values		712 ± 2	1.52 ± 0.05			
METHYL TETRADECANOATE (Methyl Myristate): Molar Mass, 242.403 g; $\text{C}_{15}\text{H}_{30}\text{O}_2$; CASRN 124-10-7						
2014-nik/pop-3		730 ± 7	1.32 ± 0.04		4d	Nikitin and Popov
METHYL HEXADECANOATE (Methyl Palmitate): Molar Mass, 270.457 g; $\text{C}_{17}\text{H}_{34}\text{O}_2$; CASRN 112-39-0						
2014-nik/pop-3		760 ± 8	1.17 ± 0.04		4d	Nikitin and Popov
METHYL OCTADECANOATE (Methyl Stearate): Molar Mass, 298.511 g; $\text{C}_{19}\text{H}_{38}\text{O}_2$; CASRN 112-61-8						
2014-nik/pop-3		785 ± 8	1.08 ± 0.04		4d	Nikitin and Popov
(E)-ETHYL 2-BUTENOATE (Ethyl <i>trans</i> -2-Butenoate): Molar Mass, 114.14 g; $\text{C}_6\text{H}_{10}\text{O}_2$; CASRN 623-70-1						
1883-paw	326.0 °C	599.2			1	Pawlewski
2014-jun/pri		593.84 ± 0.02	3.6708 ± 0.0012		2a	Juntarachat et al.
recommended values		593.8 ± 0.5	3.67 ± 0.04			
BUTYL 2-PROPENOATE (Butyl Acrylate): Molar Mass, 128.171 g; $\text{C}_7\text{H}_{12}\text{O}_2$; CASRN 141-32-2						
2005-wan/jia		597.38 ± 0.28	2.95 ± 0.02		1d	Wang et al.
2006-von/wil-2		601.2 ± 2	2.84 ± 0.06		3d	VonNiederhausern et al.
recommended values		598 ± 1	2.91 ± 0.04			
METHYL 2-METHYL-2-PROPENOATE (Methyl Metacrylate): Molar Mass, 100.117 g; $\text{C}_5\text{H}_8\text{O}_2$; CASRN 80-62-6						
2005-wan/jia		540.26 ± 0.28	3.87 ± 0.02		1d	Wang et al.
(Z)-9-METHYL OCTADECENOATE (Methyl Oleate): Molar Mass, 296.495 g; $\text{C}_{19}\text{H}_{36}\text{O}_2$; CASRN 112-62-9						
2014-nik/pop-3		777 ± 8	1.21 ± 0.04		4d	Nikitin and Popov
DIMETHYL ETHANEDIOATE (Dimethyl Oxalate): Molar Mass, 118.088 g; $\text{C}_4\text{H}_6\text{O}_4$; CASRN 553-90-2						
$T_{90} - T_{48} = 0.030$ at 533 K						
1862-reg ^d	9.48 atm		0.96		1	Regnault
1883-weg	260 °C	533			?	Weger
57-ste/kay ^r	(355 ± 7) °C, (39.3 ± 4) atm	628 ± 7	3.98 ± 0.4		1,6	Stern and Kay
2006-ver/koz		639 ± 6	4.06 ± 0.12		4d	Verevkin et al.
recommended values		637 ± 6	4.00 ± 0.15			

Table 2. continued

ref	values reported in nonstandard units	T_{90}/K	p/MPa	$\rho/\text{g}\cdot\text{cm}^{-3}$	method	authors
DIMETHYL PROPANEDIOATE (Dimethyl Malonate): Molar Mass, 132.115 g; $\text{C}_5\text{H}_8\text{O}_4$; CASRN 108-59-8						
2002-ste/chi-4		647 ± 3	3.6 ± 0.3	0.350 ± 0.015	4,6,7	Steele et al.
2006-ver/koz		649 ± 7	3.53 ± 0.11		4d	Verevkin et al.
recommended values		646 ± 4	3.56 ± 0.15	0.35 ± 0.02		
DIMETHYL BUTANEDIOATE (Dimethyl Succinate): Molar Mass, 146.142 g; $\text{C}_6\text{H}_{10}\text{O}_4$; CASRN 106-65-0						
2006-ver/koz		662 ± 7	3.06 ± 0.09		4d	Verevkin et al.
DIMETHYL PENTANEDIOATE (Dimethyl Glutarate): Molar Mass, 160.169 g; $\text{C}_7\text{H}_{12}\text{O}_4$; CASRN 1119-40-0						
2006-ver/koz		682 ± 7	2.77 ± 0.08		4d	Verevkin et al.
DIETHYL ETHANEDIOATE (Diethyl Oxalate): Molar Mass, 146.142 g; $\text{C}_6\text{H}_{10}\text{O}_4$; CASRN 95-92-1						
94-ste/chi ^c		618	2.140	0.33 ± 0.01	4,6,7	Steele et al.
2006-von/wil-2		645.8 ± 2	3.06 ± 0.06		3d	VonNiederhausern et al.
recommended values		646 ± 2	3.1 ± 0.1	0.33 ± 0.02		
DIETHYL PROPANEDIOATE (Diethyl Malonate): Molar Mass, 160.169 g; $\text{C}_7\text{H}_{12}\text{O}_4$; CASRN 105-53-3						
2004-nik/pop-1		652 ± 7	2.67 ± 0.08		4d	Nikitin et al.
DIMETHYL HEXANEDIOATE (Dimethyl Adipate): Molar Mass, 174.196 g; $\text{C}_8\text{H}_{14}\text{O}_4$; CASRN 627-93-0						
2006-ver/koz		692 ± 7	2.50 ± 0.08		4d	Verevkin et al.
DIETHYL BUTANEDIOATE (Diethyl Succinate): Molar Mass, 174.196 g; $\text{C}_8\text{H}_{14}\text{O}_4$; CASRN 123-25-1						
$T_{90} - T_{48} = 0.030$ at 663 K						
1899-rad ^f	390 °C	663			1	Radice
2004-nik/pop-1		665 ± 7	2.20 ± 0.04		4d	Nikitin et al.
recommended values		665 ± 7	2.20 ± 0.04			
DIMETHYL HEPTANEDIOATE (Dimethyl Pimelate): Molar Mass, 188.223 g; $\text{C}_9\text{H}_{16}\text{O}_4$; CASRN 1732-08-7						
2006-ver/koz		711 ± 7	2.37 ± 0.07		4d	Verevkin et al.
DIETHYL PENTANEDIOATE (Diethyl Glutarate): Molar Mass, 188.223 g; $\text{C}_9\text{H}_{16}\text{O}_4$; CASRN 818-38-2						
2004-nik/pop-1		683 ± 7	2.11 ± 0.04		4d	Nikitin et al.
DIMETHYL OCTANEDIOATE (Dimethyl Suberate): Molar Mass, 202.250 g; $\text{C}_{10}\text{H}_{18}\text{O}_4$; CASRN 1732-09-8						
2006-ver/koz		723 ± 7	2.30 ± 0.07		4d	Verevkin et al.
DIETHYL HEXANEDIOATE (Diethyl Adipate): Molar Mass, 202.250 g; $\text{C}_{10}\text{H}_{18}\text{O}_4$; CASRN 141-28-6						
2004-nik/pop-1		696 ± 7	2.05 ± 0.04		4d	Nikitin et al.
DIMETHYL DECANEDIOATE (Dimethyl Sebacate): Molar Mass, 230.304 g; $\text{C}_{12}\text{H}_{22}\text{O}_4$; CASRN 106-79-6						
2006-ver/koz		742 ± 7	2.08 ± 0.06		4d	Verevkin et al.
1,1-DIACETOXYETHANE: Molar Mass, 146.142 g; $\text{C}_6\text{H}_{10}\text{O}_4$; CASRN 542-10-9						
2006-von/wil-2		618.2 ± 2	2.97 ± 0.06		3d	VonNiederhausern et al.
ETHERS						
DIMETHYL ETHER (Methoxymethane): Molar Mass, 46.069 g; $\text{C}_2\text{H}_6\text{O}$; CASRN 115-10-6						
2001-kud/amb		400.2 ± 0.1	5.34 ± 0.05	0.275 ± 0.01	10	Kudchadker et al.
2004-wu/liu		400.378 ± 0.005	5.356 ± 0.010	0.272 ± 0.002	2,6,7	Wu et al.
2005-yas/uch		399.63 ± 0.01	5.268 ± 0.0005	0.271 ± 0.001	2	Yasumoto et al.
2007-ihm/lem		400.3 ± 0.1	5.3405 ± 0.005	0.277 ± 0.003	2	Ihmels and Lemmon
2011-wu/zho	$(5.94 \pm 0.25) \text{ mol}\cdot\text{L}^{-1}$	400.3783 ± 0.01	5.33683 ± 0.050	0.2743 ± 0.012	10	Wu et al.
recommended values		400.3 ± 0.1	5.34 ± 0.02	0.274 ± 0.004		
DIETHYL ETHER (Ethoxyethane): Molar Mass, 74.123 g; $\text{C}_4\text{H}_{10}\text{O}$; CASRN 60-29-7						
$T_{90} - T_{68} = -0.039$ at 467 K						
83-hal/gun		$(466.74)^f$		0.265	7	Hales et al.
2001-kud/amb		466.7 ± 0.1	3.644 ± 0.01	0.264 ± 0.003	10	Kudchadker et al.
2011-pol/abd		466.815 ± 0.2		0.2655 ± 0.002	3	Polikhronidi et al.
2011-pol/abd		$(466.845)^u$	3.6050 ± 0.020	0.2561 ± 0.002	6	Polikhronidi et al.
recommended values		466.8 ± 0.1	3.63 ± 0.02	0.261 ± 0.003		
2-METHOXY-2-METHYLPROPANE (<i>tert</i> -Butyl Methyl Ether): Molar Mass, 88.150 g; CASRN 1634-04-4						
2001-kud/amb		497.1 ± 0.2	3.430 ± 0.01		10	Kudchadker et al.
2013-han/xia		497.0 ± 1.2	3.448 ± 0.015		2	Han et al.
recommended values		497.1 ± 0.2	3.43 ± 0.01			
1,2-DIMETHOXYETHANE (Ethylene Glycol Dimethyl Ether): Molar Mass, 90.122 g; $\text{C}_4\text{H}_{10}\text{O}_2$; CASRN 110-71-4						
2001-kud/amb		540 ± 3	3.90 ± 0.1	0.293 ± 0.02	10	Kudchadker et al.
2011-ihm/ste		535.32 ± 0.1	3.827 ± 0.010	0.2911 ± 0.005	2	Ihmels et al.
recommended values		537 ± 3	3.88 ± 0.1	0.292 ± 0.008		
DIPROPYL ETHER (1-Propoxypropane): Molar Mass, 102.177 g; $\text{C}_6\text{H}_{14}\text{O}$; CASRN 111-43-3						
$T_{90} - T_{68} = -0.040$ at 530.6 K						
74-amb/bro		530.56 ± 0.15	3.028 ± 0.003		1	Ambrose et al.
2000-lia/ma		532.14 ± 0.6	5.25 ± 0.02		1	Liang et al.
recommended values		531 ± 1	3.03 ± 0.05			

Table 2. continued

ref	values reported in nonstandard units	T_{90}/K	p/MPa	$\rho/\text{g}\cdot\text{cm}^{-3}$	method	authors
2-METHOXY-2-METHYLBUTANE: Molar Mass, 102.177 g; $\text{C}_6\text{H}_{14}\text{O}$; CASRN 994-05-8						
2001-kud/amb		535 ± 1	3.20 ± 0.05	0.273 ± 0.02	10	Kudchadker et al.
2002-ste/chi-1		537 ± 2	3.55 ± 0.3	0.264 ± 0.01	4,6	Steele et al.
recommended values		535 ± 1	3.20 ± 0.05	0.27 ± 0.02		
DIHEXYL ETHER (1-Hexoxyhexane): Molar Mass, 186.339 g; $\text{C}_{12}\text{H}_{26}\text{O}$; CASRN 112-58-3						
2014-nik/pop-1		665 ± 7	1.44 ± 0.04		4d	Nikitin et al.
DIOCTYL ETHER (1-Octoxyoctane): Molar Mass, 242.447 g; $\text{C}_{16}\text{H}_{34}\text{O}$; CASRN 629-82-3						
2014-nik/pop-1		723 ± 7	1.19 ± 0.04		4d	Nikitin et al.
DIDECYL ETHER (1-Decoxydecane): Molar Mass, 298.555 g; $\text{C}_{20}\text{H}_{42}\text{O}$; CASRN 2456-28-2						
2014-nik/pop-1		768 ± 8	1.03 ± 0.03		4d	Nikitin et al.
AROMATIC C–H–O COMPOUNDS						
PHENOL (Hydroxybenzene): Molar Mass, 94.113 g; $\text{C}_6\text{H}_6\text{O}$; CASRN 108-95-2						
2001-kud/amb		694.2 ± 0.2	5.93 ± 0.05		10	Kudchadker et al.
2007-nik/pop		695 ± 7	5.67 ± 0.17		4d	Nikitin et al.
recommended values		694.2 ± 0.2	5.93 ± 0.05			
1,3-BENZENEDIOL: Molar Mass, 110.112 g; $\text{C}_6\text{H}_6\text{O}_2$; CASRN 108-46-3						
2006-von/wil-1		836 ± 2	6.24 ± 0.12		3d	VonNiederhausern et al.
PHENYLMETHANOL: Molar Mass, 108.140 g; $\text{C}_7\text{H}_8\text{O}$; CASRN 100-51-6						
90-amb/ghi		715 ± 3	4.3 ± 0.1		1	Ambrose and Ghiassee
2014-nik/pop-3		713 ± 7	4.33 ± 0.1		4d	Nikitin and Popov
recommended values		714 ± 3	4.3 ± 0.1			
2-PHENYL-1-ETHANOL: Molar Mass, 122.167 g; $\text{C}_8\text{H}_{10}\text{O}$; CASRN 60-12-8						
2006-von/wil		723.5 ± 2	3.99 ± 0.08		3d	VonNiederhausern et al.
2007-nik/pop		720 ± 7	3.85 ± 0.12		4d	Nikitin et al.
recommended values		724 ± 2	3.95 ± 0.08			
3-PHENYL-1-PROPANOL: Molar Mass, 136.194 g; $\text{C}_9\text{H}_{12}\text{O}$; CASRN 122-97-4						
2007-nik/pop		732 ± 7	3.46 ± 0.10		4d	Nikitin et al.
4-PHENYL-1-BUTANOL: Molar Mass, 150.221 g; $\text{C}_{10}\text{H}_{14}\text{O}$; CASRN 3360-41-6						
2007-nik/pop		746 ± 7	3.07 ± 0.09		4d	Nikitin et al.
4-(METHYLPHENYL)METHANOL: Molar Mass, 122.167 g; $\text{C}_8\text{H}_{10}\text{O}$; CASRN 589-18-4						
2014-nik/pop-3		719 ± 7	3.61 ± 0.1		4d	Nikitin and Popov
ACIDS						
PHENYLMETHANOIC (Benzoic) ACID: Molar Mass, 122.123 g; $\text{C}_7\text{H}_6\text{O}_2$; CASRN 65-85-0						
2014-nik/pop-3		755 ± 8	4.53 ± 0.1		4d	Nikitin and Popov
PHENYLETHANOIC ACID: Molar Mass, 136.150 g; $\text{C}_8\text{H}_8\text{O}_2$; CASRN 103-82-2						
2006-nik/pop-1		766 ± 8	3.95 ± 0.14		4d	Nikitin et al.
2-METHYLBENZOIC (<i>o</i> -Toluic) ACID: Molar Mass, 136.150 g; $\text{C}_8\text{H}_8\text{O}_2$; CASRN 118-90-1						
2014-nik/pop-3		763 ± 8	3.92 ± 0.12		4d	Nikitin and Popov
3-METHYLBENZOIC (<i>m</i> -Toluic) ACID: Molar Mass, 136.150 g; $\text{C}_8\text{H}_8\text{O}_2$; CASRN 99-04-7						
2014-nik/pop-3		771 ± 8	3.85 ± 0.12		4d	Nikitin and Popov
4-METHYLBENZOIC (<i>p</i> -Toluic) ACID: Molar Mass, 136.150 g; $\text{C}_8\text{H}_8\text{O}_2$; CASRN 99-94-5						
2014-nik/pop-3		775 ± 8	3.80 ± 0.10		4d	Nikitin and Popov
3-PHENYLPROPANOIC ACID: Molar Mass, 150.177 g; $\text{C}_9\text{H}_{10}\text{O}_2$; CASRN 501-52-0						
2006-nik/pop-1		776 ± 8	3.46 ± 0.12		4d	Nikitin et al.
4-PHENYLBUTANOIC ACID: Molar Mass, 164.204 g; $\text{C}_{10}\text{H}_{12}\text{O}_2$; CASRN 1821-12-1						
2006-nik/pop-1		783 ± 8	3.21 ± 0.11		4d	Nikitin et al.
5-PHENYLPENTANOIC ACID: Molar Mass, 178.231 g; $\text{C}_{11}\text{H}_{14}\text{O}_2$; CASRN 2270-20-4						
2006-nik/pop-1		790 ± 8	2.95 ± 0.10		4d	Nikitin et al.
6-PHENYLHEXANOIC ACID: Molar Mass, 192.258 g; $\text{C}_{12}\text{H}_{16}\text{O}_2$; CASRN 5581-75-9						
2006-nik/pop-1		794 ± 8	2.60 ± 0.09		4d	Nikitin et al.
7-PHENYLHEPTANOIC ACID: Molar Mass, 206.285 g; $\text{C}_{13}\text{H}_{18}\text{O}_2$; CASRN 40228-90-8						
2006-nik/pop-1		798 ± 8	2.47 ± 0.09		4d	Nikitin et al.
2-CARBOXYBENZALDEHYDE (Phthalaldehydic Acid): Molar Mass, 150.133 g; $\text{C}_8\text{H}_6\text{O}_3$; CASRN 119-67-5						
2014-nik/pop-3		864 ± 9	4.80 ± 0.15		4d	Nikitin and Popov
METHYL BENZOATE: Molar Mass, 136.150 g; $\text{C}_8\text{H}_8\text{O}_2$; CASRN 93-58-3						
2002-ste/chi-1		702 ± 1	4.02 ± 0.2	0.334 ± 0.01	4,6,7	Steele et al.
2007-nik/pop-1		695 ± 7	3.46 ± 0.10		4d	Nikitin et al.
recommended values		700 ± 5	3.6 ± 0.2			
ETHYL BENZOATE: Molar Mass, 150.177 g; $\text{C}_9\text{H}_{10}\text{O}_2$; CASRN 93-89-0						
2007-nik/pop-1		700 ± 7	3.05 ± 0.09		4d	Nikitin et al.
PROPYL BENZOATE: Molar Mass, 164.204 g; $\text{C}_{10}\text{H}_{12}\text{O}_2$; CASRN 2315-68-6						
2007-nik/pop-1		710 ± 7	2.60 ± 0.08		4d	Nikitin et al.

Table 2. continued

ref	values reported in nonstandard units	T_{90}/K	p/MPa	$\rho/\text{g}\cdot\text{cm}^{-3}$	method	authors
2007-nik/pop-1	BUTYL BENZOATE: Molar Mass, 178.231 g; $\text{C}_{11}\text{H}_{14}\text{O}_2$; CASRN 136-60-7					
		725 ± 7	2.40 ± 0.07		4d	Nikitin et al.
2007-nik/pop-1	PENTYL BENZOATE: Molar Mass, 192.258 g; $\text{C}_{12}\text{H}_{16}\text{O}_2$; CASRN 2049-96-9					
		736 ± 7	2.19 ± 0.07		4d	Nikitin et al.
2007-nik/pop-1	HEXYL BENZOATE: Molar Mass, 206.285 g; $\text{C}_{13}\text{H}_{18}\text{O}_2$; CASRN 6789-88-4					
		748 ± 7	1.98 ± 0.06		4d	Nikitin et al.
2002-wil/von	PHENYL ETHANOATE: Molar Mass, 136.150 g; $\text{C}_8\text{H}_8\text{O}_2$; CASRN 122-79-2					
		685.7 ± 2	3.59 ± 0.07		3d	Wilson et al.
2006-nik/pop	DIMETHYL-1,2-BENZENEDICARBOXYLATE (Dimethyl Phthalate): Molar Mass, 194.186 g; $\text{C}_{10}\text{H}_{10}\text{O}_4$; CASRN 131-11-3					
		772 ± 9	2.77 ± 0.1		4d	Nikitin et al.
2006-nik/pop	DIETHYL-1,2-BENZENEDICARBOXYLATE (Diethyl Phthalate): Molar Mass, 222.240 g; $\text{C}_{12}\text{H}_{14}\text{O}_4$; CASRN 84-66-2					
		776 ± 9	2.26 ± 0.08		4d	Nikitin et al.
2006-nik/pop	DIPROPYL-1,2-BENZENEDICARBOXYLATE (Dipropyl Phthalate): Molar Mass, 250.294 g; $\text{C}_{14}\text{H}_{18}\text{O}_4$; CASRN 131-16-6					
		784 ± 9	1.90 ± 0.07		4d	Nikitin et al.
2006-nik/pop	DIBUTYL-1,2-BENZENEDICARBOXYLATE (Dibutyl Phthalate): Molar Mass, 278.348 g; $\text{C}_{16}\text{H}_{22}\text{O}_4$; CASRN 84-74-2					
		797 ± 9	1.66 ± 0.06		4d	Nikitin et al.
2006-nik/pop	DIPENTYL-1,2-BENZENEDICARBOXYLATE (Dipentyl Phthalate): Molar Mass, 306.402 g; $\text{C}_{18}\text{H}_{26}\text{O}_4$; CASRN 84-74-2					
		811 ± 9	1.43 ± 0.05		4d	Nikitin et al.
2006-nik/pop	DIHEXYL-1,2-BENZENEDICARBOXYLATE (Diethyl Phthalate): Molar Mass, 334.456 g; $\text{C}_{20}\text{H}_{30}\text{O}_4$; CASRN 84-75-3					
		817 ± 9	1.34 ± 0.05		4d	Nikitin et al.
2006-nik/pop	DIHEPTYL-1,2-BENZENEDICARBOXYLATE (Diheptyl Phthalate): Molar Mass, 362.510 g; $\text{C}_{22}\text{H}_{34}\text{O}_4$; CASRN 3648-21-3					
		830 ± 9	1.24 ± 0.04		4d	Nikitin et al.
2006-nik/pop	DIOCTYL-1,2-BENZENEDICARBOXYLATE (Dioctyl Phthalate): Molar Mass, 390.564 g; $\text{C}_{24}\text{H}_{38}\text{O}_4$; CASRN 117-84-0					
		840 ± 9	1.08 ± 0.04		4d	Nikitin et al.
2006-nik/pop	Bis(2-ETHYLHEXYL)-1,2-BENZENEDICARBOXYLATE (Bis(2-Ethylhexyl) Phthalate): Molar Mass, 390.564 g; $\text{C}_{24}\text{H}_{38}\text{O}_4$; CASRN 117-81-7					
		835 ± 9	1.07 ± 0.04		4d	Nikitin et al.
2006-nik/pop	DINONYL-1,2-BENZENEDICARBOXYLATE (Dinonyl Phthalate): Molar Mass, 418.618 g; $\text{C}_{26}\text{H}_{42}\text{O}_4$; CASRN 84-76-4					
		858 ± 9	1.02 ± 0.04		4d	Nikitin et al.
2006-nik/pop	DIDECYL-1,2-BENZENEDICARBOXYLATE (Didecyl Phthalate): Molar Mass, 446.672 g; $\text{C}_{28}\text{H}_{46}\text{O}_4$; CASRN 84-77-5					
		870 ± 10	0.94 ± 0.03		4d	Nikitin et al.
2006-nik/pop	DIUNDECYL-1,2-BENZENEDICARBOXYLATE (Diundecyl Phthalate): Molar Mass, 474.726 g; $\text{C}_{30}\text{H}_{50}\text{O}_4$; CASRN 3648-20-2					
		886 ± 10	0.89 ± 0.03		4d	Nikitin et al.
2001-kud/amb	BENZALDEHYDE: Molar Mass, 106.124 g; $\text{C}_7\text{H}_6\text{O}$; CASRN 100-52-7					
		695 ± 10	4.7 ± 1			Kudchacker et al.
2014-nik/pop-3		692 ± 7	4.41 ± 0.1		4d	Nikitin and Popov
	recommended values	693 ± 7	4.41 ± 0.1			
2014-nik/pop-3	4-METHYLBENZALDEHYDE: Molar Mass, 120.151 g; $\text{C}_7\text{H}_6\text{O}$; CASRN 104-87-0					
		712 ± 7	3.80 ± 0.1		4d	Nikitin and Popov
2006-von/wil-2	1,2-EPOXY-2-METHYLPROPANE (2,2-Dimethyloxirane): Molar Mass, 72.107 g; $\text{C}_4\text{H}_8\text{O}$; CASRN 558-30-5					
		499.9 ± 2	4.35 ± 0.09		3d	VonNiederhausern et al.
96-wil/wil	γ -BUTYROLACTONE: Molar Mass, 86.090 g; $\text{C}_4\text{H}_6\text{O}_2$; CASRN 96-48-0					
		731.0 ± 0.2	5.131 ± 0.034		2a,c	Wilson et al.
2006-von/wil-2		732.5 ± 2	5.10 ± 0.10		3d	VonNiederhausern et al.
	recommended values	731.3 ± 0.04	5.11 ± 0.04			
2002-wil/von	PROPYLENE CARBONATE: Molar Mass, 102.089 g; $\text{C}_4\text{H}_6\text{O}_3$; CASRN 108-32-7					
		762.7 ± 2	4.14 ± 0.08		3d	Wilson et al.
ORGANIC SULFUR (C + H + S) COMPOUNDS						
2000-lia/ma	DIMETHYL SULFIDE: Molar Mass, 62.130 g; $\text{C}_2\text{H}_6\text{S}$; CASRN 75-18-3					
		499.21	5.34		1	Liang et al.
2001-tso/amb		503 ± 1	5.53 ± 0.10	0.305 ± 0.005	10	Tsonopoulos and Ambrose
	recommended values	503 ± 1	5.53 ± 0.10	0.305 ± 0.005		
2001-tso/amb	DIMETHYL DISULFIDE (2,3-Dithiabutane): Molar Mass, 94.190 g; $\text{C}_2\text{H}_6\text{S}_2$ CASRN 624-92-0					
		615 ± 10			10	Tsonopoulos and Ambrose
2006-von/wil-2		607.8 ± 2	5.07 ± 0.09		3d	VonNiederhausern et al.
	recommended values	608 ± 3	5.07 ± 0.10			
2006-von/wil	1-DODECANETHIOL: Molar Mass, 202.400 g; $\text{C}_{12}\text{H}_{26}\text{S}$; CASRN 112-55-0					
		733.7 ± 2	1.81 ± 0.04		3d	VonNiederhausern et al.
2006-von/wil	METHYLTHIOBENZENE (Thioanisole): Molar Mass, 124.201 g; $\text{C}_7\text{H}_8\text{S}$; CASRN 100-68-5					
		705.9 ± 2	4.05 ± 0.08		3d	VonNiederhausern et al.
ORGANIC FERRUM (C + H + Fe) COMPOUNDS						
2012-nik/pop	Bis(η -CYCLOPENTADIENYL) IRON (Ferrocene): Molar Mass, 186.035 g; $\text{C}_{10}\text{H}_{10}\text{Fe}$; CASRN 102-54-5					
		785 ± 8	3.61 ± 0.10		4d	Nikitin and Popov

Table 2. continued

ref	values reported in nonstandard units	T_{90}/K	p/MPa	$\rho/\text{g}\cdot\text{cm}^{-3}$	method	authors
1-ACETYLFERROCENE: Molar Mass, 228.072 g; $\text{C}_{12}\text{H}_{12}\text{FeO}$; CASRN 1271-55-2						
2012-nik/pop		847 ± 13	3.28 ± 0.13		4d	Nikitin and Popov
BUTYLFERROCENE: Molar Mass, 242.143 g; $\text{C}_{14}\text{H}_{18}\text{Fe}$; CASRN 31904-29-7						
2012-nik/pop		784 ± 12	2.18 ± 0.09		4d	Nikitin and Popov
BENZOYLFERROCENE: Molar Mass, 290.143 g; $\text{C}_{17}\text{H}_{14}\text{FeO}$; CASRN 1272-44-2						
2012-nik/pop		886 ± 13	1.59 ± 0.06		4d	Nikitin and Popov
ORGANIC COMPOUNDS CONTAINING NITROGEN						
BENZONITRILE: Molar Mass, 103.124 g; $\text{C}_7\text{H}_5\text{N}$; CASRN 100-47-0						
94-ste/chi		(699.4) ^v	4.215		6	Steele et al.
2006-mar/you		700 ± 15	4.2 ± 0.5		10	Marsh et al.
recommended values		700 ± 15	4.2 ± 0.5			
1-METHYLPYRROLIDINE: Molar Mass, 85.150 g; $\text{C}_5\text{H}_{11}\text{N}$; CASRN 120-94-5						
2011-ihm/ste		548.62 ± 0.1	4.004 ± 0.010	0.2818 ± 0.005	2	Ihmels et al.
PIPERAZINE: Molar Mass, 86.138 g; $\text{C}_4\text{H}_{10}\text{N}_2$; CASRN 110-85-0						
2006-von/wil-1		656.3 ± 2	5.42 ± 0.11		3d	VonNiederhausern et al.
2006-mar/you		659 ± 3	5.6 ± 0.3	0.32 ± 0.03	10	Marsh et al.
recommended values		657 ± 3	5.5 ± 0.2	0.32 ± 0.03		
2,6-DIMETHYLQUINOLINE: Molar Mass, 157.216 g; $\text{C}_{11}\text{H}_{11}\text{N}$; CASRN 877-43-0						
2007-chi/joh		786 ± 2	3.480	0.3075	4,6,7	Chirico et al.
ORGANIC COMPOUNDS CONTAINING HALOGENS						
CHLOROTRIFLUOROMETHANE (R-13): Molar Mass, 104.456 g; CClF_3 CASRN 75-72-9						
$T_{90} - T_{68} = -0.007\text{K}$ at 302 K						
75-mur/skr		301.91 ± 0.03			1	Muratov and Skripov
76-vit/kol	$(28.78 \pm 0.01)^\circ\text{C}$	301.92	3.873 ± 0.002		1	Vitkalov et al.
2007-mar/abr		301.95 ± 0.05	3.88 ± 0.01	0.580 ± 0.002	10	Marsh et al.
recommended values		301.95 ± 0.05	3.88 ± 0.01	0.580 ± 0.002		
CHLORODIFLUOROMETHANE (R-22): Molar mass 86.465 g; CHClF_2 ; CASRN 75-45-6						
92-goo/def		(369.275) ^{uv}	4.9871 ± 0.0012		6	Goodwin et al.
2007-mar/abr		369.25 ± 0.15	4.99 ± 0.01	0.523 ± 0.005	10	Marsh et al.
recommended values		369.25 ± 0.15	4.99 ± 0.01	0.523 ± 0.005		
DIFLUOROMETHANE (R-32): Molar Mass, 52.023 g; CH_2F_2 ; CASRN 75-10-5						
98-shi/zhu		351.295 ± 0.010	5.785 ± 0.002	0.425 ± 0.003	2,6	Shi et al.
2002-pit/str		351.25 ± 0.01	5.790 ± 0.01		2	Pitschmann et al.
2007-mar/abr		351.25 ± 0.03	5.783 ± 0.005	0.425 ± 0.004	10	Marsh et al.
recommended values		351.25 ± 0.03	5.783 ± 0.005	0.425 ± 0.004		
PENTAFLUOROETHANE (R125): Molar Mass, 120.020 g; C_2HF_5 ; CASRN 354-33-6						
2002-pit/str		339.13 ± 0.01	3.630 ± 0.01		2	Pitschmann et al.
2007-mar/abr		339.40 ± 0.07	3.63 ± 0.01	0.570 ± 0.004	10	Marsh et al.
2010-soo/thé		339.26 ± 0.02	3.619 ± 0.001		2	Soo et al.
2011-soo		339.26 ± 0.03	3.619 ± 0.001		2	Soo
recommended values		339.30 ± 0.05	3.625 ± 0.006	0.570 ± 0.004		
1,1,1,2-TETRAFLUOROETHANE (R134a): Molar Mass, 102.030 g; $\text{C}_2\text{H}_2\text{F}_4$; CASRN 811-97-2						
92-goo/def		(374.255) ^x	4.0551 ± 0.0003		6	Goodwin et al.
96-gre/zhe		374.25	4.059	0.5091	1a	Grebekov et al.
2000-yat/hor		374.107 ± 0.02^y			7	Yata et al.
2003-poo/del		374.12	4.045		1	Poot and de Loos
2004-uch/yas		374.13 ± 0.01	4.053 ± 0.0005	0.508 ± 0.001	2	Uchida et al.
2007-mar/abr		374.18 ± 0.08	4.055 ± 0.006	0.512 ± 0.003	10	Marsh et al.
2010-soo/thé		374.13 ± 0.03	4.044 ± 0.001		2	Soo et al.
2011-soo		374.23 ± 0.03	4.044 ± 0.001		2	Soo
recommended values		374.14 ± 0.05	4.050 ± 0.006	0.510 ± 0.004		
1,1,1-TRIFLUOROETHANE (R-143a): Molar Mass, 84.040 g; $\text{C}_2\text{H}_3\text{F}_3$; CASRN 420-46-2						
2002-pit/str		345.88 ± 0.01	3.775 ± 0.01		2	Pitschmann et al.
2007-mar/abr		345.86 ± 0.02	3.768 ± 0.006	0.434 ± 0.003	10	Marsh et al.
recommended values		345.86 ± 0.02	3.768 ± 0.006	0.434 ± 0.003		
1,1-DIFLUOROETHANE (Ethylidene Fluoride, R-152a): Molar mass 66.050 g; $\text{C}_2\text{H}_4\text{F}_2$; CASRN 75-37-6						
96-gre/zhe		386.57	4.517	0.3650	1a	Grebekov et al.
2007-mar/abr		386.43 ± 0.04	4.52 ± 0.01	0.368 ± 0.002	10	Marsh et al.
recommended values		386.43 ± 0.04	4.52 ± 0.01	0.368 ± 0.002		
HEXAFLUOROETHANE (R-116): Molar Mass, 138.012 g; C_2F_6 ; CASRN 76-16-4						
2006-lem/spa		293.03	3.048	0.614	8	Lemmon and Span
2007-mar/abr		293.02 ± 0.02	3.04 ± 0.02	0.619 ± 0.01	10	Marsh et al.
recommended values		293.02 ± 0.02	3.04 ± 0.02	0.619 ± 0.01		

Table 2. continued

ref	values reported in nonstandard units	T_{90}/K	p/MPa	$\rho/\text{g}\cdot\text{cm}^{-3}$	method	authors
OCTAFLUOROPROPANE (R-218): Molar mass 188.017 g; C_3F_8 ; CASRN 76-19-7						
96-gre/zhe		345.05	2.671	0.6280	1a	Grebenkov et al.
2007-mar/abr		345.0 ± 0.2	2.66 ± 0.02	0.627 ± 0.005	10	Marsh et al.
recommended values		345.0 ± 0.2	2.66 ± 0.02	0.627 ± 0.005		
1,1,1,2,3,3,3-HEPTAFLUOROPROPANE (R227ea): Molar Mass, 170.027 g; C_3HF_7 ; CASRN 431-89-0						
2006-frö/bot		374.90 ± 0.05			2	Fröba et al.
2007-mar/abr		375.02 ± 0.05	2.930 ± 0.005	0.595 ± 0.005	10	Marsh et al.
recommended values		374.96 ± 0.05	2.930 ± 0.005	0.595 ± 0.005		
1,1,1,2,3,3-HEXAFLUOROPROPANE (R-236ea): Molar mass 152.037 g; $\text{C}_3\text{H}_2\text{F}_6$; CASRN 431-63-0						
2007-mar/abr		412.44 ± 0.02	3.42 ± 0.04	0.565 ± 0.003	10	Marsh et al.
2008-gru/kha		412.21 ± 0.05		0.560 ± 0.002	7	Gruzdev et al.
recommended values		412.44 ± 0.02	3.42 ± 0.04	0.565 ± 0.003		
1,1,1,3,3-PENTAFLUOROPROPANE (R245fa): Molar Mass, 134.047 g; $\text{C}_3\text{H}_3\text{F}_5$; CASRN 460-73-1						
2004 gre/bel		427.16	3.651	0.516	8	Grebenkov et al.
2007-mar/abr		427.20 ± 0.08	3.639 ± 0.004	0.517 ± 0.008	10	Marsh et al.
2010-soo/thé		427.15 ± 0.01	3.657 ± 0.001		2	Soo et al.
2011-soo		427.15 ± 0.03	3.657 ± 0.001		2	Soo
2015-hig/hay		427.01 ± 0.02	3.651	0.522 ± 0.003	2	Higashi et al.
recommended values		427.10 ± 0.10	3.655 ± 0.004	0.520 ± 0.004		
DECAFLUOROBUTANE (R610): Molar Mass, 238.024 g; C_4F_{10} ; CASRN 355-25-9						
2007-mar/abr		386.2 ± 0.3	2.32 ± 0.02	0.63 ± 0.01	10	Marsh et al.
2010-soo/thé		386.41 ± 0.02	2.299 ± 0.001		2	Soo et al.
2011-soo		386.41 ± 0.03	2.299 ± 0.001		2	Soo
recommended values		386.33 ± 0.05	2.302 ± 0.006	0.63 ± 0.01		
1,1,1,3,3-PENTAFLUOROBUTANE (R365mfc): Molar Mass, 148.074 g; $\text{C}_4\text{H}_3\text{F}_5$; CASRN 406-58-6						
2007-mar/abr		460 ± 2			10	Marsh et al.
2010-soo/thé		459.91 ± 0.02	3.200 ± 0.001		2	Soo et al.
2011-soo		459.91 ± 0.12	3.200 ± 0.001		2	Soo
recommended values		459.9 ± 0.2	3.20 ± 0.01			
OCTAFLUOROOCTANE (Perfluorooctane): Molar mass 438.052 g; C_8F_{18} ; CASRN 307-34-6						
2007-mar/abr		498.2 ± 0.4	1.50 ± 0.05	0.61 ± 0.01	10	Marsh et al.
2009-kha/sta		496.98 ± 0.05		0.596 ± 0.003	7	Khairulin and Stankus
recommended values		497.0 ± 0.2	1.50 ± 0.05	0.600 ± 0.005		
HEXAFLUOROPROPENE (R1216) Molar Mass, 150.021 g; C_3F_6 ; CASRN 116-15-4						
2007-dup		358.95			?	Du Pont
2010-coq/ram		358.8 ± 0.1	3.129 ± 0.001	0.57903	1,7	Coquelet et al.
2010-soo/thé		359.02 ± 0.02	3.138 ± 0.001		2	Soo et al.
2011-soo		359.02 ± 0.03	3.138 ± 0.001		2	Soo
recommended values		359.00 ± 0.08	3.134 ± 0.005	0.579 ± 0.002		
3,3,3-TRIFLUORO-1-PROPENE (R1234zf): Molar mass 96.051 g; $\text{C}_3\text{H}_3\text{F}_3$; CASRN 677-21-4						
2007-mar/abr		377.8 ± 0.8	3.65 ± 0.08	0.46 ± 0.01	10	Marsh et al.
2013-bro/din		(378.59)	3.631		6	Brown et al.
recommended values		377.8 ± 0.8	3.65 ± 0.08	0.46 ± 0.01		
(E)-1-CHLORO-3,3,3-TRIFLUOROPROPENE (HCFO-1233zd(E)): Molar Mass, 130.494 g; $\text{C}_3\text{H}_2\text{ClF}_3$; CASRN 102687-65-0						
2012-hul/bas		438.75 ± 0.05	3.7721 ± 0.030		1,6	Hulse et al.
(E)-1,3,3,3-TETRAFLUOROPROPENE (HFO-1234ze(E)): Molar Mass, 114.041 g; $\text{C}_3\text{H}_2\text{F}_4$; CASRN 29118-24-9						
2010-hig/tan		382.51 ± 0.01	3.632 ± 0.003	0.486 ± 0.003	2,7	Higashi et al.
2010-tan/tak		(382.51) ^{ac}	3.632 ± 0.003		6	Tanaka et al.
2014-jun/val		382.39 ± 0.01	3.6728 ± 0.0002		2a	Juntarachat et al.
recommended values		382.4 ± 0.02	3.640 ± 0.020	0.486 ± 0.003		
(Z)-1,3,3,3-TETRAFLUOROPROPENE (HFO-1234ze(Z)): Molar Mass, 114.041 g; $\text{C}_3\text{H}_2\text{F}_4$; CASRN 29118-25-0						
2015-hig/hay		423.27 ± 0.03	3.533 ± 0.003	0.470 ± 0.005	2,7	Higashi et al.
2,3,3,3-TETRAFLUORO-1-PROPENE (HFO-1234yf): Molar Mass, 114.041 g; $\text{C}_3\text{H}_2\text{F}_4$; CASRN 754-12-1						
2009-hul/sin	94.8 °C	367.95 ± 0.2	3.260		1,6	Hulse et al.
2010-din/pol		(368.15) ^{aa}	3.3895		6	Di Nicola et al.
2010-tan/hig		367.85 ± 0.01	3.382 ± 0.003	0.478 ± 0.003	2,6	Tanaka and Higashi
2010-fuk		367.86 ± 0.02	3.378 ± 0.004	0.476 ± 0.006	?	Fukushima
2011-ric/mcl		(367.85) ^{bb}	3.3822	0.47555	8	Richter et al.
2014-jun/val		368.75 ± 0.04	3.3829 ± 0.0012		2a	Juntarachat et al.
recommended values		367.85 ± 0.06	3.381 ± 0.004	0.476 ± 0.002		
HEXAFLUOROBENZENE: Molar Mass, 186.054 g; C_6F_6 ; CASRN 392-56-3						
2006-sta/kha		516.66 ± 0.05		0.5505 ± 0.002	7	Stankus and Khairulin
2007-mar/abr		516.7 ± 0.1	3.28 ± 0.01	0.552 ± 0.003	10	Marsh et al.

Table 2. continued

ref	values reported in nonstandard units	T_{90}/K	p/MPa	$\rho/\text{g}\cdot\text{cm}^{-3}$	method	authors
HEXAFLUOROBENZENE: Molar Mass, 186.054 g; C_6F_6 ; CASRN 392-56-3						
recommended values		516.7 ± 0.1	3.28 ± 0.01	0.552 ± 0.003		
PENTAFLUOROBENZENE: Molar Mass, 168.064 g; C_6HF_5 ; CASRN 363-72-4						
2007-mar/abr		530.9 ± 0.1	3.53 ± 0.03	0.518 ± 0.003	10	Marsh et al.
2009-kha/sta		531.03 ± 0.05		0.5191 ± 0.002	7	Khairulin and Stankus
2010-sta/kha		531.042 ± 0.05		0.51883 ± 0.002	7	Stankus and Khairulin
recommended values		531.0 ± 0.1	3.53 ± 0.03	0.518 ± 0.003		
1,4-DICHLOROBENZENE: Molar Mass, 184.865 g; $\text{C}_6\text{H}_4\text{Cl}_2$; CASRN 106-46-7						
$T_{90} - T_{68} = -0.047$ at 669 K						
49-gol/rav	396 °C	669			1	Golik et al.
ORGANIC COMPOUNDS CONTAINING HALOGENS + O						
TRIFLUORO(TRIFLUOROMETHYL)OXIRANE (1,2-Epoxy-1,1,2,3,3,3-hexafluoropropane, HFPO):						
Molar Mass, 166.020 g; $\text{C}_3\text{F}_6\text{O}$; CASRN 428-59-1						
95-sin/mik	(86.5 ± 0.1) °C	359.6	2.93 ± 0.01	0.570 ± 0.006	1,6,7	Sinityn et al.
2008-dup		395.15	2.896	0.59	?	Du Pont
2010-soo/thé		359.48 ± 0.02	2.936 ± 0.001		2	Soo et al.
2011-soo		359.48 ± 0.03	2.936 ± 0.001		2	Soo
2011-dic/bel		359.3 ± 0.1	2.930 ± 0.001	0.58418	2,7	Dicko et al.
recommended values		359.4 ± 0.1	2.930 ± 0.007	0.585 ± 0.008		
ORGANIC COMPOUNDS CONTAINING Si + O						
TETRAISOPROPOXYLSILANE: Molar Mass, 264.437 g; $\text{C}_{12}\text{H}_{28}\text{O}_4\text{Si}$; CASRN 1992-48-9						
2012-nik/pop-1		587 ± 6	1.31 ± 0.04		4d	Nikitin et al.
ORGANIC COMPOUNDS CONTAINING Ti + O						
TETRAETHOXYTITANIUM: Molar Mass, 228.111 g; $\text{C}_8\text{H}_{20}\text{O}_4\text{Ti}$; CASRN 3087-36-3						
2010-nik/pop		647 ± 6	1.40 ± 0.04		4d	Nikitin et al.
TETRAPROPOXYTITANIUM: Molar Mass, 284.219 g; $\text{C}_{12}\text{H}_{28}\text{O}_4\text{Ti}$; CASRN 3087-37-4						
2010-nik/pop		685 ± 7	1.15 ± 0.04		4d	Nikitin et al.
TETRAISOPROPOXYTITANIUM: Molar Mass, 284.219 g; $\text{C}_{12}\text{H}_{28}\text{O}_4\text{Ti}$; CASRN 546-68-9						
2012-nik/pop-1		641 ± 6	1.15 ± 0.04		4d	Nikitin et al.
TETRABUTOXYTITANIUM: Molar Mass, 340.327 g; $\text{C}_{16}\text{H}_{36}\text{O}_4\text{Ti}$; CASRN 5593-70-4						
2010-nik/pop		719 ± 7	1.02 ± 0.03		4d	Nikitin et al.
TETRA- <i>TERT</i> -BUTOXYTITANIUM: Molar Mass, 340.327 g; $\text{C}_{16}\text{H}_{36}\text{O}_4\text{Ti}$; CASRN 3087-39-6						
2012-nik/pop-1		657 ± 7	1.07 ± 0.03		4d	Nikitin et al.
TETRAPENTOXYTITANIUM: Molar Mass, 396.435 g; $\text{C}_{20}\text{H}_{44}\text{O}_4\text{Ti}$; CASRN 10585-24-7						
2010-nik/pop		751 ± 8	0.91 ± 0.03		4d	Nikitin et al.
TETRAHEXOXYTITANIUM: Molar Mass, 452.543 g; $\text{C}_{24}\text{H}_{52}\text{O}_4\text{Ti}$; CASRN 7360-52-3						
2010-nik/pop		780 ± 8	0.78 ± 0.02		4d	Nikitin et al.
TETRAHEPTOXYTITANIUM: Molar Mass, 508.651 g; $\text{C}_{28}\text{H}_{60}\text{O}_4\text{Ti}$; CASRN 22063-06-5						
2010-nik/pop		808 ± 8	0.73 ± 0.02		4d	Nikitin et al.
TETRAOCTOXYTITANIUM: Molar Mass, 564.759 g; $\text{C}_{32}\text{H}_{68}\text{O}_4\text{Ti}$; CASRN 3061-42-5						
2010-nik/pop		831 ± 8	0.66 ± 0.02		4d	Nikitin et al.
TETRANONOXYTITANIUM: Molar Mass, 620.867 g; $\text{C}_{36}\text{H}_{76}\text{O}_4\text{Ti}$; CASRN 6167-42-6						
2010-nik/pop		855 ± 9	0.63 ± 0.02		4d	Nikitin et al.
TETRADECOXYTITANIUM: Molar Mass, 676.975 g; $\text{C}_{40}\text{H}_{84}\text{O}_4\text{Ti}$; CASRN 83877-93-4						
2010-nik/pop		882 ± 9	0.61 ± 0.02		4d	Nikitin et al.

^aThe critical temperatures were taken from TRC Thermodynamic Tables [2000-trc]. ^bThe critical temperatures were taken from Ambrose and Tsonopoulos [95-amb/tso]. ^cThe critical temperature was taken from Haynes and Goodwin [82-hay/go]. ^dThe critical temperatures were taken from Brunner [88-bru]. ^eThe critical temperature was taken from Ambrose [80-amb]. ^fThe critical temperature was taken from Hugill and McGlashan [78-hug/mcg]. ^gThe critical temperature was taken from Angus et al. [80-ang/arm]. ^hThe critical temperature was taken from Ambrose et al. [60-amb/cox]; the critical pressure and density were obtained by fitting, respectively, the vapor pressure and liquid density data of Forziati et al. [50-for/cam]. ⁱThe critical temperature was taken from Polt et al. [92-pol/pla]. ^jThe critical temperature was taken from Tsonopoulos and Ambrose [95-tso/amb]. ^kIn [87-dau/jal] a graduated Bourdon gauge, presumably in psi was used. They stated the uncertainty in psi but reported the pressure in MPa. The critical temperature was taken from [78-amb/tow]. ^lThe critical temperature was measured by VonNiederhausen et al. [2000-von/wil]. ^mThe critical temperature was measured by Nikitin et al. [95-nik/pav]. ⁿMixture of isomers. ^oAn ampule with a compound under study was immersed for 60 s into a molten salt bath with near-critical temperature. ^pLinde and Manning [62-lin/man] stated that Regnault [1862-reg] did not purport to determine the critical point of dimethylethanediatoate and that values given in that reference were wrongly quoted in the *International Critical Tables* [28-ict] as the critical temperature and the critical pressure. ^qStern and Kay [57-ste/kay] found that the compound decomposed above 200 °C. At T_c they observed a p_c of 55.4 atm, "indicating the presence of gaseous decomposition products". The reported p_c was therefore obtained by extrapolation of a simple equation fitted to vapor pressures measured at temperatures up to 200 °C. ^rSteele et al. [94-ste/chi] values of T_c , p_c , and ρ_c give a low value of the compression (compressibility) factor, and their critical temperature is lower than that given by Stern and Kay [57-ste/kay] and Verevkin et al. [2006-ver/koz] for dimethylethanediatoate. ^sRadice [1899-rad] reported that diethylbutanediatoate decomposed, as would be expected. ^tThe critical temperature was calculated from the measured values of saturated-liquid and -vapor densities using a scaling equation. ^uThe critical temperature was measured by Guye and Mallet [02-guy/mal]. ^vThe critical temperature was taken from [80-amb] and converted into ITS-90. ^wThe critical temperature was taken from [91-mor/war] and converted into ITS-90. ^xThe critical temperature was calculated assuming that the critical density $\rho_c = 0.511 \text{ g cm}^{-3}$. ^yThe critical temperature was taken from [2010-hig/tan]. ^zThe critical temperature was taken from [2008-min/spa]. ^{aa}The critical temperature was taken from [2008-min/spa]. ^{bb}The critical temperature was taken from Tanaka and Higashi [2010-tan/hig]. ^{cc}The critical temperature was taken from [2006-bue/wag-1]. ^{dd}Method of measurement not stated. ^{ee}Molar masses are based on the following relative atomic masses: hydrogen, 1.008; carbon, 12.011; nitrogen, 14.007; oxygen, 15.999; fluorine, 18.998; silicon, 28.085; sulfur, 32.06; chlorine, 35.45; titanium, 47.867; iron, 55.845 [2011-wie/cop].

Table 3. Key to Methods of Critical Point Determination

1. visual - in glass tube
2. visual - in cell with windows
3. non-visual - pVT measurement or vapor pressure measurement
4. other non-visual measurement
5. critical pressure measurement combined with vapor pressure measurement up to the critical point
6. critical pressure by extrapolation of vapor pressure curve
7. orthobaric density measurements
8. equation of state, thermodynamic study
9. calculation from another physical property
10. literature survey
 - a with stirring
 - b instrumental detection of critical point
 - c special feature of apparatus
 - d decomposition at critical temperature

■ SELECTION OF BEST VALUES

Normal Alkanes. The recommended values of Wagner and de Reuck [96-wag/der] are identical with those of Setzmann and Wagner [91-set/wag], which were recommended in Part 2 so those recommendations are retained. The first few members of the normal alkane family have been measured by a variety of methods with high precision. For example, Wagner's group [86-kle/wag] used the pVT method for methane, augmented by vapor pressure measurements (see page 532 of Part 2). The method given in Table 2 of Part 2 is "3, 5" but it should probably be "3, 5, 7" because, as stated on page 532, the "final value of the critical density was determined from the rectilinear diameter." The availability of such data has made it possible to make highly precise recommendations with the equation of state method, such as those of [91-set/wag]. For ethane the [86-kle/wag] approach, including the two equations given on page 532, was used for ethane by Funke et al. [2002-fun/kle] who recommended the values $T_c = (305.322 \pm 0.01) \text{ K}$; $P_c = (4.8722 \pm 0.0011) \text{ MPa}$, and $\rho_c = (0.20618 \pm 0.00015) \text{ g}\cdot\text{cm}^{-3}$. The same values were used in the equation of state of Bücker et al. [2006-bue/wag]. The 2002 recommendations of Wagner's group and the 1995 Part 2 recommendations are in good agreement, as they are also in agreement with the "most comprehensive and accurate pVT study" of Douslin and Harrison [73-dou/har]. The only difference, and it is a small one, is in the ρ_c value of $0.2066 \text{ g}\cdot\text{cm}^{-3}$ by [73-dou/har, 95-amb/tso] compared with $0.2062 \text{ g}\cdot\text{cm}^{-3}$ by [2002-fun/kle]. The latter value is preferred in view of the agreement with the EOS recommendation of [2006-bue/wag]. There have been many measurements on propane since the recommendations given in Part 2. The selection was based on the values of [2001-yos, 2004-hig, 2005-yas/uch, 2008-gil/oti, 2011-soo]. The average values for T_c (369.83 K) and p_c (4.248 MPa) agree with the Part 2 recommendations, but the recommended value for ρ_c ($0.2213 \text{ g}\cdot\text{cm}^{-3}$) is slightly higher than the Part 2 recommendation of $(0.220 \pm 0.003) \text{ g}\cdot\text{cm}^{-3}$. Accordingly, the new recommendation for ρ_c is $(0.221 \pm 0.003) \text{ g}\cdot\text{cm}^{-3}$. For butane the recommended values were based on [95-deá/vic, 96-war, 2001-hor/fis, 2005-yas/uch, 2011-soo], which led to the new recommendations for T_c and p_c as $(425.10 \pm 0.1) \text{ K}$ and $(3.792 \pm 0.01) \text{ MPa}$. ρ_c remains unchanged at $(0.228 \pm 0.003) \text{ g}\cdot\text{cm}^{-3}$. The more recent values listed in Table 2 for pentane give relatively high averages for T_c (470.1 K) and p_c (3.413 MPa) and the three ρ_c values are all from the same laboratory.

The average of 22 T_c values for hexane reported in Part 2 (from 1960 to 1992) and 10 values given in Table 2 (excluded [13-sch]) confirmed the 1995 recommendation of 507.6 K. Similarly, the more limited ρ_c values also confirmed the 1995 recommendation of $0.234 \text{ g}\cdot\text{cm}^{-3}$. However, the average of 15 p_c values from Part 2 and the 10 more recent values in Table 2 resulted in a slightly higher value for p_c . For the higher n -alkanes up to dodecane the more recent measurements confirm the previous recommendations with either no change or very minor changes to the recommendations. The majority of measurements of n -alkanes from C_{22} to C_{60} were by Nikitin and co-workers using the pulse heating method, a method suitable for compounds that decompose below the critical temperature.

There have been a number of new measurements on branched and cycloalkanes that serve to confirm the previous recommendations. Most of the new recommendations differ by less than the previously estimated uncertainties. Squalane (2,6,10,15,19,23-hexamethyltetracosane) is the exception where critical values of $(795.9 \pm 1) \text{ K}$ [2000-von/wil] and $(822 \pm 12) \text{ K}$ [2005-nik/pop] have been measured. VonNiederhausen et al. [2000-von/wil] presented vapor pressure data for squalane over the range 628.8 K to 795.9 K. Both the experimental data and the results of the fit with the Riedel equation are presented in Table 2 of [2000-von/wil]. Extrapolating with the Riedel equation to the T_c recommended in [2005-nik/pop], 822 K, gives a p_c of 875 kPa rather than the 2005 recommendation of 700 kPa. On the other hand, if we accept the vapor pressure of [2000-von/wil] and the p_c of [2005-nik/pop] this gives a $T_c = 807.5 \text{ K}$. The simulation of Zhuravlev et al. [97-zhu/sie] gave a $T_c = (807 \pm 6) \text{ K}$ but a second simulation by the same group predicted a $T_c = (851 \pm 2) \text{ K}$ [2002-zhu/mar] which is likely too high. Nikitin and Popov [2005-nik/pop] have argued that there was likely decomposition present in the measurements of VonNiederhausen et al. [2000-von/wil] that would have led to a lower observed critical temperature. More weight is given to the [2005-nik/pop] critical temperature and the recommended critical pressure is based on the vapor pressure values from [2000-von/wil] extrapolated to the recommended critical temperature. A new vapor pressure fit should be made with the Wagner equation.

Cycloalkanes. The 1996 recommendations for cyclohexane were based primarily on the two sets of measurements by Teja and co-workers. The values are now slightly revised based on the average of the values reported more recently by [96-dau], [2004-mor/lui], [2006-naz/sar], and [2008-gil/oti] and the critical pressure based on [96-dau], [2000-ewi/och], [2003-nik/pop], and [2008-gil/oti]. For the higher cycloalkanes there have been minor changes to those recommended in [96-dau]. For propylcyclohexane the value from [2003-nik/pop] is about 7 K below that from [2004-mor/lui]. The recommended value is that reported by [2004-mor/lui] after minor rounding because of the greater uncertainty of 6 K reported by [2003-nik/pop]. For (1-methylethyl)cyclohexane the value of T_c in [2004-mor/lui] is 3 K lower than that reported by [2009-ihm/wol] (claimed uncertainties are 0.2 and 0.1 K, respectively). Both are static measurements; however, [2009-ihm/wol] measured the vapor pressure close to the critical temperature as well as the critical density so more weight is given to that work. For butylcyclohexane T_c from [2004-mor/lui] is higher than T_c from [2009-ihm/wol] by 4 K and similarly more weight is given to [2009-ihm/wol]. T_c for 1,3,5-trimethyltricyclo[3.3.1.1(3,7)]decane is about 5 K lower than that of 1,3-dimethyltricyclo[3.3.1.1(3,7)]decane which has one less C

atom. One would expect 1,3,5-trimethyltricyclo[3.3.1.1(3,7)]decane to have a higher T_c .

Unsaturated Aliphatic Hydrocarbons. For the 1-alkenes up to heptene the changes are minor and are within the previous estimated uncertainties. T_c and p_c for 1-alkenes from 1-tridecene to 1-eicosane have been reported by [99-nik/pop] with relatively high uncertainties. Many compounds in this category have only T_c measured by [2004-mor/lui] with uncertainties ranging from 0.2 to 2 K but in light of the discrepancies noted above, these uncertainties have been increased. For cyclohexene the p_c value reported by [96-ste/chi-1], 4.905 MPa, is clearly much too high; it should be lower than the value for cyclopentane, 4.80 MPa. The p_c value of [92-ma/fan] is preferred.

Aromatic Hydrocarbons. There have been a considerable number of measurements on a number of aromatic hydrocarbons with carbon number up to C_{10} and they have confirmed the previous recommendations within the uncertainties stated. For 1,3,5-triethylbenzene, the values of p_c reported by 97-ste/chi was derived by fitting their vapor pressure measurements up to 534.7 K to the Wagner equation and extrapolating to their T_c . Measurements on the high carbon number aromatic compounds have been reported by Nikitin and co-workers, by Steele et al., and by Nesterov et al. generally with considerably greater uncertainties.

Alkanols, Alkanals, and Alkanones. There have been a considerable number of measurements on the critical properties of the lower straight chain, branched and unsaturated alkanols but these have not resulted in any major changes to the previous recommendations. In the number of cases the uncertainty in the recommendations has improved. There have been a number of new studies on the alkane diols by various authors and most measurements are usually in agreement within the authors assigned uncertainties, the major discrepancy being 1,3-butanediol where the value by [96-ste/wil] is 16 K lower than the value of [2006-von/wil-2]. The latter value is preferred. [2002-ma/gao] has measured a number of alkanals and the results for T_c for butanal are lower than the recommended value by 15 K, hence the uncertainties given in Table 1 to the other alkanals reported by [2002-ma/gao] are much higher than the authors assigned uncertainties.

Acids, Esters, and Ethers. Again Nikitin and co-workers have made extensive measurements on T_c for high molar mass acids and where there are measurements by other authors, the results for T_c agree within the assigned uncertainties. For 1-methylethyl methanoate, the result of [95-ma/wan] for T_c is 14 K higher than the two measurements by [2011-sam/naz] which are preferred. Wilson et al. in [96-wil/wil] and [2002-wil/von] measured T_c and p_c for both 2-(2-ethoxyethoxy)ethyl ethanoate and 2-(2-butoxyethoxy)ethyl ethanoate and the T_c results differ by 10 and 14 K respectively while the p_c results differ by 1 MPa for 2-(2-butoxyethoxy)ethyl ethanoate. Surprisingly while the discrepancy is given in a table there is no comment on the difference. The later work is preferred. For dimethyl ethandioate the results of [1883-weg] are clearly wrong but the results for T_c of [57-ste/kay] and [2006-ver/koz] disagree by 11 K but they agree within the combined uncertainties. [57-ste/kay] noted decomposition of the sample. The technique used by [2006-ver/koz] minimizes the effects of decomposition on T_c so their results are preferred. For diethylethanedioate the T_c and p_c values reported by [94-ste/chi] are lower than the values reported in [2006-von/wil-2] by 28 K and 1 MPa, respectively, and again there is no discussion in the later

reference. The results of [2006-von/wil-2] are preferred because their technique involves a low residence time while [94-ste/chi] used a rapid scan heat capacity measurement where decomposition was noted. In the previous recommendation for 1,2-dimethoxymethane considerable weight was placed on [99-mor/lui] to give the recommended T_c of (540 ± 3) K when the three other values ranged between (536 and 539) K. The new measurements reported in [2011-ihm/ste] leads to a lower recommended T_c .

Other Compounds. The majority of compounds in this class are the halogenated alkanes where there remains considerable activity in high accuracy measurements of their critical properties with most new measurements confirming the previous recommendations within their stated uncertainties. The only exceptions are octadecafluorooctane where the new value for T_c is based on [2009-kha/sta].

AUTHOR INFORMATION

Corresponding Author

*E-mail: ken.marsh@canterbury.ac.nz.

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Notes

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#Deceased 1st February 2015.

DEDICATION

This submission is dedicated to Dr. Tony Goodwin who died on December 12, 2014 and Dr. Douglas Ambrose who died on February 1, 2015.

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- 97-ste/chi-1 Steele, W. V.; Chirico, R. D.; Knipmeyer, S. E.; Nguyen, A. *J. Chem. Eng. Data* **1997**, 42, 1021–1036 (1,4-bis(1,1-dimethylethyl)benzene, cyclohexanol)
- 97-ste/chi-2 Steele, W. V.; Chirico, R. D.; Cowell, A. B.; Knipmeyer, S. E.; Nguyen, A. *J. Chem. Eng. Data* **1997**, 42, 1053–1066 (4-methyl-1-pentene)

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- 97-zhu/sie Zhuravlev, N. D.; Siepmann, J. I. *Fluid Phase Equilib.* **1997**, *134*, 55–61
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- 99-nik/pop Nikitin, E. D.; Popov, A. P. *Fluid Phase Equilib.* **1999**, *166*, 237–243 (1-tridecene to 1-eicosene)
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- 2001-tso/amb Tsonopoulos, C.; Ambrose, D. *J. Chem. Eng. Data* **2001**, *46*, 480–485 (Part 8 of this series)
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- 2002-ste/chi Steele, W. V.; Chirico, R. D.; Knipmeyer, S. E.; Nguyen, A. *J. Chem. Eng. Data* **2002**, *47*, 648–666 (butylbenzene, (1-methylpropyl)benzene, (1,1-dimethylethyl)benzene)
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- 2003-nik/pop Nikitin, E. D.; Popov, A. P.; Bogatishcheva, N. S. *J. Chem. Eng. Data* **2003**, *48*, 1137–1140 (cyclohexane, methylcyclohexane to butylcyclohexane)
- 2003-poo/del Poot, W.; de Loos, T. W. *Fluid Phase Equilib.* **2003**, *210*, 69–75 (1,1,1,2-tetrafluoroethane)
- 2003-spa/wag Span, R.; Wagner, W. *Int. J. Thermophys.* **2003**, *24*, 41–109 (methane, ethane, propane, butane, 2-methylpropane, pentane, hexane, heptane, octane, cyclohexane, ethene)
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- 2004-chi/ste Chirico, R. D.; Steele, W. V. *J. Chem. Thermodyn.* **2004**, *36*, 633–643 (1,1'-bicyclopentyl)
- 2004-gre/bel Grebenkov, A. J.; Beliyeva, O. V.; Klepatski, P. M.; Saplitza, V. V.; Timofeyev, B. D.; Tsurbelev, V. P.; Zayats, T. A. *Thermophysical Properties of R245fa*. ASHRAE Research Project 1256-RP, Final Report; Joint Institute for Power and Nuclear Research - Sosny, 99 Akademik Krasin St., Minsk, 220109, Republic of Belarus, Minsk, September 30, 2004 (1,1,1,3,3-pentafluoropropane)
- 2004-hig Higashi, Y. *Fluid Phase Equilib.* **2004**, *219*, 99–103 (propane)
- 2004-hu/qin Hu, T.; Qin, Z.; Wang, G.; Hou, X.; Wang, J. J. *Chem. Eng. Data* **2004**, *49*, 1809–1814 (ethanol, ethyl ethanoate)
- 2004-mor/lui Morton, D. W.; Lui, M. P. W.; Tran, C. A.; Young, C. L. *J. Chem. Eng. Data* **2004**, *49*, 283–285 (2-methylbutene, 2,3-dimethyl-1-butene, 3,3-dimethyl-1-butene, 2,3-dimethyl-2-butene, 2-methyl-1-pentene, 4-methyl-1-pentene, 2-methyl-2-pentene, (Z)-4-methyl-2-pentene, 2,3-dimethyl-1-pentene, 4,4-dimethyl-1-pentene, (Z)-2-hexene, (Z)-3-hexene, (E)-2-hexene, (E)-3-hexene, 2-methyl-1-heptene, 5-methyl-1-hexene, (Z)-2-heptene, (E)-2-heptene, (E)-3-heptene, 2-methyl-1-hexene, 2-methyl-2-heptene, (E)-2-octene, (E)-4-octene, cyclohexane, cyclooctane, ethylcyclohexane, propylcyclohexane, (1-methylethyl)cyclohexane, butylcyclohexane, (2-methylpropyl)cyclohexane, (1,1-dimethylethyl)cyclohexane)
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- 2004-wu/liu Wu, J.; Liu, Z.; Wang, B.; Pan, J. J. *Chem. Eng. Data* **2004**, *49*, 704–708 (dimethyl ether)
- 2005-chi/ste Chirico, R. D.; Steele, W. V. *J. Chem. Eng. Data* **2005**, *50*, 1052–1059 (diphenylmethane)
- 2005-ewi/och Ewing, M. B.; Ochoa, J. C. S. *J. Chem. Eng. Data* **2005**, *50*, 1543–1547 (heptane)
- 2005-ihm/fis Ihmels, E. C.; Fischer, K.; Gmehling, J. *Fluid Phase Equilib.* **2005**, *228–229*, 155–171 (1-butene, (Z)-2-butene, (E)-2-butene, 2-methyl-1-propene)
- 2005-nik/pop Nikitin, E. D.; Popov, A. P. *Fluid Phase Equilib.* **2005**, *237*, 16–20 (squalane)
- 2005-yas/uch Yasumoto, M.; Uchida, Y.; Ochi, K.; Furuya, T.; Otake, K. *J. Chem. Eng. Data* **2005**, *50*, 596–602 (propane, butane, 2-methylpropane, dimethyl ether)
- 2005-wan/jia Wang, X.; Jia, Q.; Gao, J.; Xia, S.; Ma, P. J. *Chem. Ind. Eng. (China)*, **2005**, *56* (8), 1385–1388 (1-heptene, butyl 2-propenoate, methyl 2-methyl-2-propenoate, 5-hexen-2-one, 2-propen-1-ol)
- 2005-zha/qin Zhang, R.; Qin, Z.; Wang, G.; Dong, M.; Hou, X.; Wang, J. J. *Chem. Eng. Data* **2005**, *50*, 1414–1418 (cyclohexane)
- 2005-zha/qin-1 Zhang, Z.; Qin, Z.; Wang, G.; Wang, J. J. *Fuel Chem. Technol.*, **2005**, *33* (1), 96–100 (methylbenzene)

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- 2006-bue/wag-1 Bucker, D.; Wagner, W. *J. Phys. Chem. Ref. Data* **2006**, *35*, 929–1019 (butane, 2-methylpropane, 2-methylpropane)
- 2006-ewi/och Ewing, M. B.; Ochoa, J. C. S. *J. Chem. Thermodyn.* **2006**, *38*, 283–288 (hexane)
- 2006-ewi/och-1 Ewing, M. B.; Ochoa, J. C. S. *J. Chem. Thermodyn.* **2006**, *38*, 289–295 (pentane)
- 2006-frö/bot Fröba, A. P.; Botero, C.; Leipertz, A. *Int. J. Thermophys.* **2006**, *27*, 1609–1625 (1,1,1,2,3,3,3-heptafluoropropane)
- 2006-hig Higashi, Y. *J. Chem. Eng. Data* **2006**, *51*, 406–408 (2-methylpropane)
- 2006-ji/qin Ji, X.; Qin, Z.; Wang, G.; Dong, M.; Hou, X.; Wang, J. *J. Chem. Eng. Data* **2006**, *51*, 1156–1160 (propane)
- 2006-kur Kurzeja, N., Private communication, Lehrstuhl für Thermodynamik, Ruhr-Universität Bochum, 2006 (propene)
- 2006-lem/spa Lemmon, E. W.; Span, R. *J. Chem. Eng. Data* **2006**, *51*, 785–850 (2,2-dimethylpropane, hexafluoroethane, 2-propanone)
- 2006-mar/you Marsh, K. N.; Young, C. L.; Morton, D. W.; Ambrose, D.; Tsonopoulos, C. *Chem. Eng. Data* **2006**, *51*, 305–314 (Part 9 of this series)
- 2006-mas/hon Masui, G.; Honda, Y.; Uematsu, M. *J. Chem. Thermodyn.* **2006**, *38*, 1711–1716 (2-methylpropane)
- 2006-naz/sar Nazmutdinov, A. G.; Sarkisova, V. S.; Vodenkova, N. N.; Nesterov, I. A.; Nesterova T. N. *Petro. Chem.* **2006**, *46* (6), 428–434 (cyclohexane, 1,3-dimethyltricyclo-[3.3.1.1(3,7)]decane, 1,3,5-trimethyltricyclo-[3.3.1.1(3,7)]decane)
- 2006-naz/sar-1 Nazmutdinov, A. G.; Sarkisova, V. S.; Flekina, E. V.; Stepanov, V. N. *Izvestiya Samarskogo Nauchnogo Zentra Rossiiskoi Akademii Nauk* **2006**, *8* (3), 652–657 (heptane, benzene)
- 2006-nes/nes Nesterov, I. A.; Nesterova, T. N. Nazmutdinov, A. G.; Vodenkova, N. N.; Novozhenina, T. P. *Rus. J. Phys. Chem.* **2006**, *80*, 1809–1815 (1,3-bis(1,1-dimethylethyl)benzene, 1,4-bis(1,1-dimethylethyl)benzene, 1,3,5-tris(1,1-dimethylethyl)benzene, 1,3-bis(1,1-dimethylethyl)-5-methylbenzene)
- 2006-nik/pop Nikitin, E. D.; Popov, A. P.; Yatluk, Y. G. *J. Chem. Eng. Data* **2006**, *51*, 1326–1330 (dimethyl-1,2-benzenedicarboxylate to diocetyl-1,2-benzenedicarboxylate, diundecyl-1,2-benzenedicarboxylate, bis(2-ethylhexyl)-1,2-benzenedicarboxylate)
- 2006-nik/pop-1 Nikitin, E. D.; Popov, A. P.; Yatluk, Y. G. *J. Chem. Eng. Data* **2006**, *51*, 1335–1338 (phenylethanoic, 3-phenylpropanoic, 4-phenylbutanoic, 5-phenylpentanoic, 6-phenylhexanoic, 7-phenylheptanoic acids)
- 2006-pol/rad Polikhronidi, N. G.; Radzhabova, L. M.; Rasulov, A. R.; Stepanov, G. V. *High Temp.* **2006**, *44*, 507–512 (methanol)
- 2006-sta/kha Stankus, S. V.; Khairulin, R. A. *Int. J. Thermophys.* **2006**, *27* (4), 1110–1122 (hexafluorobenzene)
- 2006-ver/koz Verevkin, S. P.; Kozlova, S. A.; Emel'yanenko, V. N.; Nikitin, E. D.; Popov, A. P.; Krasnykh, E. L. *J. Chem. Eng. Data* **2006**, *51*, 1896–1905 (dimethylethanedioate to dimethyl decanedioate)
- 2006-von/wil VonNiederhausern, D. M.; Wilson, G. M.; Giles, N. F. *J. Chem. Eng. Data* **2006**, *51*, 1982–1985 (naphthalene, 2-phenyl-1-ethanol, 1-(1,1-dimethylethoxy)-2-propanol, 2-propene-1-ol, 1-dodecanethiol, methylthiobenzene)
- 2006-von/wil-1 VonNiederhausern, D. M.; Wilson, G. M.; Giles, N. F. *J. Chem. Eng. Data* **2006**, *51*, 1986–1989 (1,6-hexanediol, 1,3-benzenediol, piperazine)
- 2006-von/wil-2 VonNiederhausern, D. M.; Wilson, G. M.; Giles, N. F. *J. Chem. Eng. Data* **2006**, *51*, 1990–1995 (1,4-bis(1-methylethyl)benzene, γ -butyrolactone, 4-oxa-1,7-heptanediol, 1,3-butanediol, 1,2-epoxy-2-methylpropane, diethylethanedioate, butyl propenoate, 1,1-dimethylethyl ethanoate, 1,1-diacetoxyethane, dimethyl disulfide)
- 2007-baz/abd Bazaev, A. R.; Abdulagatov, I. M.; Bazaev, E. A.; Abdurashidova, A. A.; Ramazanov, A. E. *J. Supercrit. Fluids* **2007**, *41*, 217–226 (methanol)
- 2007-baz/abd-1 Bazaev, A. R.; Abdulagatov, I. M.; Bazaev, E. A.; Abdurashidova, A. *Int. J. Thermophys.* **2007**, *28*, 194–219 (ethanol)
- 2007-chi/joh Chirico, R. D.; Johnson, R. D. III; Steele, W. V. *J. Chem. Thermodyn.* **2007**, *39*, 698–711 (2,6-dimethylquinoline)
- 2007-dup DuPont. Technical information, dupont fluorointermediates: hexafluoropropylene. Technical report, DuPont, USA, 2007. http://www2.dupont.com/FluoroIntermediates/en_US/assets/downloads/h96535.pdf. (hexafluoropropene)
- 2007-ihm/lem Ihmels, E. C.; Lemmon, E. W. *Fluid Phase Equilib.* **2007**, *260*, 36–48 (dimethyl ether)
- 2007-li/qin Li, J.; Qin, Z.; Wang, G.; Dong, M.; Wang, J. *J. Chem. Eng. Data* **2007**, *52*, 1736–1740 (2-methylpropane, 2,2,4-trimethylpentane, 1-butene)
- 2007-mar/abr Marsh, K. N.; Abramson, A.; Ambrose, D.; Morton, D. W.; Nikitin, E.; Tsonopoulos, C.; Young, C. L. *J. Chem. Eng. Data* **2007**, *52*, 1509–1538 (Part 10 of this series)
- 2007-miy/uem H. Miyamoto, H.; Uematsu, M. *J. Chem. Thermodyn.* **2007**, *39*, 827–832 (butane)
- 2007-nik/pop Nikitin, E. D.; Popov, A. P.; Yatluk, Y. G. *J. Chem. Eng. Data* **2007**, *52*, 315–317 (phenol, 2-phenyl-1-ethanol, 3-phenyl-1-propanol, 4-phenyl-1-butanol)
- 2007-nik/pop-1 Nikitin, E. D.; Popov, A. P.; Yatluk, Y. G. *J. Chem. Eng. Data* **2007**, *52*, 1336–1339 (methyl benzoate to hexyl benzoate)

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- 2008-abd/pol Abdulagatov, I. M.; Polikhronidi, N. G.; Bruno, T. J.; Batyrova, R. G.; Stepanov, G. V. *Fluid Phase Equilib.* **2008**, *263*, 71–84 (methylbenzene)
- 2008-dup DuPont. Push Bulletin, Dupont FluoroIntermediates: Hexafluoropropylene oxide. Technical report, DuPont, USA, 2008. http://www2.dupont.com/FluoroIntermediates/en_US/assets/downloads/k05132.pdf (trifluoro(trifluoromethyl)oxirane)
- 2008-gil/oti Gil, L.; Otín, S. F.; Embid, J. M.; Gallardo, M. A.; Blanco, S.; Artal, M.; Velasco I. J. *Supercrit. Fluids* **2008**, *44*, 123–138 (ethane, propane, hexane, cyclohexane, methanol, ethanol, 1-propanol, 1-butanol, 1-heptanol)
- 2008-gru/kha Gruzdev, V. A.; Khairulin, R. A.; Komarov, S. G.; Stankus, S. V. *Int. J. Thermophys.* **2008**, *29*, 546–556 (1,1,1,2,3,3-hexafluoropropane)
- 2008-hon/sat Honda, Y.; Sato T.; Uematsu M. *J. Chem. Thermodyn.* **2008**, *40*, 208–211 (propane)
- 2008-ihm/wol Ihmels, C.; Wollmann, R.; Laue, S.; Hieronimus, N. DIPPR Project 851: Final Report for 2007; Oct. 2008 (ethylcyclohexane, (Z)-1,3-dimethylcyclohexane, (E)-1,2-dimethylcyclohexane, (1-methylethyl)cyclohexane)
- 2008-min/spa Minor, B.; Spatz, M. *International Refrigeration and Air Conditioning Conference at Purdue*, July 14–17 2008, paper 2349, pp 1–8 (2,3,3,3-tetrafluoro-1-propene)
- 2008-naz/ale Nazmutdinov, A. G.; Alekina, E. V.; Nesterova, T. N. *Russ. J. Phys. Chem. A* **2008**, *82*, 1857–1862 (octane, decane, methylbenzene, methanol, ethanol, 1-butanol, 2-propanol, 2-butanone)
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- 2009-amb/tso Ambrose, D.; Tsonopoulos, C.; Nikitin, E. D. *J. Chem. Eng. Data* **2009**, *54*, 669–689 (Part 11 of this series)
- 2009-hul/sin Hulse, R.; Singh, R.; Pham, H. *Proceedings of the Third IIR Conference on Thermophysical Properties and Transfer Processes of Refrigerants*, paper 178, Boulder, CO, 2009 (2,3,3,3-tetrafluoro-1-propene (HFO-1234yf))
- 2009-ihm/wol Ihmels, C.; Wollmann, R.; Steinmann, M.; Hieronimus, N. DIPPR Project 851: Final Report for 2008; Oct. 2009 (butylcyclohexane, (1,1-dimethylethyl)cyclohexane)
- 2009-ihm/wol-1 Ihmels, C.; Wollmann, R.; Steinmann, M.; Fattroth, J. DIPPR Project 851: Final Report for 2009; Oct 2009 (butylbenzene, (1-methylpropyl)benzene, (1,1-dimethylethyl)benzene)
- 2009-kha/sta Khairulin, R. A.; Stankus, S. V. *Vestnik SibGUTI*, **2009**, *3*, 117–122 (octadecafluorooctane, pentafluorobenzene)
- 2009-lem/mcl Lemmon, E. W.; McLinden, M. O.; Wagner, W. J. *Chem. Eng. Data* **2009**, *54*, 3141–3180 (propane)
- 2010-ale/naz Alekina, E. V.; Nazmutdinov, A. G.; Sarkisova, V. S. *Izv. Vyssh. Uchebn. Zaved. Khim. Khim. Tekhnol.* **2010**, *53* (9), 28–32 (methanol, ethanol, 1-propanol, 2-propanol, 1-hexanol)
- 2010-che/yao Chen, Z.; Yao, Z.; Zhu, F.; Cao, K.; Li, Y.; Huang Z. *J. Chem. Eng. Data* **2010**, *55*, 2004–2007 (ethene, propene)
- 2010-coq/ram Coquelet, C.; Ramjugernath, D.; Madani, H.; Valtz, A.; Naidoo, P.; Meniai, A. H. *J. Chem. Eng. Data* **2010**, *55*, 2093–2099 (hexafluoropropene)
- 2010-din/pol Di Nicola, G.; Polonara, F.; Santori, G. J. *Chem. Eng. Data* **2010**, *55*, 201–204 (2,3,3,3-tetrafluoro-1-propene (HFO-1234yf))
- 2010-fuk Fukushima, M. *Proceedings of Japan Symposium on Thermophysical Properties 2010*, 31, 58–60 (2,3,3,3-tetrafluoro-1-propene (HFO-1234yf))
- 2010-hig/tan Higashi, Y.; Tanaka, K.; Ichikawa, T. *J. Chem. Eng. Data* **2010**, *55*, 1594–1597 ((E)-1,3,3,3-tetrafluoropropene)
- 2010-nik/pop Nikitin, E. D.; Popov, A. P.; Yatluk, Y. G.; Simakina, V. A. *J. Chem. Eng. Data* **2010**, *55*, 178–183 (tetraethoxy- to tetradecoxytitaniums)
- 2010-ras/abd Rasulov, S. M.; Abdulagatov, I. M. *J. Chem. Eng. Data* **2010**, *55*, 3247–3261 (pentane)
- 2010-soo/thé Soo, Ch.-B.; Théveneau, P.; Coquelet, C.; Ramjugernath, D.; Richon, D. *J. Supercritical Fluids* **2010**, *55*, 545–553 (propane, butane, pentane, hexane, ethanol, 1-propanol, trifluoro(trifluoromethyl)oxirane, hexafluoropropene, pentafluoroethane, 1,1,1,2-tetrafluoroethane, 1,1,1,3,3-pentafluoropropane, 1,1,1,3,3-pentafluorobutane, decafluorobutane)
- 2010-sta/kha Stankus, S. V.; Khairulin, R. A. *Int. J. Thermophys.* **2010**, *31*, 2078–2085 (pentafluorobenzene)
- 2010-tan/hig Tanaka, K.; Higashi, Y. *Int. J. Refrig.* **2010**, *33*, 474–479 (2,3,3,3-tetrafluoropropene (HFO-1234yf))
- 2010-tan/tak Tanaka, K.; Takahashi, G.; Higashi, Y. *J. Chem. Eng. Data* **2010**, *55*, 2169–2172 ((E)-1,3,3,3-tetrafluoro-1-propene)
- 2011-dic/bel Dicko, M.; Belaribi-Boukais, G.; Coquelet, C.; Valtz, A.; Belaribi, F. B.; Naidoo, P.; Ramjugernath, D. *Ind. Eng. Chem. Res.* **2011**, *50*, 4761–4768 (trifluoro(trifluoromethyl)oxirane)
- 2011-ihm/ste Ihmels, C.; Stenmann, M.; Hinnners, K. DIPPR Project 851: Final Report for 2011; Nov. 2011 ((Z)-1,2-dimethylcyclohexane, (E)-1,4-dimethylcyclohexane, 1,2-dimethoxyethane, 1-methylpyrrolidine)

- 2011-naz/sam Nazmutdinov, A. G.; Samarov, A. A.; Nesterova, T. N. *Izv. Vyssh. Uchebn. Zaved. Khim. Khim. Tekhnol.* **2011**, *54* (12), 36–39 (ethyl methanoate, propyl methanoate, butyl methanoate, pentyl methanoate, hexyl methanoate, heptyl methanoate, octyl methanoate)
- 2011-pol/abd Polikhronidi, N. G.; Abdulagatov, I. M.; Batyrova, R. G.; Stepanov, G. V.; Ustuzhanin, E. E.; Wu, J. T. *Int. J. Thermophys.* **2011**, *32*, 559–595 (diethyl ether)
- 2011-ric/mcl Richter, M.; McLinden, M. O.; Lemmon, E. W. *J. Chem. Eng. Data* **2011**, *56*, 3254–3264 (2,3,3,3-tetrafluoro-1-propene (HFO-1234yf))
- 2011-sam/naz Samarov, A. A.; Nazmutdinov, A. G.; Nesterova, T. N. *Izv. Vyssh. Uchebn. Zaved. Khim. Khim. Tekhnol.* **2011**, *54* (12), 40–42 (1-methylethyl methanoate, 2-methylpropyl methanoate)
- 2011-soo Soo, Ch.-B. Thesis, Paris Institute of Technology, Paris, France, 2011 (propane, butane, pentane, hexane, ethanol, 1-propanol, hexafluoropropene, pentafluoroethane, 1,1,1,2-tetrafluoroethane, 1,1,1,3,3-pentafluoropropane, 1,1,1,3,3-pentafluorobutane, decafluorobutane, trifluoro(trifluoromethyl) oxirane)
- 2011-wie/cop Wieser, M. E.; Coplen, T. B. *Pure Appl. Chem.* **2011**, *83*, 359–396
- 2011-wu/zho Wu, J.; Zhou, Y.; Lemmon, E. J. *J. Phys. Chem. Ref. Data* **2011**, *40*, 023104/1–023104/16 (dimethyl ether)
- 2012-gil/bla Gil, L.; Blanco, S. T.; Rivas, C.; Laga, E.; Fernandez, J.; Artal, M.; Velasco, I. *J. Supercrit. Fluids* **2012**, *71*, 26–44 (hexane, methanol, ethanol, 1-propanol, 1-butanol)
- 2012-hul/bas Hulse, R. J.; Basu, R. S.; Singh, R. R.; Thomas, R. H. P. *J. Chem. Eng. Data* **2012**, *57*, 3581–3586 ((E)-1-chloro-3,3,3-trifluoropropene)
- 2012-jun/mor Juntarachat, N.; Moreno, P. D. B.; Bello, S.; Privat, R.; Jaubert, J.-N. *J. Supercrit. Fluids* **2012**, *68*, 25–30 (pentane, heptane, decane)
- 2012-nik/pop Nikitin, E. D.; Popov, A. P. *Fluid Phase Equilib.* **2012**, *324*, 13–16 (ferrocene, 1-acetylferrocene, butylferrocene, benzoylferrocene)
- 2012-nik/pop-1 Nikitin, E. D.; Popov, A. P.; Yatluk, Yu. G.; Simakina, V. A. *Fluid Phase Equilib.* **2012**, *331*, 1–5 (tetraisopropoxysilane, tetraisopropoxytitanium, tetra-*tert*-butoxytitanium)
- 2012-tho/lem Thol, M.; Lemmon, E. W.; Span, R. *High Temp.-High Pressures* **2012**, *41*, 467–486 (benzene)
- 2013-bro/din Brown, J. S.; Di Nicola, G.; Fedele, L.; Bobbo, S.; Zilio, C. *Fluid Phase Equilib.* **2013**, *351*, 48–52 (3,3,3-trifluoro-1-propene)
- 2013-han/xia Han, K.; Xia, S.; Ma, P.; Yan, F.; Liu, T. *J. Chem. Thermodyn.* **2013**, *62*, 111–117 (hexane, heptane, cyclohexane, methanol, ethanol, 1-propanol, 2-methoxy-2-methylpropane)
- 2013-jun/bel Juntarachat, N.; Bello, S.; Privat, R.; Jaubert, J.-N. *J. Chem. Eng. Data* **2012**, *58*, 671–676 (propane, butane)
- 2013-nik/pop Nikitin, E. D.; Popov, A. P.; Bogatishcheva, N. S.; Kuznetsov, V. A. *Fluid Phase Equilib.* **2013**, *355*, 40–45 (1,3-propanediol, 1,4-butanediol, 1,5-pentanediol, 1,6-hexanediol, 1,8-octanediol, 1,9-nonanediol, 1,10-decanediol, 1,12-dodecanediol)
- 2013-nik/pop-1 Nikitin, E. D.; Popov, A. P.; Bogatishcheva, N. S. *Fluid Phase Equilib.* **2013**, *358*, 296–300 (1-methylnaphthalene, 2-methylnaphthalene, 1,3-dimethylnaphthalene, 1,4-dimethylnaphthalene, 2-ethylnaphthalene)
- 2013-ras/rad Rasulov, S. M.; Radzhabova, L. M.; Abdulagatov, I. M.; Stepanov, G. V. *Fluid Phase Equilib.* **2013**, *337*, 323–353 (1-butanol)
- 2014-jun/pri Juntarachat, N.; Privat, R.; Coniglio, L.; Jaubert, J.-N. *J. Chem. Eng. Data* **2014**, *59*, 3205–3219 (ethyl ethanoate, ethyl propionate, ethyl butanoate, ethyl pentanoate, ethyl hexanoate, ethyl heptanoate, ethyl octanoate, (E)-ethyl-2-butenate)
- 2014-jun/val Juntarachat, N.; Valtz, A.; Coquelet, C.; Privat, R.; Jaubert, J.-N. *Int. J. Refrig.* **2014**, *47*, 141–152 ((E)-1,3,3,3-tetrafluoroprop-1-ene (R1234ze(E)), 2,3,3,3-tetrafluoroprop-1-ene (R1234yf))
- 2014-liu/fen Liu, Q.; Feng, X.; Zhang, K.; An, B.; Duan, Y. *Fluid Phase Equilib.* **2014**, *382*, 260–269 (2-methylpropane)
- 2014-liu/zha Liu, Y.; Zhang, Y.; He, M.-g.; Xin, N. *J. Chem. Eng. Data* **2014**, *59*, 3852–3857 (hexane, heptane, octane, nonane, decane)
- 2014-nes/vos Nesterova, T. N.; Vostrikov, S. V.; Nesterov, I. A.; Nazmutdinov, A. G.; Sosin, S. E. *Fluid Phase Equilib.* **2014**, *368*, 14–38 (2-methylheptane, 2-methyloctane)
- 2014-nik/pop-1 Nikitin, E. D.; Popov, A. P.; Bogatishcheva, N. S. *J. Chem. Thermodyn.* **2014**, *68*, 288–292 (dihexyl ether, dioctyl ether, didecyl ether)
- 2014-nik/pop-2 Nikitin, E. D.; Popov, A. P. *Fluid Phase Equilib.* **2014**, *379*, 191–195 (nonadecane to tetracosane, hexacosane, octacosane, triacontane, hexatriacontane, tetracontane, tetratriacontane, hexacontane)
- 2014-nik/pop-3 Nikitin, E. D.; Popov, A. P. *Fluid Phase Equilib.* **2014**, *380*, 11–17 (benzoic acid, 2-methylbenzoic acid, 3-methylbenzoic acid, 4-methylbenzoic acid, phenylmethanol, 4-(methylphenyl)methanol, benzaldehyde, 4-methylbenzaldehyde, 2-carboxybenzaldehyde, 2,2-dimethyl-1,3-propanediol, methyl tetradecanoate, methyl hexadecanoate, methyl octadecanoate, (Z)-9-methyl octadecanoate)
- 2015-hig/hay Higashi, Y.; Hayasaka, S.; Shirai, C.; Akasaka, R. *Int. J. Refrig.* **2015**, *52*, 100–108 ((Z)-1,3,3,3-tetrafluoroprop-1-ene (R 1234ze(Z)), 1,1,1,3,3-pentafluoropropane (R 245fa))
- 2015-nik/pop Nikitin, E. D.; Popov, A. P. *J. Chem. Thermodyn.* **2015**, *80*, 124–127 (acenaphthene, fluorene, anthracene, phenanthrene, pyrene)

2015-nik/pop-1 Nikitin, E. D.; Popov, A. P. *Fuel* **2015**, 153, 634–639 (methyl hexanoate to methyl dodecanoate)